

The Architecture of Compliance: Optimal Regulation of Tokenized Securities

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Abstract

How should a tokenized security be regulated? The usual question is *stringency*: how tightly to screen. We argue the deeper question is *architecture*: where compliance binds. A *gate* excludes bad actors in advance. But it destroys composability, the source of a token's value. A *ledger* instead deters misconduct after the fact. On-chain transparency makes this cheap and accurate. It leaves composability intact. In our model the optimal architecture *flips* with traceability. Opaque assets are gated; transparent ones are policed. Past a threshold, the ledger dominates outright. Gating can also trigger adverse-selection collapse. So optimal regulation is an open, policed market.

Keywords: tokenized securities, blockchain, optimal regulation, composability, ex-post enforcement, market microstructure, public goods.

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1 Introduction

A security that has been “tokenized” is recorded and transferred as an entry on a programmable, shared ledger rather than in the books of a custodian. Issuers and regulators agree on one constraint: tokenization does not repeal securities law. The U.S. Securities and Exchange Commission put the point bluntly in early 2026: tokenizing a security does not change its legal status, and the issuer remains responsible for the same investor-protection and market-integrity obligations as before ([Securities and Exchange Commission, 2026](#)). The live policy question is how to discharge those obligations on an asset that anyone, anywhere, can in principle hold and move at machine speed.

The question is not hypothetical. Tokenized real-world assets on public chains have grown several-fold in a year, reaching roughly \$30 billion by early 2026. They are issued now not by crypto ventures but by the largest asset managers (BlackRock, Franklin Templeton, Circle, Apollo) with tokenized Treasuries and money-market funds leading and tokenized equities following ([RWA.xyz et al., 2025](#)). The tension this paper studies is already visible in how these assets are built. The dominant institutional products are *permissioned*. BlackRock’s tokenized Treasury fund restricts holders to whitelisted qualified purchasers, and the permission travels with the token: every protocol that would integrate it must confine itself to the same whitelisted set. That is the opposite of free composability. The DeFi-native issuers make the reverse choice, favoring composability and open access at the cost of screening. The market has thus split along the margin we formalize, and it has done so with no theory of which choice is right, or when. The result is a market that is large but thin. Industry trackers place the value locked inside permissioned venues at many multiples of the value freely circulating on-chain, alongside persistently low secondary-market volumes and long holding periods ([RWA.xyz et al., 2025](#)). Meanwhile the illicit share that ex-post enforcement must target sits below one percent of on-chain volume ([Chainalysis, 2025](#)).

We take the question from the ground up. Start with an *unregulated* tokenized market: open to anyone, with neither screening nor enforcement, in which legitimate investors and bad actors trade side by side and the harm the bad actors impose goes unchecked. The design question is then direct. Starting from there, what regulation maximizes welfare? This is a different starting point from the one the literature has taken. The closest prior work begins from a specific regulatory *proposal*, screening participants until the expected flow of illicit trading falls below a cap, and evaluates its consequences. It shows that the screen thins the market and, pushed far enough, makes tokenization unviable ([Cartea, Drissi, and Saleh, 2026](#)). We instead hand the regulator the full set of instruments the technology supplies and solve for the best one. The distinction matters, because the optimal regulation is not the instrument prior work evaluates. Screening is inherited from intermediated finance, where compliance has always meant vetting who may transact. It carries a clear prediction: a market open to anyone is exposed to everyone, so the more dangerous the environment, the tighter the screen should be. For a transparent

tokenized asset the optimum runs the other way. The screen should be *looser* than that logic implies, not tighter; for a sufficiently transparent asset it should be zero; and tightening the screen as the logic recommends can collapse the very market it is meant to protect.

What reverses the conventional answer is a force of market microstructure, not of cryptography. Liquidity in a securities market is priced by adverse selection: market makers widen the spread against the risk of trading with someone better informed (Glosten and Milgrom, 1985). On a tokenized asset, the screening instrument is *coupled* to that adverse-selection problem through *composability*: the token's capacity to serve as a building block in the rest of the on-chain economy. The coupling runs in a chain. Screening who may hold a token strips it of composability, because the permission propagates to every protocol and counterparty that touches it. Lost composability lowers the value of holding the token, so legitimate traders leave. Their exit raises the toxic share of the remaining order flow and widens the spread. The wider spread drives out still more traders. So the instrument meant to clean the market can instead unravel it: an adverse-selection collapse set off by the act of compliance itself. Detecting and punishing misconduct *ex post*, by reading the asset's permanent public history in the manner of Becker (1968), escapes the trap. It conditions nothing on who may participate, so it leaves composability, and the market, intact. That asymmetry is where the whole argument turns. It makes the two ways of enforcing compliance not substitutes of differing cost but instruments of different kind, and it is special to an asset whose compliance and whose liquidity are written on the same ledger.

The distinction tokenization draws is therefore one of *architecture*, where in the system compliance binds, rather than of degree. In the language of law and economics, this is the choice between two ways of protecting an entitlement. A *property rule* bars anyone from taking it without permission secured in advance; a *liability rule* lets them act and imposes damages after the fact (Calabresi and Melamed, 1972). The classic finding is that liability rules fit harmful externalities, where consent is impractical and the state sets the price of a violation, while property rules fit transferable possessions (Kaplow and Shavell, 1996). That finding already tilts toward ex-post enforcement for the misconduct at issue here. What makes the tilt decisive is the technology of the asset itself: its transparency makes the liability rule cheap to run, while its composability makes the property rule of exclusion uniquely expensive. In intermediated finance, compliance is enforced *at intermediaries*: broker-dealers run KYC, exchanges surveil trading, transfer agents maintain the register. A token can dissolve those intermediaries. But it offers two enforcement substitutes that intermediated finance does not. First, compliance can be *embedded in the asset*: a transfer-restriction rule coded into the token travels with it, enforced by the protocol rather than by a gatekeeper. Second, the asset's entire history is *permanently and publicly auditable*: every transfer is observable forever, so misconduct can be detected and punished ex post at a fraction of the cost of off-chain investigation. The IMF's 2026 assessment of tokenized finance captures both features and draws the supervisory implication we formalize

below: when “trading, settlement, custody, and compliance are embedded in code, supervision must extend beyond market participants to the design, governance, and resilience of market infrastructures themselves” (International Monetary Fund, 2026).

These two features define two enforcement architectures, and the choice between them is the subject of this paper.

We call ex-ante, embedded permissioning the *gate*. A gate of intensity g (with $g = 0$ fully open and $g = 1$ fully closed) screens entrants: it admits fewer bad actors, at the cost of also excluding some legitimate ones and, this is the feature unique to tokenization, of destroying *composability*. A token whose transfers are whitelisted cannot freely serve as collateral in a permissionless lending protocol, cannot trade against an open automated market maker, and cannot move across venues without re-clearing the whitelist. Composability is the property that makes a token more than a database row; embedding a gate in the asset is the act of giving it up. The crucial asymmetry with intermediated finance is here: when a broker runs KYC on a client, the asset the client buys is no less usable anywhere else; when a gate is written into a token, the asset is less usable everywhere else. The reason is that permissioning is *contagious*: because exposure to the token is transitive through the composition graph (a protocol that holds it, and a user who draws against that protocol, both acquire its exposure) the whitelist must hold throughout the token’s composition closure, so a gate fragments the asset into a permissioned sub-network. We derive this in §2.2 rather than assume it, and show in §6.4 that the results need only that the gate be more contagious than the ledger, which the argument guarantees. The gate’s cost is borne in the coin of liquidity and capital formation.

We call ex-post detect-and-punish enforcement the *ledger*. Enforcement of intensity e (the resources committed to ex-post detection), operating on a ledger of transparency θ (the asset’s traceability, from $\theta = 0$ opaque to $\theta = 1$ transparent), detects a bad actor’s misconduct with some probability and imposes a penalty; in the manner of Becker (1968), the expected penalty deters entry. The ledger reduces illicit activity not by excluding anyone but by making misconduct unprofitable in expectation. It therefore leaves composability untouched: a permissionless, fully composable token can be policed ex post because its history is public. The ledger’s cost is borne in the coin of enforcement resources, and on-chain transparency makes those resources go further than any off-chain regime could.

We write down a self-contained model (its own agents, its own microfoundation of liquidity, its own welfare criterion) in which a planner chooses the gate g and the ledger intensity e for an asset of given traceability θ . We deliberately do not inherit the equilibrium machinery of the existing literature; the point is to ask the architecture question from primitives. Bad actors with heterogeneous private benefits decide whether to participate and offend; legitimate investors trade in a market whose liquidity is set, in the manner of Glosten and Milgrom (1985), by the share of toxic order flow they face and by the composability premium the token earns. The planner maximizes legitimate surplus net of the external harm bad actors cause and the

resource cost of enforcement.

Four results organize the paper.

1. *Two margins, two currencies (Propositions 2–3)*. The gate and the ledger are substitutes in suppressing illicit activity: each lowers the mass of active bad actors, and each thereby raises legitimate surplus by cleaning the order flow. They are not substitutes in cost. The first-order conditions are identical in form save for one block of terms: the gate’s marginal cost contains the loss of composability, the exclusion of legitimate users, and the thinning of the legitimate base; the ledger’s marginal cost is the pure resource cost of enforcement. Optimal regulation equates the social value of cleaning the market to a *different* marginal cost for each instrument, and leans on whichever is cheaper.

2. *The architecture flips (Proposition 5)*. Because tokenization simultaneously raises the price of the gate (composability is valuable and is what the gate destroys) and lowers the price of the ledger (transparency makes detection cheap), the optimal mix is not invariant to the technology. As traceability θ rises, the optimal enforcement intensity rises and the optimal gate falls; above a threshold $\bar{\theta}$ the optimal gate is *exactly zero*. Opaque securities (paper certificates, intermediated book-entry, privacy-preserving chains) are optimally gated. Transparent tokenized securities are optimally left permissionless and policed ex post. The optimal regulation of a transparent security is the opposite of the optimal regulation of an opaque one.

3. *Tokenization dominates, it does not merely trade off (Proposition 6)*. The stringency framing implies a frontier: more compliance means less liquidity. That frontier is a property of the gate technology. The ledger technology has a different, better frontier. With sufficient traceability, and a gate that is not a near-perfect screen, the entire attainable liquidity–compliance frontier under the ledger lies strictly above and to the left of the *best* outcome attainable under the gate: more liquidity *and* less illicit activity than any gated regime can reach. The naive intuition that permissionless markets must be more lawless has the comparative static backwards. A transparent market can be policed without being gated, and a market that is policed but not gated keeps the liquidity that gating would have destroyed. Whether the planner spends this “transparency dividend” on liquidity, on compliance, or on both depends on how harmful the bad actors are; that the dividend exists does not.

4. *Regulate the standard, not the venue (Proposition 7)*. Composability has a second, less obvious consequence. Because a token admitted on any one venue can transact across all venues, the effective stringency of a gate is set by the *laxest* venue that admits the asset: gating is a weakest-link public good. Venue-by-venue regulation therefore fails, each venue free-rides on others’ screening, the equilibrium gate is inefficiently low, and any single permissive venue undermines compliance system-wide. Ex-post enforcement suffers no such failure, because there is one shared ledger and one enforcement authority whose reach is global by construction. The implication is that effective ex-ante compliance, if used at all, must be imposed at the level of the *token standard* rather than the venue, while the natural instrument for a composable

system is ledger enforcement. This is the formal content of the supervisory shift “beyond market participants to . . . market infrastructures” that policy bodies have begun to articulate.

A fifth result tempers the first four and reconciles them with practice. Detection probabilities are bounded by traceability, and penalties are bounded by wealth, so a hard core of high-benefit, judgment-proof actors cannot be deterred by any feasible ledger. When the harm such actors do is severe enough, a *small* residual gate is optimal even at full traceability, reserved for this undeterrable tail (Proposition 8). This explains why even the most transparent tokenized markets keep narrow permissioning, such as the screening of sanctioned addresses, rather than going fully open. The residual gate is not a vestige of the stringency regime; it is the optimal handling of the actors the ledger cannot reach.

For policy, the four results compose into a single inversion of current practice. Today most tokenized securities are issued into permissioned, whitelisted venues, compliance by exclusion, the stringency prescription ported intact from intermediated finance. For a transparent asset the model says this is close to backwards. Keep access *permissionless* and police the shared ledger *ex post*. When ex-ante screening is used at all, write it into the *token standard* rather than leaving it to individual venues, where it is a weakest-link public good. And reserve the gate for the narrow, undeterrable tail rather than applying it to the market at large. The supervisory object thus shifts: from vetting who may trade, to designing the standard and resourcing the enforcement that a public ledger makes cheap. This is the same shift, beyond participants to infrastructures, that regulators have begun to articulate without a model to ground it.

These results describe what a planner would choose. The second half of the paper takes the model to the institutions that actually choose, and the architecture thesis sharpens at each step. The gate and the ledger are set not by one planner but by two agents: the issuer permissions, the regulator polices. The issuer captures only part of the surplus its market generates, so it over-permissions. The walled garden that dominates real tokenized markets is its rational response to liability, not evidence that gating is efficient. Ex-post enforcement crowds out the issuer’s private gating, so a regulator induces open markets by building enforcement rather than by mandating openness. Whether enforcement crowds gating out or *in* depends on the liability regime the regulator writes; and inducing openness requires committing to enforcement, which the regulator’s capacity or reputation supplies. Finally, once the issuer also chooses how transparent to be, transparency, openness, and enforcement reinforce one another. The market can then lock into a self-confirming opaque, walled configuration that only committed enforcement dislodges. The unifying thesis is that optimal regulation of a tokenized security is not the calibration of a screening level. It is the selection, through credible enforcement, of a transparent, open, policed equilibrium: an architecture the regulator’s credibility calls into being, not a rule it applies to assets it is handed.

The nearest prior model of this question is [Cartea, Drissi, and Saleh \(2026\)](#). They take a regulatory proposal as given (a cap on illicit trading, discharged by ex-ante screening, the issuer’s

only compliance instrument) and evaluate its consequences. In that setting, tighter regulation necessarily thins the market and, beyond a point, makes tokenization non-viable: compliance is bought with liquidity because screening is the sole lever, and the policy conclusion is that regulation imperils tokenization. Our point of departure is to add the instrument the on-chain environment actually supplies (*ex-post enforcement on the public ledger*) and the conclusion inverts. The stringency–liquidity trade-off is not a law but an artifact of attending only to the gate. Their single-instrument setting is the $e = 0$ restriction of ours, where the same trade-off reappears: tighter gating thins the legitimate base and, pushed far enough, makes the market non-viable. Our contribution is to characterize what becomes available once that restriction is lifted. For an asset transparent enough to be policed, the ledger *dominates*: it attains higher welfare and strictly more liquidity at the same compliance (Proposition 18). A permissionless market is thus more liquid *and* no less compliant than any gated one, and the optimal gate falls to zero. The policy reading reverses with it. Where the single-instrument model concludes that regulation endangers tokenization, ours shows that regulation *fitted to the technology*, open access policed ex post, makes tokenization viable precisely where screening would have killed it. This is why the first-order object is the enforcement *architecture*, not the level of a screening instrument, and why a regulator’s question should be where compliance binds rather than how strict to set the screen. We depart from the DEX literature in the same direction. [Harvey, Hasbrouck, and Saleh \(2026\)](#) examine the risks, benefits, and oversight of decentralized trading venues, and [Hasbrouck, Rivera, and Saleh \(2025\)](#) model the economics of liquidity provision on an automated market maker. Both take the decentralized venue as the object of study, whereas we ask how a regulator discharges compliance on such a venue, and supply the enforcement architecture that regulates the *ledger* rather than the intermediary, not merely feasible but welfare-dominant.

The coupling is the paper’s main contribution, and it is more than a formalization of the fact that gating costs liquidity. Its sharp consequence is that the two instruments, though both suppress illicit activity, are *not substitutes*: the ledger is unraveling-safe and the gate is not, so no quantity of gating delivers the market quality enforcement delivers (Proposition 18). This is the precise sense in which the regulatory problem is one of architecture and not of degree, and the flip is its corollary rather than its substance. Both halves of the mechanism are derived from primitives, not assumed, composability from the contagion of permissioning (Lemma 1), the liquidity feedback from adverse-selection microstructure (Lemma 3), so that the reversal of the stringency prescription is a property of the environment.

In treating enforcement as a choice between exclusion and deterrence we draw on the economics of public enforcement ([Becker, 1968](#)) and the judgment-proof problem ([Shavell, 1986](#)); in modeling composability as a cross-venue externality with weakest-link structure we draw on the theory of public goods ([Hirshleifer, 1983](#)). The liquidity block builds on adverse-selection microstructure ([Glosten and Milgrom, 1985](#); [Kyle, 1985](#)). Relative to the broader economics of

blockchains (Cong and He, 2019), our contribution is to show that the features usually cataloged as regulatory *problems* (disintermediation, permissionless access, public transaction histories) are, taken together, the ingredients of a *better* enforcement technology, and to characterize the regulation that exploits it.

Most of all, the paper extends the theory of financial intermediation to a setting where the friction that rationalizes the intermediary is relaxed. Classical theory explains the intermediary as a response to information and monitoring costs. Intermediaries arise to monitor borrowers on investors' behalf, as *delegated monitors* (Diamond, 1984); and, where outcomes are costly to observe, to economize on the costly *state verification* that contracting otherwise requires (Townsend, 1979). The enforcement institutions of securities markets (the broker-dealer who runs know-your-customer checks, the exchange that surveils trading, the transfer agent who keeps the register) are the compliance embodiment of this logic. Conduct is policed *at intermediaries* because verifying it is costly and screening must happen at a gate. Tokenization relaxes that friction: a permanent, public record makes conduct cheap to verify, so the monitoring that intermediation theory delegates to an institution can instead be performed ex post on the ledger. In these terms, our gate-versus-ledger choice is the choice between two architectures, ex-ante screening at a gatekeeper, or its post-intermediation substitute, ex-post detection on a transparent ledger. The central result is that the substitute dominates once the asset is transparent enough that verification is cheap. The paper thus supplies a theory of *when the intermediary's compliance function is optimally disintermediated*: a question the intermediation literature, holding the verification friction fixed, did not need to ask. The claim is confined to that function. A ledger does not replicate the liquidity and maturity transformation, certification, or risk-bearing that intermediaries also provide, so disintermediating screening complements the intermediary rather than replacing it wholesale. The argument also connects back to the market-microstructure tradition from which we borrow the adverse-selection engine (Glosten and Milgrom, 1985; Kyle, 1985). There the intermediary is the market maker who *prices* the adverse selection that asymmetric information creates. Here the prior question is whether an asset's own transparency can substitute for the screening intermediaries traditionally supply, and the toxicity the market maker prices is what the ledger, by deterring informed misconduct, reduces at its source.

In this respect the paper is a closer counterpart to the recent literature on intermediation in decentralized markets than to the classical theory, and it complements that literature on two margins. The first is the market-making technology. In decentralized markets the specialist is replaced by an automated market maker, and a body of work characterizes the adverse-selection cost its liquidity providers bear. This is the automated-market-maker form of the toxic arbitrage that Foucault, Kozhan, and Tham (2017) model for dealer markets, characterized in the equilibrium pool-provision analyses of Lehar and Parlour (2025) and Capponi and Jia (2025). That literature treats informed order flow as a fixed feature of the environment and redesigns

the *pool* to withstand it. We instead treat that flow as the conduct of bad actors and ask how *enforcement* reduces it, market design and regulation as complementary responses to the same adverse selection our microstructure block prices. The second margin is the issuer. The capture rate ω (the share of its market’s surplus the issuer captures) and the platform’s permissioning belong to the industrial organization of token issuance, where platforms set fees and subsidize participation (Cong, Li, and Wang, 2021), and where Sockin and Xiong (2023) show that tokenization can itself serve as a *commitment device* that keeps a platform from exploiting its users. Our decentralized model is the compliance-side counterpart. The walled garden is the issuer’s permissioning strategy and ω its fee markup, and the regulator’s credible enforcement is the commitment device that resolves the issuer’s coordination failure: a compliance analogue of the user-protection commitment that tokenization supplies in that work. Relative to both strands the paper’s addition is the same: the enforcement architecture. Neither the market-design nor the token-IO literature asks *where* compliance should bind, and that is the question that decides whether a decentralized market ends up open and policed or walled and opaque.

One form of on-chain misconduct bears directly on the enforcement side of our argument, and the finance literature has already modeled its economics. *Maximal extractable value* (MEV), front-running and sandwich attacks executed by paying for favorable transaction ordering, is the on-chain version of latency arbitrage. Budish, Cramton, and Shim (2015) show that the continuous limit order book invites just this: a race to exploit stale prices that taxes ordinary traders and that competition only intensifies rather than competes away. The race generates the socially excessive speed investment of Biais, Foucault, and Moinas (2015) and the arms-race tax quantified by Aquilina, Budish, and O’Neill (2022). Its market-quality cost is the *toxic arbitrage* of Foucault, Kozhan, and Tham (2017): traders who pick off stale quotes raise the adverse selection that dealers face and impair liquidity. This is the toxicity our microstructure block prices, now arriving through the order-sequencing channel rather than the asymmetric-information channel. The on-chain realization is documented in the security literature (Daian et al., 2020), where the public mempool plays the role of the latency channel. It bears on our argument in three ways. First, it widens the cast of bad actors the ledger is meant to deter: the searcher who sandwiches an honest trade is the on-chain toxic arbitrageur, an offender ex post detection and punishment can target. Second, it is the empirical realization of the one channel in our model where transparency turns against itself, because the public mempool is a disclosure that arms the attacker: the visibility that lets a regulator detect misconduct ex post also lets a front-runner commit it ex ante. Third, the remedy is *architectural* in our sense rather than a restriction on who may trade. The market-design response in finance is the frequent batch auction of Budish, Cramton, and Shim (2015), which converts competition on speed into competition on price. Its on-chain analogues are fair-ordering consensus and encrypted mempools that conceal pending orders from searchers while preserving settlement: the latter being the sequencing analogue of our zero-knowledge result, revealing to the party who must

verify and concealing from the party who would exploit. The convergence is the point: in the order-flow literature as in ours, the misconduct problem is addressed by the architecture of disclosure and ordering, not by the stringency of access.

The remainder of the paper proceeds as follows. Section 2 sets up the model. Section 3 proves the core results: the architecture flip, the dominance of the ledger, and the residual gate for the undeterrable tail. Section 4 establishes the three load-bearing assumptions on which those results rest. Section 5 illustrates the flip in a calibrated example, and Section 6 derives the interaction among the levers and shows the flip is globally monotone, bounded below by a residual gate only under two precise conditions. Section 7 replaces the reduced-form liquidity block with an explicit Glosten–Milgrom market and endogenous participation, deriving a breakdown threshold and showing the ledger, not the gate, to be the instrument that can prevent collapse. Section 8 maps the model’s parameters to observables and states the predictions that distinguish the coupling mechanism from the stringency view. Section 9 embeds the model in a system of many issuers sharing one network and one ledger, where the gate is socially futile and the decentralized equilibrium over-gates and under-enforces. Section 10 draws the policy implications and Section 11 concludes. Three companion papers extend the analysis. Companion A develops the dynamics of the toxicity trap and the tipping role of enforcement. Companion B recasts composability as a percolation network and solves the general equilibrium of many issuers sharing one ledger. Companion C studies the political economy of who gates and who polices. Proofs are in the Appendix.

2 The Model

The economics the model must capture is a three-cornered tension. A regulator wants to keep bad actors (fraudsters, manipulators, money launderers) out of a securities market, and has two broad ways to do it: screen them at the door, or catch them after the fact. Screening is the familiar instrument of securities regulation; we call it a *gate*. Catching offenders after the fact by reading a permanent, auditable record of who transacted with whom is an instrument that public blockchains make newly cheap; we call it a *ledger* regime. The two would be near-interchangeable but for one difference tokenization creates: a token that screens who may hold it cannot be freely combined with the rest of the on-chain economy, so the gate sacrifices *composability* in a way the ledger does not. Both instruments act on a market that is, in the sense of Glosten and Milgrom (1985), an adverse-selection market: liquidity providers widen spreads against the risk of facing a better-informed counterparty, so the presence of bad actors and the departure of legitimate ones both raise the cost of trading.

The regulator’s problem is one of design relative to the *unregulated* market: at $g = e = 0$ neither instrument operates, access is open to all and no misconduct is detected, so this laissez-faire allocation is the status quo against which any regulation is measured, and the regulator’s

task is to choose how far to depart from it, and in which direction. The model that follows is the smallest structure containing these three forces: the regulator’s two instruments, the composability cost that separates them, and the adverse-selection market they act on. Two choices are worth flagging at the outset. First, every technology is *derived from primitives*: the detection probability, the gate’s error rates, the composability premium, and the spread are not posited functional forms but consequences of an explicit microfoundation, so that the central results are properties of the environment rather than modeling conveniences. Second, the economic argument lives in the two instruments and their asymmetry (§2.2); a reader following the economics can treat the equilibrium construction (§§2.3–2.7) as standard and return to it. The economy is static and all agents are risk-neutral.

2.1 Agents and endowments

A unit-priced risky security has uncertain fundamental value

$$V \in \{v_0 - \delta, v_0 + \delta\}, \quad \Pr(V = v_0 + \delta) = \Pr(V = v_0 - \delta) = \frac{1}{2},$$

where $v_0 > 0$ is the prior mean and $\delta > 0$ measures the security’s informational sensitivity: the rent available to an agent who learns or manipulates V . Two unit-measure-normalized continua of agents trade it. *Legitimate investors*, of mass $\mu > 0$, trade the security to realize private gains from trade: hedging, rebalancing, meeting liquidity needs. *Bad actors*, of mass $\beta > 0$, can extract value from the market through fraud, manipulation, or trading on illicitly obtained information, and impose social harm when they do. Identities are pseudonymous: an address does not reveal its owner’s type. On a tokenized ledger, however, the *transaction history* attached to an address is public and permanent: the property we make precise through the traceability parameter below, and the feature the ledger instrument exploits.

A bad actor who acts (“offends”) obtains a private benefit $b \geq 0$ and imposes social harm $\eta > 0$. Benefits are heterogeneous and privately known, drawn independently across bad actors from an exponential distribution with rate $\rho > 0$; we write its survival function

$$\bar{S}(x) \equiv \Pr(b \geq x) = e^{-\rho x}, \quad x \geq 0.$$

The harm decomposes as $\eta = \eta_I + \eta_E$, where $\eta_I \geq 0$ is harm to legitimate investors *inside* the market and $\eta_E \geq 0$ is harm to third parties *outside* it. As shown in §2.4, an offending bad actor’s only effect inside the market is to act as an informed (toxic) counterparty; the internal harm η_I is therefore not a separate primitive but is the equilibrium adverse-selection loss, which is endogenous to the model. The external harm η_E (the social cost of illicit finance, laundering, sanctions evasion) enters welfare directly.

A legitimate investor i has a private gains-from-trade value $x_i \geq 0$ for executing a trade, drawn independently from an exponential distribution with rate $\xi > 0$ and survival function

$\bar{G}(x) = e^{-\xi x}$. This value captures the surplus the investor realizes from rebalancing toward a desired position; it is the source of the “noise” (liquidity-motivated) order flow in the market.

2.2 The regulator’s instruments and the asset’s traceability

A *regime* is a pair (g, e) applied to a security whose ledger has *traceability* $\theta \in [0, 1]$. Traceability is a design property of the token and its chain, the probability that a given transaction is publicly attributable to its originator, and is taken as exogenous through §3, then endogenized in a companion paper. The intensity $g \in [0, 1]$ is the strength of an ex-ante *gate*; the intensity $e \geq 0$ is the level of ex-post *ledger* enforcement. We now specify each technology and derive its functional form.

Detection technology. We microfound the probability that an offense is detected. Conditional on an offense, two independent events must both occur for the authority to detect and sanction it. First, the offending transaction must be *attributable*: the on-chain trail must be public and linkable to the perpetrator, which occurs with probability θ and fails with probability $1 - \theta$ when obscured by privacy tooling or an off-chain leg. Attributability is a property of the asset and is independent of enforcement effort; no quantity of effort detects an unattributable offense. Second, conditional on attributability, the authority must *discover* the offense among the mass of on-chain activity. Enforcement effort e generates investigative leads according to a Poisson process of intensity e , and the offense is discovered if at least one lead lands on it, which occurs with probability $1 - e^{-e}$. Detection requires both, so by independence the detection probability is

$$p(e, \theta) = \theta(1 - e^{-e}) \in [0, \theta]. \quad (1)$$

This derived technology has the properties the analysis uses: p is strictly increasing and strictly concave in effort ($p_e = \theta e^{-e} > 0$, $p_{ee} = -\theta e^{-e} < 0$), strictly increasing and linear in traceability ($p_\theta = 1 - e^{-e} > 0$), and bounded above by the attributability ceiling θ , which no effort can exceed.

The gate. The gate is an admission test applied at entry, embedded in the token standard so that it binds wherever the token trades. Its intensity g indexes a one-parameter family of admission policies with two error rates. A bad actor is (correctly) rejected at rate sg and so is admitted with probability

$$a_B(g) = 1 - sg, \quad s \in (0, 1],$$

while a legitimate investor is (wrongly) rejected at rate λg and admitted with probability

$$a_L(g) = 1 - \lambda g, \quad \lambda \in (0, s).$$

The restriction $0 < \lambda < s$ makes the test *informative*: raising intensity rejects bad actors faster than legitimate ones. The restriction $s \leq 1$ admits the possibility of a perfect screen only in the limit; for $s < 1$ the gate is imperfect and, as the analysis shows, cannot substitute fully for the ledger.

Composability, microfounded. The distinctive cost of an asset-embedded gate is the loss of *composability*: the capacity of the token to serve as a component in other on-chain protocols. We derive this cost, and the asymmetry between the two instruments, rather than assume it. Let the on-chain ecosystem be a unit measure of potential *composition partners*: lending protocols that might hold the token as collateral, automated market makers that might quote it, vaults that might wrap it, other tokens it might pair with. A holder who can compose the token with a subset of partners of measure n obtains composability value $v(n)$. Under a constant-elasticity (Dixit–Stiglitz) love-of-variety aggregator over a continuum of composition services, this value is $v(n) = u_1 n^\alpha$ with $\alpha \in (0, 1)$, strictly increasing and strictly concave, the marginal value of breadth largest when little remains.¹

The two instruments act on the reachable set in structurally different ways, and this is the model’s central asymmetry. A *gate* is a permissioning predicate embedded in the token: a whitelist that its transfer function consults. Composition with partner P then requires P to satisfy the predicate; and because exposure is transitive, it requires P ’s counterparties to satisfy it too. If a lending protocol holds the gated token as collateral, a borrower who draws against that collateral acquires the token’s economic exposure, so admitting an un-whitelisted borrower would let an excluded actor obtain the token through P : defeating the gate. Compliance therefore propagates: the whitelist must hold throughout the token’s *composition closure*. Permissioning is contagious. A gate of intensity g —interpreted as the fraction of partners that will not or cannot join the whitelist, whether for privacy, jurisdiction, or the cost of subjecting a smart contract and its users to KYC—shrinks the reachable set to measure $\varphi = 1 - g$, and the premium is $v(\varphi) = u_1 \varphi^\alpha$.

A *ledger* regime is a predicate of the opposite logical type. Detection reads the shared history and punishes violations *ex post*; it conditions nothing on who may interact with the token *ex ante*. Any partner may compose; an offending composition is caught after the fact through the audit trail, not blocked before it. A postcondition does not propagate through the composition closure, because it imposes no requirement a partner or its counterparties must satisfy to interact. Hence enforcement leaves the reachable set intact. We record the asymmetry the rest of the paper turns on.

Lemma 1 (Composability asymmetry). *Composability is $\varphi = 1 - g$ with premium $u_1 \varphi^\alpha$. The gate destroys composability, $\partial \varphi / \partial g = -1 < 0$; the ledger does not, $\partial \varphi / \partial e = \partial \varphi / \partial \theta = 0$. This is*

¹Standard CES result: utility from a continuum of measure n of symmetric varieties with elasticity of substitution $1/(1 - \alpha)$ is proportional to n^α . The concavity $\alpha < 1$ is what makes the gate’s marginal cost rise steeply as $g \rightarrow 1$ and delivers interior optimal architectures.

not a parametric choice but a consequence of the logical type of each instrument: the gate is a precondition on interaction, which propagates through the token's composition closure, whereas detection is a postcondition, which does not. The crucial contrast with intermediated finance is the same fact stated for the legacy system: a broker's KYC is a precondition attached to a client, not to the asset, so it does not propagate to the asset's other uses; an embedded gate is a precondition attached to the asset, so it does.

The microfoundation also disciplines the extreme. The clean form $\partial\varphi/\partial e = 0$ is the limit in which the ledger's transparency requirements impose no composition friction of their own. Section 6 relaxes it (auditable enforcement may demand standardized interfaces, excluding some bespoke compositions) and shows that what the results require is only that the gate be *more* contagious than the ledger, which the logical-type argument guarantees: a precondition excludes a partner and its entire downstream closure, a postcondition at most the compositions that cannot meet a reporting standard.

Penalty. A detected offender pays a penalty $K > 0$ (a fine, disgorgement, or asset seizure). Penalties are transfers from offenders to the public purse and so do not enter welfare directly; they affect the planner only through deterrence, as in [Becker \(1968\)](#). Operating the ledger costs the authority $\frac{c_e}{2}e^2$ in real resources (forensic labor, chain-analysis capacity), convex in effort; operating the gate costs $c_g g$, with $c_g \geq 0$ (set to zero in the baseline).

2.3 Bad actors: the offending decision and deterrence

A bad actor first faces the gate and is admitted with probability $a_B(g)$. An admitted bad actor of type b then chooses whether to offend. Offending yields b but incurs the expected penalty $p(e, \theta) K$; not offending yields zero.² Hence the bad actor offends if and only if

$$b \geq \kappa(e, \theta) \equiv p(e, \theta) K = \theta(1 - e^{-e})K,$$

where κ is the *deterrence threshold*. The mass of *active* bad actors, admitted and offending, is the admission rate times the offending probability times the population:

$$B(g, e; \theta) = a_B(g) \beta \bar{S}(\kappa(e, \theta)) = (1 - sg) \beta e^{-\rho\kappa(e, \theta)}. \quad (2)$$

Equation (2) is the model's behavioral core: bad activity is suppressed along two margins, a *multiplicative exclusion* margin $1 - sg$ operated by the gate and a *deterrence* margin $e^{-\rho\kappa}$ operated by the ledger. The following lemma records the comparative statics used throughout; all proofs are in the Appendix.

²In the baseline an admitted bad actor's outside option is normalized to zero and the penalty is a fixed K . A companion paper enriches the decision by allowing the offender to invest in evading detection and by making the penalty confiscatory.

Lemma 2 (Suppression margins). *B is strictly decreasing in each of g , e , and θ , with*

$$B_g = -s\beta e^{-\rho\kappa} < 0, \quad B_e = -\rho\kappa_e B < 0, \quad B_\theta = -\rho\kappa_\theta B < 0,$$

where $\kappa_e = \theta K e^{-e} > 0$ and $\kappa_\theta = K(1 - e^{-e}) > 0$. The gate's suppression $-B_g$ is independent of e in level but its offending content $\bar{S}(\kappa)$ falls as e or θ rises; the ledger's suppression $-B_e$ is increasing in θ for fixed e whenever $\rho\kappa < 1$.

2.4 The market: adverse selection and the spread

Liquidity is determined by adverse selection in the sequential-trade framework of [Glosten and Milgrom \(1985\)](#). Trade occurs through a competitive, risk-neutral market-making sector that posts a bid and an ask and earns zero expected profit. In a trading round a single agent arrives and submits a market order for one unit. The arriving agent is an active bad actor, who, having chosen to offend, does so *by trading*: such an agent is informed (knows V) or extracts the rent δ by manipulation, and trades in the profitable direction, or a participating legitimate investor, who trades for liquidity reasons and is equally likely to buy or sell. Let π denote the probability that the arriving order comes from an active bad actor: the *toxicity* of order flow.

Lemma 3 (The adverse-selection spread). *With a competitive risk-neutral market-making sector and a toxic order-flow share π , the unique zero-profit quotes are the ask $A = v_0 + \delta\pi$ and the bid $\underline{A} = v_0 - \delta\pi$, so the half-spread, the per-trade cost borne by a legitimate investor, is*

$$\varsigma(\pi) = \delta\pi.$$

The half-spread is the informational rent δ scaled by the toxic share π . The toxic share depends on how many legitimate investors participate. Write $N(g) \equiv a_L(g)\mu = (1 - \lambda g)\mu$ for the mass of *admitted* legitimate investors, and let $L \leq N$ be the mass who actually trade. Then

$$\pi = \frac{B}{B + L}.$$

In the baseline we assume *full participation*, $L = N$, so that π equals the population toxicity

$$\tau(g, e) \equiv \frac{B}{N + B},$$

and the half-spread is $\varsigma = \delta\tau$. Writing $\sigma \equiv \delta$ for the informational-rent parameter, the per-trade cost is $\sigma\tau$, the reduced form used in the core analysis. Full participation is the relevant regime when the composability premium is large enough that every admitted investor's value covers the spread; §7 characterizes exactly when it holds, shows that the market otherwise breaks down, and proves existence and uniqueness of the equilibrium with endogenous L .

Remark 1 (Why η_I is not double-counted). The internal harm bad actors do to legitimate

investors is the adverse-selection loss in Lemma 3: each toxic encounter transfers informational rent from liquidity providers (passed through to investors via the spread) to the offender. The parameter $\sigma = \delta$ therefore is the internal-harm channel, and we do not add η_I separately to welfare.

2.5 Legitimate investors: participation and surplus

An admitted legitimate investor with gains-from-trade value x trades iff the value, augmented by the composability benefit of holding the token, covers the half-spread. We let composability enter as a reduction in the effective cost of participating, so the investor trades iff $x \geq \varsigma - \psi(\varphi)$, where $\psi(\varphi) = u_1 \varphi^\alpha$ is the composability premium of §2.2.³ Under full participation, aggregate legitimate surplus is the admitted mass times the per-capita net value:

$$S_L(g, e) = N(g) \cdot \max\{\pi_{\text{pc}}(g, e), 0\}, \quad \pi_{\text{pc}}(g, e) = u_0 + u_1 \varphi(g)^\alpha - \sigma \tau(g, e), \quad (3)$$

where $u_0 > 0$ is a baseline participation value and the three terms are, respectively, the baseline value, the composability premium, and the adverse-selection cost. We read S_L as the social value the market creates for legitimate users: gains from trade, liquidity services, and the capital formation they support. The gate lowers S_L through three channels, exclusion ($N \downarrow$), composability ($\varphi \downarrow$), and, by thinning the legitimate base, toxicity ($\tau \uparrow$); the ledger raises S_L through one channel, deterrence ($\tau \downarrow$), and touches no other.

2.6 Timing and equilibrium

The extensive form is as follows. (0) Nature draws the asset's traceability θ and the regulator commits to a regime (g, e) . (1) Nature draws each agent's type (bad actors' benefits b , legitimate investors' values x) and the fundamental V , observed by informed agents. (2) The gate admits each agent with probability $a_B(g)$ or $a_L(g)$ according to type. (3) Admitted bad actors choose whether to offend; admitted legitimate investors choose whether to trade. (4) Trade occurs through competitive market making; payoffs and harm are realized; detected offenders pay the penalty.

For a fixed regime (g, e) and traceability θ , the continuation from stage (1) defines an equilibrium of the agents' subgame.

Definition 1 (Equilibrium). Given (g, e, θ) , an equilibrium consists of an offending decision for each admitted bad actor, a participation decision for each admitted legitimate investor, and market-maker quotes, such that: (a) each admitted bad actor of type b offends iff $b \geq \kappa(e, \theta)$; (b) each admitted legitimate investor of type x trades iff $x \geq \varsigma^* - \psi(\varphi)$; (c) quotes are competitive

³In the full-participation baseline the premium ψ is large relative to the spread, so the participation constraint never binds ($L = N$) and the term plays no role in the core results; it becomes active in §7. We carry it here so that the baseline is the genuine special case of the general model rather than a separate model.

and break even given the resulting toxic share $\pi^* = B/(B + L^*)$, so $\varsigma^* = \delta\pi^*$ by Lemma 3, with L^* the participating mass from (b). The induced aggregates are $B(g, e)$ in (2), participation L^* , and legitimate surplus S_L .

In the full-participation baseline of §2.5–§2.4, the composability premium covers the spread for every admitted investor, so $L^* = N$, $\pi^* = \tau$, and $\varsigma^* = \sigma\tau$; the equilibrium is then unique and in closed form. When participation responds to the spread, conditions (b)–(c) form a fixed point whose existence and uniqueness, and the possibility of adverse-selection breakdown, are established in Proposition 15 of §7. The planner in (5) below chooses (g, e) anticipating this equilibrium.

2.7 Welfare and the planner’s problem

Welfare is the sum of legitimate surplus and the public purse, net of external harm and resource costs. Because the penalty K is a transfer (paid by detected offenders, received by the authority) it cancels from the social sum and influences welfare only through the deterrence it produces. We do not credit bad actors’ private benefits b in the objective: illicit gains are given no social weight, the standard normative stance in the economics of enforcement.⁴ Welfare is therefore

$$W(g, e; \theta) = S_L(g, e) - \eta_E B(g, e) - \frac{c_e}{2} e^2 - c_g g. \quad (4)$$

The planner chooses the regime to maximize welfare, taking the asset’s traceability as given:

$$(g^*(\theta), e^*(\theta)) \in \arg \max_{(g, e) \in [0, 1] \times \mathbb{R}_+} W(g, e; \theta), \quad W^*(\theta) \equiv W(g^*(\theta), e^*(\theta); \theta). \quad (5)$$

Proposition 1 (Existence). *For every $\theta \in [0, 1]$ the program (5) has a solution. Any solution also has $e^*(\theta) \leq \bar{e} \equiv (\eta_E + \sigma)\rho K\beta/c_e$, so the effective domain is the compact rectangle $[0, 1] \times [0, \bar{e}]$.*

The bound $\bar{e}$ follows because the marginal benefit of enforcement, $MV \cdot (-B_e) = MV \rho \kappa_e B$, is at most $(\eta_E + \sigma)\rho K\beta$ (using $MV = \sigma N^2/(N + B)^2 + \eta_E \leq \eta_E + \sigma$, $\kappa_e \leq K$, and $B \leq \beta$) while its marginal cost $c_e e$ grows without bound; beyond $\bar{e}$ the objective is strictly decreasing in e , so the supremum is attained on the compact rectangle $[0, 1] \times [0, \bar{e}]$, on which W is continuous. The next section characterizes the solution.

Remark 2 (Allocation of authority). Problem (5) is stated for a single planner to isolate the architecture question. In application the instruments have distinct owners: traceability θ and

⁴This is a normalization, not a substantive restriction. Crediting a fraction $\omega \in [0, 1]$ of offenders’ benefits adds $\omega(1 - sg)\beta \int_{\kappa}^{\infty} b \rho e^{-\rho b} db$ to welfare: the realized illicit gains of the admitted, offending mass. This raises the social cost of *suppressing* bad activity for both instruments, but asymmetrically: the gate forgoes offenders at their *average* benefit $\mathbb{E}[b \mid b \geq \kappa] = \kappa + 1/\rho$, whereas the ledger deters them at the *margin*, benefit κ , the lowest above the threshold. Because the average strictly exceeds the margin, crediting illicit gains penalizes the gate more than the ledger and so, if anything, *reinforces* the architecture comparison rather than disturbing it. All results below hold for ω below a threshold, beyond which weighting illicit gains so heavily would undercut the case for any enforcement.

the gate standard are properties of the token, set by the issuer under regulatory mandate; ledger enforcement e is supplied by a public authority; venue operators may add local gates. Section 3 shows this allocation is not innocuous, composability makes a venue-chosen gate a weakest-link public good, under-provided relative to (5), which is why the optimal object of regulation is the standard, not the venue.

3 Analysis

The analysis is a single economic argument in four steps. We first show that the two instruments buy the same good (a cleaner, less toxic market) but pay for it in different currencies: the gate in lost liquidity and composability, the ledger in enforcement resources (§3.1–3.2). Written into the planner’s first-order conditions, this asymmetry is the engine of everything that follows. We then establish the central comparative static: as an asset grows more traceable, the optimal policy shifts away from the gate and toward the ledger: the architecture *flips* (§3.4). The remaining results draw out the consequences. That the ledger dominates the gate as a technology for suppressing bad activity, that gating across competing venues is a weakest-link public good best regulated at the token standard rather than the venue, and that a residual gate survives only for an undeterrable tail of offenders. Each result is stated as a proposition and then read back in economic terms.

Throughout, subscripts denote partial derivatives, and we use the suppression derivatives of Lemma 2 together with $\tau_B = N/(N+B)^2 > 0$, $\tau_N = -B/(N+B)^2 < 0$, and $N_g = -\lambda\mu < 0$. We maintain that at the relevant allocations the legitimate market is active, $\pi_{pc} > 0$, so $S_L = N\pi_{pc}$ is differentiable; this holds in the calibration of Section 5. Complete proofs are in the Appendix.

3.1 Two margins, two currencies

Proposition 2 (Gate–ledger duality). *Both instruments reduce active bad activity and both raise legitimate surplus through the toxicity channel:*

$$B_g < 0, \quad B_e < 0, \quad \left. \frac{\partial S_L}{\partial(-B)} \right|_{\tau\text{-channel}} = \sigma N \tau_B > 0.$$

The instruments differ in exactly one respect. The gate has a direct liquidity cost (it excludes legitimate investors, destroys composability, and thins the legitimate base)

$$\left. \frac{\partial S_L}{\partial g} \right|_{\text{direct}} = \underbrace{-\lambda\mu \pi_{pc}}_{\text{exclusion}} \underbrace{-Nu_1\alpha\varphi^{\alpha-1}}_{\text{composability}} \underbrace{-\sigma\lambda\mu \frac{NB}{(N+B)^2}}_{\text{base-thinning}} < 0,$$

whereas the ledger has no direct liquidity effect: $\partial S_L/\partial e|_{\text{direct}} = 0$, and its only cost is the resource cost $c_e e$.

The proposition formalizes the sense in which the two instruments “pay in different currencies.” Each buys the same good (a cleaner market, fewer toxic counterparties, less external harm) but the gate pays in liquidity and composability while the ledger pays in enforcement resources. The proof is a direct differentiation (Appendix A.4).

3.2 The compliance-cost asymmetry

Define the *social marginal value of suppressing bad activity*,

$$\text{MV}(g, e) \equiv \sigma N \tau_B + \eta_E = \frac{\sigma N^2}{(N + B)^2} + \eta_E > 0,$$

the sum of the marginal reduction in adverse-selection cost (toxicity channel) and the marginal reduction in external harm. The first-order conditions of (5) can then be written in a form that exposes the asymmetry.

Proposition 3 (The compliance-cost asymmetry). *An interior optimum (g^*, e^*) satisfies*

$$(ledger) \quad \text{MV} \cdot (-B_e) = c_e e^*, \tag{6}$$

$$\begin{aligned} (gate) \quad \text{MV} \cdot (-B_g) &= \underbrace{Nu_1 \alpha \varphi^{\alpha-1} + \lambda \mu \pi_{pc} + \sigma \lambda \mu \frac{NB}{(N+B)^2}}_{\equiv C_g(g^*, e^*) \text{ (composability + exclusion + base-thinning)}} + c_g. \\ &\equiv C_g(g^*, e^*) \text{ (composability + exclusion + base-thinning)} \end{aligned} \tag{7}$$

Both conditions equate the same social value MV times the instrument’s marginal suppression of bad activity to that instrument’s marginal cost. The ledger’s marginal cost is the pure resource cost $c_e e^*$. The gate’s marginal cost C_g additionally contains the composability cost, the exclusion cost, and the base-thinning cost, none of which appears in the ledger’s condition. Tokenization raises C_g (through the composability premium u_1 and its concavity α) and lowers the cost of the ledger (through traceability θ , which raises $-B_e$ for given e).

Equation (7) minus (6) is the whole paper in one line: at the optimum the planner pushes each instrument until its marginal social value equals its marginal cost, and the gate’s marginal cost is strictly larger by the composability-and-exclusion block. When that block is large (when composability is valuable, as it is for a token) the gate is pushed less far and the ledger further. The proof rearranges the first-order conditions (Appendix A.5).

3.3 Regularity from primitives

The comparative statics require a second-order condition and the signs of three cross-partial. Rather than assume them, we show that all but one reduce to a transparent primitive restriction. Let $\text{MV} \equiv \sigma N^2 / (N + B)^2 + \eta_E$ denote the marginal social value of suppressing a unit of bad activity (the toxicity benefit plus external harm averted).

The cross-partial carry the signs that drive the flip: enforcement and traceability are complements in producing deterrence, and each is a substitute for the gate, because all three compress the same pool of active bad actors B . These signs are not assumed but follow from a single condition on primitives. That bad actors are no more numerous than legitimate investors.

Proposition 4 (Regularity from primitives). *Suppose $\beta \leq \mu$. Then on the active-market region $\{\pi_{pc} > 0\}$:*

- (i) *toxicity is bounded, $\tau \leq \frac{\beta}{\mu + \beta} \leq \frac{1}{2}$, with the bound attained at $(g, e) = (0, 0)$;*
- (ii) *the gate is a substitute for both instruments: $W_{ge} < 0$ and $W_{g\theta} < 0$;*
- (iii) *enforcement is strictly concave: $W_{ee} < 0$;*
- (iv) *if in addition $\rho K \leq 1$ (or, more weakly, $\rho \kappa < 1$ at the optimum, which holds whenever the offending fraction exceeds e^{-1}) then traceability and enforcement are complements: $W_{e\theta} > 0$.*

The only element of the second-order condition not implied by $\beta \leq \mu$ and $\rho K \leq 1$ is that the Hessian is negative definite at the interior optimum. Since $W_{ee} < 0$, this is equivalent by the Schur complement to local concavity of the concentrated objective $V(g) \equiv \max_e W(g, e)$ at g^* :

$$V''(g^*) = \frac{\det \mathbf{H}}{W_{ee}} = \frac{W_{gg}W_{ee} - W_{ge}^2}{W_{ee}} < 0.$$

Welfare is not globally concave in g (V'' is positive at intermediate gate intensities, where the composability and toxicity channels reinforce) so this is necessarily a local condition. It holds throughout the calibrated region.

Proposition 4 replaces a blanket regularity assumption with a primitive sufficient condition, $\beta \leq \mu$, that delivers the toxicity bound, both substitution signs, the complementarity, and the concavity of enforcement as theorems. What remains, local concavity of the gate's concentrated payoff at the optimum, cannot be made global, because the gate's love-of-variety benefit and its toxicity effect jointly render W non-concave at moderate g ; this is the irreducible content of the following maintained assumption.

Assumption 1 (Regularity). At every interior solution of (5) the concentrated objective is locally concave, $V''(g^*) < 0$; equivalently, the Hessian $\mathbf{H} = \begin{pmatrix} W_{gg} & W_{ge} \\ W_{ge} & W_{ee} \end{pmatrix}$ is negative definite.

For completeness we record the cross-partial signs and the exact conditions behind Proposition 4(ii)–(iv); the proof is in Appendix A.6.

Lemma 4 (Cross-partial). *At any interior solution: $W_{e\theta} > 0$ if $\rho \kappa < 1$; $W_{ge} < 0$ if $\sigma N^2(N - B)/(N + B)^3 + \eta_E \geq 0$; and $W_{g\theta} < 0$ under the same inequality. Each holds whenever $\tau \leq \frac{1}{2}$ (in particular whenever $\beta \leq \mu$), and the enforcement-cost terms reinforce both substitution signs.*

3.4 The architecture flips

We now vary traceability and trace the optimal architecture. The next result is the paper’s central claim: the optimal regime is not a fixed stringency but a function of the technology, and it inverts as the technology changes.

Proposition 5 (The architecture flip). *Suppose $\beta \leq \mu$ and that $\rho\kappa < 1$ at the relevant optima (a primitive sufficient condition is $\rho K \leq 1$; the calibration satisfies the former with $\rho K > 1$)⁵, and maintain Assumption 1. By Proposition 4 the cross-partials then satisfy $W_{ge} < 0$, $W_{g\theta} < 0$, $W_{e\theta} > 0$ and $W_{ee} < 0$. Then:*

- (i) Enforcement rises, the gate recedes. *On any interval of interior optima, $e^*(\theta)$ is strictly increasing and $g^*(\theta)$ is strictly decreasing in θ .*
- (ii) A permissionless threshold exists. *There is a traceability threshold $\bar{\theta} \in (0, 1)$ such that the optimal gate is exactly zero for all $\theta \geq \bar{\theta}$: optimal regulation of a sufficiently transparent security uses no ex-ante gate.*
- (iii) Global monotonicity under harm dominance. *If the external-harm channel dominates the adverse-selection channel at the margin (e.g. σ small or η_E large), $g^*(\theta)$ is nonincreasing throughout $[0, 1]$, so the optimal architecture shifts monotonically from gate-reliant to ledger-reliant as traceability rises.*

Proof sketch. For (i), differentiate the first-order conditions $W_g = W_e = 0$ in θ :

$$\mathbf{H}(dg^*/d\theta, de^*/d\theta)^\top = -(W_{g\theta}, W_{e\theta})^\top.$$

Under Assumption 1 the Hessian $\mathbf{H}$ is negative definite, so by Cramer’s rule the signs of $dg^*/d\theta$ and $de^*/d\theta$ are determined by the cross-partials. Lemma 4 gives $W_{g\theta} < 0$, $W_{e\theta} > 0$, $W_{ge} < 0$, which yield $dg^*/d\theta < 0$ and $de^*/d\theta > 0$. For (ii)–(iii), define the marginal value of the first unit of gate at $g = 0$ under optimal enforcement, $\Delta(\theta) \equiv W_g(0, \hat{e}(\theta); \theta) = \text{MV } s\beta\bar{S}(\kappa^*) - C_g$; the gate is unused iff $\Delta(\theta) \leq 0$. As θ rises, $\hat{e}(\theta)$ rises (part (i) at $g = 0$), κ^* rises, and the deterrence factor $\bar{S}(\kappa^*)$ falls, so the benefit falls while $\text{MV} \leq \sigma + \eta_E$ is bounded; the cost C_g is governed by its composability and exclusion components, nondecreasing in θ at $g = 0$ (the base-thinning component is second-order and need not be signed), and Proposition 11 establishes that the net effect is a strictly decreasing Δ whenever $\eta_E \geq \sigma/27$ and a nonincreasing optimal gate for every $\eta_E > 0$; hence Δ crosses zero once. With $\Delta(0) > 0$ (a gate is optimal when the ledger is inert, given harm warrants compliance) and $\Delta(1) < 0$ (the residual deterrence floor $e^{-\rho K}$ undercuts the gate’s cost), the intermediate value theorem gives a unique $\bar{\theta}$, and $g^* = 0$ beyond it. Complete argument in Appendix A.7. $\square$

⁵In the calibration $\rho\kappa < 1$ holds throughout the gated interval except a negligible final slice, $\theta \in (0.656, 0.663)$, immediately before the flip at $\bar{\theta} \approx 0.665$; direct computation confirms g^* and e^* remain strictly monotonic through this slice as well, so part (i) is not an artifact of the sufficient condition’s local slack.

The economics is transparent. Raising traceability does two things to the gate's value, both downward. It makes the ledger a better substitute, so the planner deters more ($e^* \uparrow$) and there are fewer active bad actors left for the gate to exclude. And because deterrence has already thinned the offending population, the marginal bad actor the gate would exclude is less likely to have offended anyway ($\bar{S}(\kappa^*) \downarrow$). Meanwhile the gate's cost, the composability it destroys, does not depend on traceability at all. So the gate's benefit-to-cost ratio falls monotonically in θ , and past $\bar{\theta}$ the optimal gate is zero. The same logic runs in reverse for opaque assets: when θ is low the ledger is impotent, the gate is the only instrument that works, and optimal regulation is heavily gated. *The optimal regulation of a transparent security and the optimal regulation of an opaque one are qualitatively opposite.*

3.5 Tokenization dominates the gate technology

The stringency view holds that compliance and liquidity trade off along a frontier. They do, along the gate's frontier. The ledger has a different frontier, and with enough traceability it is not merely a different point on the same frontier but a strictly better frontier.

For a technology with traceability θ , let the attainable set be

$$\mathcal{A}(\theta) = \{(S_L(g, e), B(g, e)) : g \in [0, 1], e \geq 0\},$$

and let $\mathcal{A}_{\text{gate}} = \{(S_L(g, 0), B(g, 0)) : g \in [0, 1]\}$ be the gate-only technology (the ledger inert, as under opacity). We say technology X *dominates* technology Y if every point on Y 's efficient frontier is weakly Pareto-dominated (weakly higher S_L , weakly lower B) by a point attainable under X , and some point strictly so.

Proposition 6 (Frontier domination).

- (i) Traceability expands the attainable set: $\theta' > \theta \Rightarrow$ every (S_L, B) attainable under θ is attainable under θ' , so $\mathcal{A}(\theta) \subseteq \mathcal{A}(\theta')$ in the efficient sense.
- (ii) Sufficient traceability dominates the gate outright: there is $\theta_0 < 1$ such that for all $\theta \geq \theta_0$ the ledger technology dominates the gate technology, and in particular the welfare-optimal gated allocation is strictly Pareto-dominated by a permissionless ($g = 0$) allocation with higher S_L and lower B .

Consequently, above θ_0 , a permissionless market policed *ex post* can be made simultaneously more liquid and more compliant than any gated market. How the planner allocates this transparency dividend (toward liquidity, toward compliance, or both) depends on the harm severity η_E ; that the dividend is available does not.

Proof sketch. (i) For fixed (g, e) , raising θ raises κ hence lowers B and (through τ) weakly raises S_L ; so each attainable pair improves or is preserved, and the efficient frontier moves outward. (ii) Evaluate the permissionless locus $g = 0$: $N = \mu$ (full legitimate mass), $\varphi = 1$ (full composability),

and $B(0, e) = \beta e^{-\rho\kappa(e, \theta)}$, which as e grows approaches the floor $\beta e^{-\rho\theta K}$. For θ near 1 this floor is small, while $S_L(0, e) \rightarrow \mu(u_0 + u_1 - \sigma\tau_{\min})$ retains the full composability premium u_1 . The best gated allocation, by contrast, must raise g to cut B , which forfeits the composability premium ($\varphi^\alpha \downarrow$) and excludes legitimate investors ($N \downarrow$). Hence for θ large there exists e' with $B(0, e') \leq B(g^{\text{gate}}, \cdot)$ and $S_L(0, e') > S_L(g^{\text{gate}}, \cdot)$: the permissionless point dominates. The threshold θ_0 is the smallest traceability at which two conditions hold together. First, the deterrence floor at full enforcement, $\beta e^{-\rho\theta K}$, must fall weakly below the gate's lowest attainable bad activity $(1-s)\beta$, so that the permissionless locus can *match* the gate on suppression; this requires $\rho\theta K \geq \ln \frac{1}{1-s}$. Second, the composability premium retained at $g = 0$ must exceed the residual-deterrence shortfall, so that the permissionless locus strictly *beats* the gate on surplus. The first condition is what makes the ranking a Pareto one rather than a trade-off; it places $\theta_0 \geq \ln \frac{1}{1-s} / (\rho K)$, finite and below one precisely when penalties and detection are strong relative to the gate's screening accuracy, $\rho K > \ln \frac{1}{1-s}$. Appendix A.8 gives the construction; Section 5 shows that in the baseline ($s = 0.85$, $\rho K = 3$, so $\rho K > \ln \frac{1}{1-s} \approx 1.9$) the *entire* $\theta = 0.9$ frontier dominates the gate optimum. $\square$

Remark 3 (Near-perfect screening). The Pareto ranking requires the ledger to *match* the gate's suppression, and this fails when the gate screens almost perfectly. As $s \rightarrow 1$ the gate drives bad activity to zero, below the ledger's irreducible deterrence floor $\beta e^{-\rho\theta K} > 0$; so when $\rho K < \ln \frac{1}{1-s}$ no traceability lets a permissionless market reach the gate's lowest B , and the two technologies trade off rather than one dominating. The comparison then reverts to the *welfare* ranking of Proposition 5: the planner weighs the composability dividend of openness against the residual harm the floor leaves, and that ranking still delivers $g^* = 0$ above $\bar{\theta}$ for moderate harm, with the residual gate of Proposition 8 reappearing for severe harm. Pareto-dominance is thus the strong form of the result, available when screening is imperfect; the welfare flip is the form that holds regardless. This is the precise boundary of the dominance claim: it is the gate's *imperfection*, not merely tokenization, that lets the ledger overtake it on every margin at once.

The result reverses a widespread intuition. Permissionless access is commonly equated with lawlessness, and gating with safety. The model says that for a transparent asset the equation is backwards: gating buys compliance by destroying liquidity, while ex-post enforcement buys the same compliance and keeps the liquidity, so the gated market is dominated. The reason permissionless markets *look* more dangerous is that in practice they are often also *unpoliced* (e low). The model separates the two: it is the absence of enforcement, not the absence of a gate, that makes a permissionless market unsafe.

3.6 Regulate the standard, not the venue

So far the market is a single venue. Composability means it is not. A token admitted on one venue can, by construction, transact against any protocol and any other venue that recognizes

the standard. This has a sharp consequence for where compliance must be enforced.

Consider $m \geq 2$ venues indexed j , each able to impose its own local gate g_j , all trading the same composable token and sharing one ledger of enforcement e . Because a bad actor admitted *anywhere* can then transact *everywhere*, the binding screen for system access is the *laxest* venue: a bad actor is blocked only if blocked at its easiest point of entry. The mass admitted is governed by $\min_j g_j$, and system-wide active bad activity is

$$B^{\text{sys}}(\mathbf{g}, e) = (1 - s \min_j g_j) \beta \bar{S}(\kappa(e, \theta)).$$

Each venue bears the composability cost of its *own* gate (its own users lose composability when it whitelists), but the suppression of B^{sys} is a public good shared by all venues, and only the minimum gate affects it.

Proposition 7 (Standard, not venue). *In the venue game with composable tokens:*

- (i) Weakest-link under-provision. *The gate is a weakest-link public good. In any Nash equilibrium no venue sets g_j above $\min_j g_j$ (a gate above the minimum incurs composability cost with zero effect on B^{sys}), and the common equilibrium gate is inefficiently low relative to the planner's: $g^{\text{Nash}} < g^*$. A single permissive venue ($g_j \rightarrow 0$) collapses system compliance regardless of others' screening.*
- (ii) Enforcement is immune. *Ledger enforcement e is global: one shared ledger, one authority, reach over all transactions. It exhibits no weakest-link structure and the decentralized choice of e does not suffer the coordination failure in (i).*
- (iii) The optimal locus. *Effective ex-ante compliance, if used at all, must be imposed as a common property of the token standard ($g_j \equiv g_{\text{std}}$ for all j , enforced by the token's code), not chosen venue-by-venue; and the natural instrument for a composable system is ledger enforcement. Venue-level gate regulation is generically dominated by standard-level rules plus ledger enforcement.*

Proof sketch. (i) Fix the others' gates with minimum $g_{-j}^{\min}$. If $g_j > g_{-j}^{\min}$, then $\min$ is unaffected by a marginal change in g_j , so $\partial B^{\text{sys}} / \partial g_j = 0$ while the composability cost is strictly positive; venue j strictly prefers to lower g_j . Hence in equilibrium all venues are at a common minimum. The venue at the minimum, when it raises its gate, confers a benefit (lower B^{sys}) shared by all m venues but internalizes only its own share and ignores external harm η_E entirely; its private first-order condition therefore stops short of the planner's, giving $g^{\text{Nash}} < g^*$. (ii) With a single shared ledger the enforcement level enters every venue's B^{sys} identically through one instrument; a single authority setting e internalizes the common effect. (iii) Imposing $g_j \equiv g_{\text{std}}$ in the token's code removes the $\min$ channel (all venues share the screen), restoring the planner's gate as implementable; combined with global e , this dominates any profile of independently chosen venue gates. Appendix A.9. $\square$

This is the formal version of the supervisory instinct that tokenized markets must be regulated at the level of their infrastructure rather than their venues (International Monetary Fund, 2026). The economics is that composability turns the gate into a public good with the worst possible aggregation technology, the weakest link, so decentralizing the gate guarantees under-provision, while the ledger, being a single shared object, is the instrument a composable system can regulate coherently. The prescription “regulate the standard, not the venue” is not an administrative preference; it is what the externality structure requires.

3.7 The undeterrable tail and the residual gate

The dominance of the ledger is bounded by two ceilings. Detection cannot exceed traceability ($p \leq \theta$), so if $\theta < 1$ some offenses go undetected no matter how large e is; and penalties cannot exceed wealth ($K \leq \bar{K}$), the judgment-proof problem of Shavell (1986). Either ceiling caps the deterrence threshold at $\bar{\kappa} = \theta K$ (or $p_{\max} \bar{K}$), so bad actors with $b > \bar{\kappa}$ are *undeterrable*: a residual mass $\beta e^{-\rho \bar{\kappa}} > 0$ offends whatever the enforcement effort. The only instrument that removes them is the gate.

Proposition 8 (Division of labor). *Suppose deterrence is capped at $\bar{\kappa} < \infty$ (because $\theta < 1$ or $K \leq \bar{K}$). Then a residual mass of bad actors is undeterrable, and there is a harm threshold $\bar{\eta}_E(\theta)$ such that for $\eta_E > \bar{\eta}_E(\theta)$ the optimal gate is strictly positive even at full enforcement and at $\theta \rightarrow 1$. The optimal architecture is then a division of labor: the ledger deters the deterrable majority, and a small gate excludes the undeterrable, judgment-proof tail.*

Proof sketch. At the enforcement ceiling, $B \rightarrow (1 - sg)\beta e^{-\rho \bar{\kappa}}$ and the planner’s problem in g has first-order benefit $MV \cdot s\beta e^{-\rho \bar{\kappa}}$ at $g = 0$, increasing in η_E through MV , against the θ -independent marginal cost $C_g(0)$. The benefit exceeds the cost (so $g^* > 0$) iff η_E exceeds a threshold $\bar{\eta}_E(\theta)$. Appendix A.10. $\square$

Proposition 8 reconciles the model with the observation that even the most transparent tokenized venues do not go fully open: they screen sanctioned addresses and known bad actors while leaving everyone else unrestricted. The model rationalizes exactly this pattern. A narrow, targeted gate is the optimal response to the hard core of high-benefit offenders whom no feasible penalty can deter; it is not a relic of the stringency regime but the efficient complement to ex-post enforcement. The residual gate is *narrow*, aimed at the tail, rather than the broad permissioning the stringency view would impose on the whole market.

4 The Three Load-Bearing Assumptions

The results so far rest on three assumptions that a skeptical reader is right to press, because the conclusion turns on them: that a gate destroys composability, that a bad actor’s two costs are distinct, and that the gate can tell bad actors from legitimate ones in the first place. Each is a

reduced form in the model above; each has a foundation, and stating it both defends the result and sharpens it.

4.1 What composability is, and why a gate restricts it

The objection to $\varphi = 1 - g$ is immediate: a permissioned system can be perfectly composable *among its permissioned members*, so why should gating destroy composability at all? The answer is that composability, in the sense that gives a token its value, is not merely interaction but *atomic, permissionless* interaction: the property that a protocol can call another and settle in a single transaction without either checking who the caller is. This is what “money legos” means, and it is what a gate breaks: a permission check inserted anywhere in a chain of calls defeats atomicity, because the chain can no longer settle as one operation if a participant midway must be screened. A gated token therefore cannot enter a permissionless protocol, which does not perform the check, and a permissioned protocol cannot atomically compose with the permissionless ecosystem outside it.

So $\varphi = 1 - g$ should be read not as composability *destroyed* but as composability *confined*: a token gated at intensity g is excluded from a fraction g of the on-chain economy, the permissionless ecosystem it can no longer atomically reach, and retains access only to the permissioned fraction $1 - g$. The premium $u_1\varphi^\alpha$ is the value of access to that fraction, concave because the marginal connection adds less than the first; the loss $u_1[1 - (1 - g)^\alpha]$ is the value of the connections to the excluded part. The objection is thus accommodated rather than refuted: the permissioned set is internally composable and the model grants it the full premium for its size, but that size is smaller, and the value foregone is access to everything outside it.

This reading connects to a network account of composability, developed in a companion paper, and inherits its sharpening. Whether the permissioned set is itself composable is not guaranteed by its being permissioned; it depends on whether the set is densely enough connected to percolate. A permissioned set below the connectivity threshold has no giant component, so even its internal composability collapses, and the loss exceeds $u_1[1 - (1 - g)^\alpha]$: the stark form of the trade-off is the below-threshold regime. The assumption $\varphi = 1 - g$ is therefore the reduced form of a network statement: gating removes a token from the permissionless giant component and confines it to a permissioned subgraph whose own composability is whatever its connectivity sustains.

4.2 Where the composability premium comes from

Section 4.1 took $u_1\varphi^\alpha$ as the value of access to a fraction φ of the composable economy; we now derive its functional form and, with it, the meaning of the curvature α , which sets the flip threshold and has so far been a free parameter. The value a tokenized security draws from composability is the value of the uses its access opens, most concretely, the protocols in which it can serve as collateral, each rehypothecation through a composable protocol generating yield.

Order these uses by the value they add and let a token with access φ reach the best φ fraction of them. If the value density across uses is $u_1 \alpha x^{\alpha-1}$ (heavy-tailed when $\alpha < 1$, a few protocols contributing most of the value) then the value of access φ is

$$\int_0^\varphi u_1 \alpha x^{\alpha-1} dx = u_1 \varphi^\alpha,$$

which is the premium. The curvature α is thus the *concentration* of composability value across protocols. A low α means being on-chain is worth most because of access to a few central protocols, so even a small exclusion forfeits a large share and the marginal cost of gating is steep; a high α means the value is spread evenly across many interchangeable venues, and gating costs less at the margin. The flip threshold inherits the reading: the more concentrated the value of composability (the lower α , the more the on-chain economy is organized around a few indispensable protocols) the dearer the composability a gate destroys, and the sooner the optimal architecture turns to the ledger. The parameter that governs when the flip occurs is thus a measurable feature of market structure.

4.3 The two harms of a bad actor

The model charges bad activity twice (once as the adverse-selection spread $\sigma\tau$ that enters legitimate traders' payoff, and once as the external harm $\eta_E B$ in welfare) and a referee is right to ask whether this is the same cost counted twice. It is not, because the two are borne by different parties and arise from different aspects of misconduct.

A bad actor plays two roles. As a *trading counterparty* it may carry private information or toxic flow, imposing adverse selection on the legitimate traders it transacts against; this is a private cost, the spread $\sigma\tau$, borne by those counterparties and internalized in their participation decision. It is why legitimate liquidity flees a toxic market, and its welfare footprint is the participation it destroys, the $N\pi_{pc}$ that shrinks as τ rises. As a *wrongdoer* the same actor launders proceeds, evades sanctions, finances crime, or defrauds parties outside the trade; this is an external cost, $\eta_E B$, borne by third parties and society, and internalized by no market participant, which is why a regulator's objective contains it when a private market's does not. The two channels do not double-count because the spread is a distortion *within* the market, falling on counterparties through reduced participation, while $\eta_E B$ is an externality *outside* it, falling on society regardless of market quality.

The distinction also clarifies what the model spans. A bad actor who trades non-toxically but launders (common for the laundering use case, where the trade itself is unremarkable) imposes $\eta_E B$ with little $\sigma\tau$; an informed manipulator imposes $\sigma\tau$ and, to the extent it harms the issuer or other investors, $\eta_E B$ as well. Setting $\eta_E = 0$ recovers a regulator who cares only about market quality, for whom the entire case rests on adverse selection; letting the bad actor's toxicity vanish isolates the pure-externality case in which the ledger's value is entirely the harm it deters. The

general specification carries both, and the flip and the residual gate are stated in terms of the harm weight η_E so that the boundary between the two regimes is explicit.

4.4 Whether the gate can tell bad from legitimate

The gate excludes bad actors at rate s and legitimate users at rate λ , and the entire value of the gate rests on $s > \lambda$. But a gate does not observe types; it observes credentials, and a skeptical reader is right to ask what delivers the asymmetry, and what happens when bad actors forge.

Ground the gate in a signal. Each agent presents an identity signal x , distributed $N(\mu_L, 1)$ for legitimate agents and $N(\mu_B, 1)$ for bad ones with $\mu_B < \mu_L$, and the gate admits when x exceeds a stringency threshold t . The exclusion rates are then $sg = \Phi(t - \mu_B)$ and $\lambda g = \Phi(t - \mu_L)$, and since $\mu_B < \mu_L$ the gate excludes bad actors more than legitimate ones, $s > \lambda$, by exactly the informativeness of the signal $\mu_L - \mu_B$. In the calibration a separation of two standard deviations yields s/λ between two and ten across stringencies: the asymmetry the model assumes is what an informative identity signal produces.

Proposition 9 (The gate’s value is eroded by credential forgery). *The gate’s screening edge $s - \lambda$ is increasing in the signal separation $\mu_L - \mu_B$. Sophisticated bad actors invest in forged credentials and synthetic identities, which shift their signal toward the legitimate distribution, $\mu_B \rightarrow \mu_L$; as they do, $s \rightarrow \lambda$ and the gate’s discriminating power vanishes: a gate that cannot tell bad from legitimate excludes both at the same rate and screens no one. The ledger detects ex post from on-chain conduct, which a forged identity cannot conceal, and is therefore robust to the forgery that erodes the gate.*

This both defends and extends the architecture result. It defends it by deriving $s > \lambda$ rather than assuming it; and it extends it by adding a margin along which the gate decays and the ledger does not. The ex-ante gate trusts an identity asserted before the fact, which a determined wrongdoer can manufacture; the ex-post ledger trusts a record of conduct accumulated after it, which a wrongdoer cannot un-write. As identity forgery grows cheaper, a direction synthetic identity has only accelerated, the screening value of the gate falls toward zero while the deterrent value of the ledger is untouched, tilting the optimal architecture further toward the ledger than the static asymmetry alone would imply.

4.5 The global structure of the optimum

A last concern is technical but real: the planner’s problem is not concave, so one should ask whether its optimum is well-defined, and what the non-concavity does to the results. It is well-defined, and the non-concavity is not a defect to be apologized for but the source of the paper’s central result.

Fix traceability and read welfare as a function of the gate, with enforcement optimized at each g . The gate pulls welfare two ways that do not net to a concave function: raising it thins

toxicity, which lifts participation, but it forfeits composability $u_1(1-g)^\alpha$ and excludes legitimate users. The composability term is concave and the exclusion linear, but the participation floor $\max(\pi_{pc}, 0)$ and the toxicity feedback render the whole non-concave, and welfare can carry two local maxima: an interior, gate-reliant optimum at positive g , and the ledger corner at $g = 0$.

Proposition 10 (The flip is a branch crossing, and the non-convexity is essential). *Welfare in the gate has at most two local maxima (an interior gate-reliant optimum $g_{\text{int}}^* > 0$ and the ledger corner $g = 0$) and the global optimum is whichever attains the higher welfare. As traceability rises, the corner’s welfare rises faster, the ledger growing more effective, until at a threshold the two branches cross and the global optimum jumps from g_{int}^* to zero. In the calibration the two maxima coexist near $\theta \approx 0.65$, with welfare 0.401 at $g \approx 0.66$ and 0.395 at $g = 0$, and the crossing falls at $\theta^* \approx 0.66$. The discontinuity of the flip, a jump in the optimal gate rather than a slide, is thus a consequence of the non-concavity, not an artifact of the solution method: a concave problem would move the optimum continuously and could not produce a flip at all.*

The non-convexity is therefore doing the paper’s central work. A smoothly concave trade-off between gating and the ledger would imply that the optimal architecture shifts gradually as an asset becomes more traceable: a little less gate, a little more ledger, with no qualitative break. The discontinuity, the sense in which the architecture genuinely *flips* from one regime to another at a threshold, exists because the two architectures are distinct local optima whose values cross rather than points on a single concave frontier. That the problem is not concave is what makes the result an architectural choice rather than a marginal adjustment.

5 A Calibrated Illustration

The propositions are qualitative. To show that the mechanisms they describe operate at sensible magnitudes (and to display the architecture flip, the frontier shift, and the residual gate quantitatively) we solve the model numerically at the calibration in Table 1. The parameters are illustrative rather than estimated; their role is to place the economy in an empirically plausible region (a sizable would-be-offender population, a meaningful composability premium, moderate external harm, costly but productive enforcement) and to verify that the comparative statics hold there. All quantities are computed on a fine grid with a Nelder–Mead refinement of the optimum.

The architecture flip. Figure 1 traces the optimal regime as traceability θ rises from 0 (an opaque asset, on which the ledger is inert) to 1 (a fully transparent one). Panel (a) confirms Proposition 5: the optimal gate $g^*(\theta)$ declines smoothly, from 0.95 at $\theta = 0$ as the ledger begins to substitute for it, and then drops to exactly zero at the threshold $\bar{\theta} \approx 0.66$, while optimal enforcement $e^*(\theta)$ rises as the gate recedes, peaks where the gate is released, and eases slightly thereafter as higher traceability makes each unit of enforcement more potent. The optimal

Table 1: Baseline calibration

Symbol	Interpretation	Value
μ	mass of legitimate investors	1.00
β	mass of would-be bad actors	0.80
ρ	rate of the benefit distribution $b \sim \text{Exp}(\rho)$	2.00
K	penalty on a detected offender	1.50
s	gate screening accuracy (bad excluded per unit g)	0.85
λ	gate false-exclusion rate (legit excluded per unit g)	0.20
σ	adverse-selection (toxicity) cost scale	2.00
u_0	baseline per-capita value of participation	0.50
u_1	composability premium	0.80
α	concavity of the composability premium	0.35
η_E	external harm per active bad actor	1.20
c_e	enforcement cost scale	0.50
c_g	gate administrative cost	0.00

architecture moves continuously from “gate, do not police” to “police, do not gate.” Panel (b) shows the consequences for the two things society cares about: as the gate is released at $\bar{\theta}$, composability is restored and legitimate surplus S_L jumps upward, while active bad activity B remains contained because enforcement has taken over the job of suppressing it. The market becomes more valuable without becoming more toxic.

The frontier shifts out. Figure 2 plots attainable combinations of liquidity (horizontal, higher is better) and bad activity (vertical, inverted so that lower, more compliant, is better). The grey curve is the gate-only frontier: it is the locus traced as the gate runs from open to closed with no ledger, and it embodies the stringency trade-off, to move down (more compliant) one must move left (less liquid). The coloured curves are the ledger frontiers at $\theta = 0.5$ and $\theta = 0.9$. They lie down and to the right of the gate frontier: the ledger technology attains combinations the gate cannot. At $\theta = 0.9$ the domination is total. The gate frontier tops out at liquidity 0.41 and cannot push bad activity below 0.12; the entire $\theta = 0.9$ ledger frontier delivers liquidity at least 0.38 *and* bad activity at most 0.054, so every point on it dominates the best the gate can do. This is Proposition 6 in the baseline: with sufficient traceability, tokenization does not present a better point on the same trade-off. It abolishes the trade-off the gate imposed and replaces it with a strictly better one.

When the gate survives. Figure 3 maps the optimal gate g^* over the plane of traceability θ (horizontal) and harm severity η_E (vertical), partitioning it into three regions. For low traceability or, at any traceability, sufficiently severe harm, the optimum is heavily gated (upper-left, red). For high traceability and moderate harm, the optimum is permissionless (lower-right, blue): the ledger does all the work and the gate is zero. Between and above lies the regime of Proposition 8: when harm is severe (here $\eta_E \gtrsim 4.25$), a substantial gate survives even at full traceability, be-

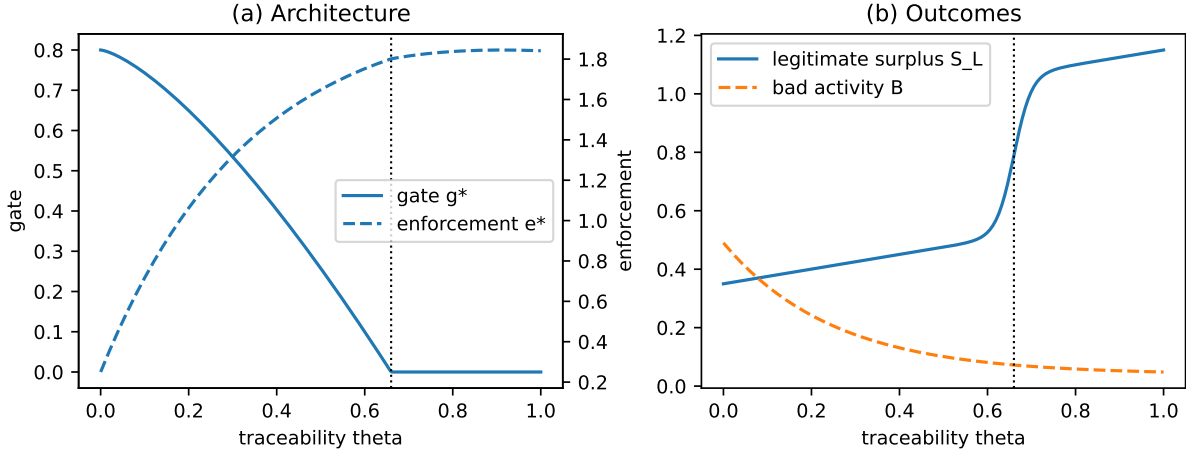


Figure 1: **The optimal regulatory architecture flips as traceability rises.** Panel (a): the optimal ex-ante gate $g^*(\theta)$ (left axis) declines and then drops to zero at the threshold $\bar{\theta} \approx 0.66$, while optimal ex-post enforcement $e^*(\theta)$ (right axis) rises as the gate recedes and eases slightly thereafter. Panel (b): under the optimal regime, legitimate surplus S_L (liquidity/capital formation) rises sharply when the gate is released and full composability is restored, while active bad activity B stays low because enforcement has substituted for exclusion.

Table 2: Optimal regime and outcomes by traceability (baseline calibration)

	gate g^*	enforcement e^*	bad activity B	toxicity τ	liquidity S_L
$\theta = 0.0$ (opaque)	0.95	0.00	0.15	0.16	0.37
$\theta = 0.5$ (partial)	0.85	0.55	0.12	0.12	0.55
$\theta = 0.9$ (transparent)	0.00	0.92	0.16	0.14	1.03

cause the undeterrable, judgment-proof tail must be excluded rather than deterred. The figure shows that “keep it permissionless” is the right prescription precisely in the empirically central case (transparent ledgers, moderate per-offender harm) while severe-harm assets retain a role for narrow, standard-level permissioning.

Optima at a glance. Table 2 reports the optimal regime and its outcomes at three traceability levels. The contrast between the opaque optimum ($\theta = 0$: a near-closed gate, no enforcement, low liquidity) and the transparent optimum ($\theta = 0.9$: no gate, active enforcement, liquidity nearly tripled at comparable bad activity) is the architecture flip in numbers. The transparency dividend here is spent overwhelmingly on liquidity: at $\theta = 0.9$ the planner accepts roughly the same bad activity as under the gate (0.16 versus 0.15) in exchange for nearly three times the legitimate surplus (1.03 versus 0.37). For more harmful assets (Figure 3) the dividend is split toward compliance instead.

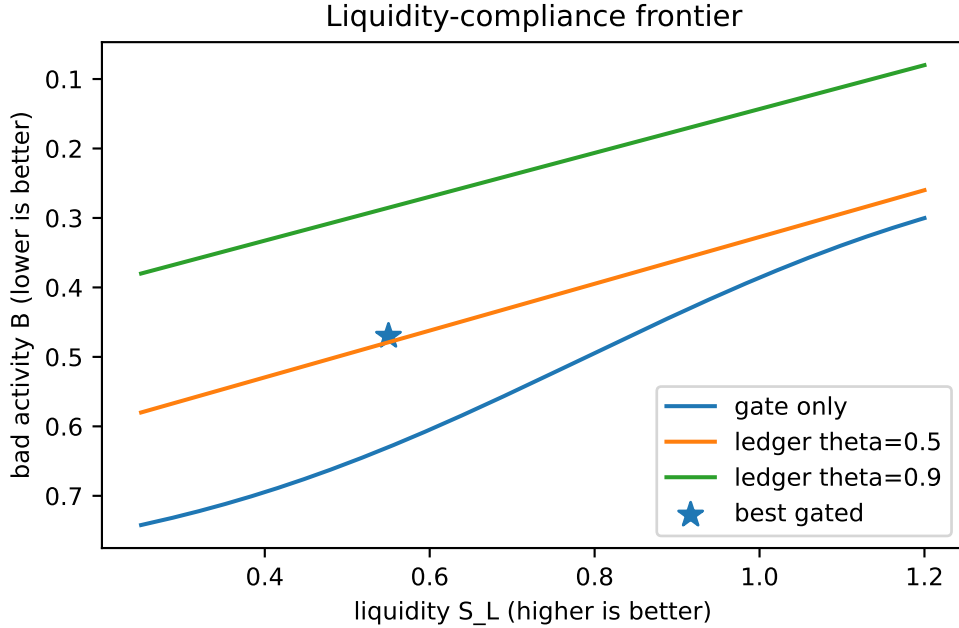


Figure 2: **Tokenization expands the liquidity–compliance frontier.** Attainable combinations of liquidity S_L (right is better) and bad activity B (axis inverted: down is better). The gate-only frontier (grey) embodies the stringency trade-off. The ledger frontiers ($\theta = 0.5$ orange, $\theta = 0.9$ blue) lie below and to the right. At $\theta = 0.9$ the entire ledger frontier dominates the best gated outcome: more liquidity *and* less bad activity than any gated regime can reach.

6 The Reinforcement Structure and the Robustness of the Flip

The flip of Section 3 is monotone: more traceability, weakly less gate. A natural concern is that this monotonicity is an artifact of the calibration. That in some configurations the architecture moves the other way, with more transparency calling for *more* permissioning. This section settles the question. We make the interaction among the three levers explicit and derive its signs from primitives; we then prove that the flip is globally monotone whenever external harm is not negligible relative to adverse selection, locate the exact knife-edge at which a reversal force appears, and characterize the two conditions under which the flip instead *stalls*, with gating persisting at full traceability.

6.1 The reinforcement structure from primitives

The three instruments (gate g , ledger e , and traceability θ) interact only through the single state variable B , the mass of active bad actors. Their interaction is the matrix of second derivatives of W , and the comparative statics of the optimum are governed by it through the system derived

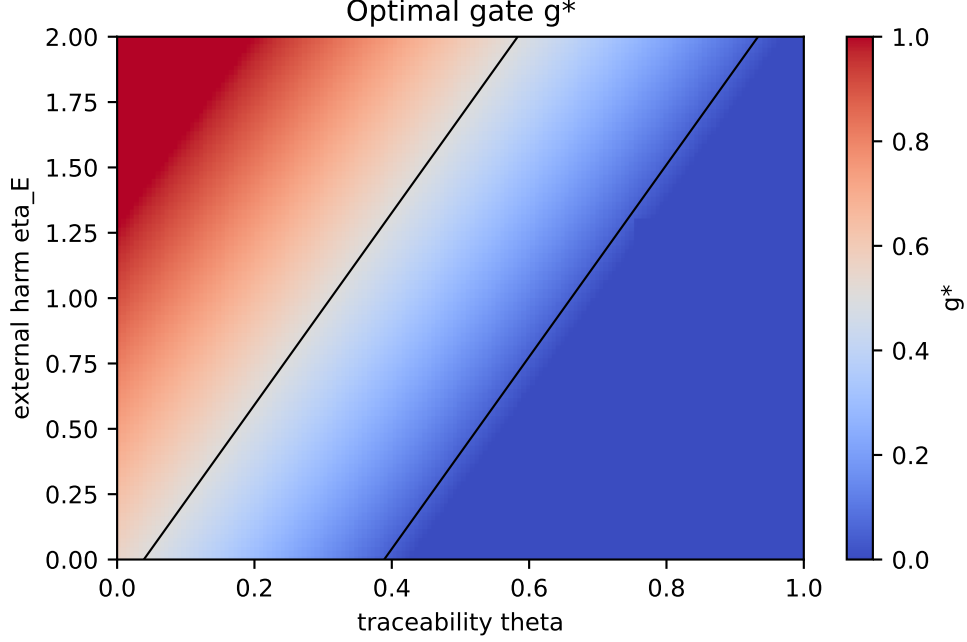


Figure 3: **Optimal architecture by technology and harm.** The optimal gate g^* over traceability θ and external harm η_E . *Gated* (red) for opaque assets or severe harm; *permissionless* (blue) for transparent assets with moderate harm, where ex-post enforcement suffices; and a region in which the *gate survives* at full traceability to exclude the undeterrable tail (Proposition 8).

in Appendix A.7,

$$\begin{pmatrix} dg^*/d\theta \\ de^*/d\theta \end{pmatrix} = -\mathbf{H}^{-1} \begin{pmatrix} W_{g\theta} \\ W_{e\theta} \end{pmatrix}, \quad \mathbf{H} = \begin{pmatrix} W_{gg} & W_{ge} \\ W_{ge} & W_{ee} \end{pmatrix}. \quad (8)$$

Proposition 4 pins the signs of this structure to a primitive restriction: under $\beta \leq \mu$, the gate is a substitute for each of the ledger and traceability ($W_{ge}, W_{g\theta} < 0$), traceability complements enforcement ($W_{e\theta} > 0$), and enforcement is concave ($W_{ee} < 0$). With $\mathbf{H}$ negative definite (Assumption 1), these signs deliver $dg^*/d\theta < 0$ and $de^*/d\theta > 0$. The reinforcement structure is therefore not assumed but follows from the single fact that the three levers compress a common pool. The question is whether its key sign ($W_{g\theta} < 0$, the substitutability of gate and traceability) can flip.

6.2 The bounded counter-force

The gate's marginal value $MV \cdot s\beta\bar{S}(\kappa) - C_g$ responds to traceability through two opposing channels. As θ rises and deterrence improves, a *deterrence-erosion* channel pushes the gate's value down: the offending fraction $\bar{S}(\kappa) = e^{-\rho\kappa}$ falls, so the gate excludes fewer actual offenders.

This is the force behind the flip. Working against it, a *toxicity-revaluation* channel pushes the value up: with fewer active bad actors the market is cleaner, so the marginal adverse-selection damage of each remaining unit of B , namely $MV = \sigma N^2/(N+B)^2 + \eta_E$, rises, and excluding one is worth more. If the upward channel dominated, the gate would rise with traceability and the flip would reverse. It cannot, except in a narrowly delimited regime, because the revaluation force is bounded.

Proposition 11 (Bounded revaluation and global monotonicity). *The strength of the toxicity-revaluation channel is bounded above by a constant independent of all other parameters:*

$$\sigma \frac{N^2(B-N)}{(N+B)^3} \leq \frac{\sigma}{27},$$

with the maximum attained at toxicity $\tau = B/(B+N) = 2/3$. Consequently:

- (i) If $\eta_E \geq \sigma/27$, then $W_{g\theta} < 0$ and $W_{ge} < 0$ at every allocation and for all parameter values (any β , any σ , any toxicity) because the same bound controls the negative part of each. Maintaining $\rho\kappa < 1$ (so that $W_{e\theta} > 0$) and Assumption 1, the flip is then globally monotone: $g^*(\theta)$ is nonincreasing and $e^*(\theta)$ nondecreasing on all of $[0, 1]$, with no restriction to $\beta \leq \mu$.
- (ii) A local reversal force, $W_{g\theta} > 0$, requires both $\eta_E < \sigma/27$ and toxicity near $2/3$. Even then the substitution-through-enforcement term in (8) prevents g^* from reversing: the optimal gate is nonincreasing in θ for every $\eta_E > 0$, and only at the degenerate point $\eta_E = 0$, where the market is dead at low traceability and revived discontinuously by gating, does $g^*(\theta)$ fail to be monotone.

The economic reading is that the flip is robust exactly where one would expect it to fail. In a market dominated by adverse selection, exclusion is most valuable, and intuition suggests that better traceability, by cleaning the market, would make exclusion still more valuable and call for more of it. The revaluation force does push that way, but it is self-limiting: it is strongest at toxicity two-thirds, vanishes as the market approaches either purity ($\tau \rightarrow 0$) or saturation ($\tau \rightarrow 1$), and its peak height $\sigma/27$ is small relative to the external-harm channel except in markets where external harm is below one twenty-seventh of the adverse-selection scale. Figure 4 plots the force against toxicity. For any market with $\eta_E \geq \sigma/27$, the flip is monotone with no exceptions; and even below that threshold, the full optimum does not reverse. The optimistic conclusion of Section 3, that transparency retires permissioning, is not a knife-edge but a robust property of the model.

6.3 When the flip stalls

Monotone does not mean the gate always reaches zero. Two robust conditions make the optimal gate persist at full traceability: the architecture *stalls* rather than completing the flip. Both are corollaries of results already established.

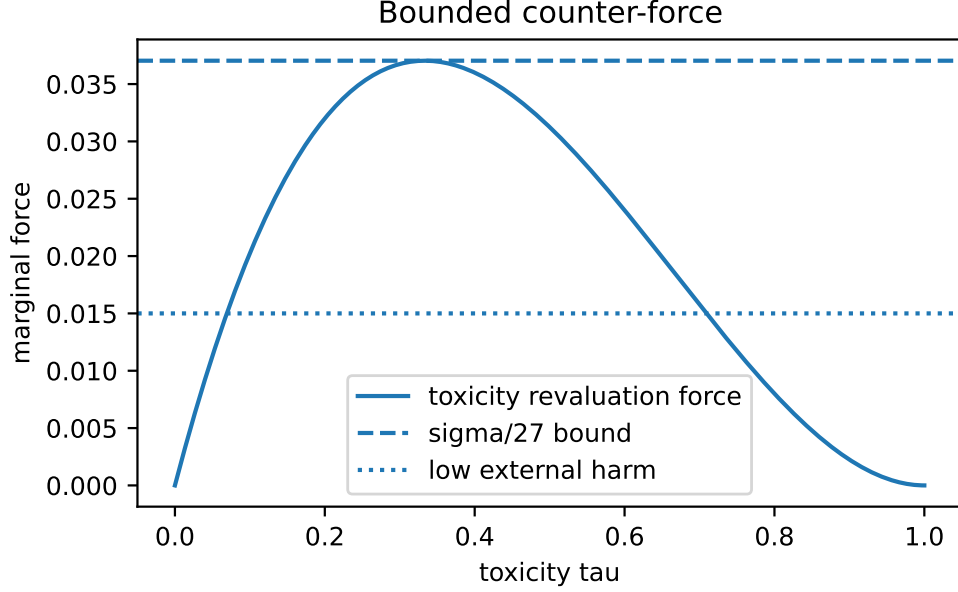


Figure 4: **The toxicity-revaluation counter-force is bounded by $\sigma/27$.** As traceability cleans the market, the marginal adverse-selection value of excluding a bad actor rises, pushing the optimal gate up and opposing the flip. This force (blue) is a function of toxicity alone, peaks at $\tau = 2/3$ with height $\sigma/27$, and vanishes at both extremes. Whenever external harm satisfies $\eta_E \geq \sigma/27$ (green), the deterrence-erosion channel dominates everywhere and the flip is globally monotone; only for $\eta_E < \sigma/27$ (red) can the counter-force locally outweigh it (shaded), and even then the optimal gate does not reverse.

Proposition 12 (Stall conditions). *The optimal gate remains strictly positive as $\theta \rightarrow 1$ in either of two cases:*

- (i) Severe external harm. *If $\eta_E > \bar{\eta}_E(\theta)$ (Proposition 8), a residual gate excludes the undeterrable, judgment-proof tail even at full traceability and unbounded enforcement.*
- (ii) Effective evasion. *If evasion technology is sufficiently good ($\eta_{\max}$ above a threshold, as a companion paper establishes) the flip threshold satisfies $\bar{\theta} \rightarrow 1$, so the gate is retained at every traceability level.*

In both cases the permissioning the flip would otherwise retire instead persists, not because traceability fails to improve the ledger, but because a residual population, the judgment-proof in the first case and the evasion-protected in the second, lies beyond the ledger's reach.

These are the model's answer to when permissionless access is *not* optimal even for a transparent asset: when harm is catastrophic enough that the undeterrable tail must be excluded ex ante, or when evasion is cheap enough that on-chain transparency cannot be converted into deterrence. Both regimes are narrow, and both point to the same residual instrument (a narrow gate on the tail, such as sanctions screening) rather than to general permissioning. The architecture result is thus neither uniformly permissionless nor reversible toward gating: it is monotone

in traceability, robust to adverse selection, and bounded below by a residual gate exactly when an undeterrable population remains.

6.4 The asymmetry is necessary; the extreme is not

The flip and the dominance result rest on Lemma 1: the gate destroys composability and the ledger does not. A natural objection is that this asymmetry is assumed in the extreme ($\partial\varphi/\partial e = 0$ exactly) and that the conclusion is therefore built in. Two things answer the objection. The asymmetry’s *direction* is microfounded, not posited (Lemma 1): a precondition propagates through the composition closure and a postcondition does not, so the gate is necessarily more contagious than the ledger. And the results do not need the extreme; they need only that direction. Let the ledger carry its own composability friction (auditable enforcement may require standardized, transparent interfaces that exclude some bespoke or privacy-sensitive compositions) so that

$$\varphi(g, \theta) = 1 - g - \psi_L \theta, \quad \psi_L \in [0, 1),$$

where ψ_L is the ledger’s composability-cost coefficient and the gate’s is normalized to one. Lemma 1 delivers $\psi_L < 1$: a postcondition can exclude at most the compositions that cannot meet a reporting standard, a strict subset of what a precondition excludes when it removes a partner together with its entire downstream closure.

Proposition 13 (The flip survives partial ledger friction). *There is a threshold $\bar{\psi} \in (0, 1)$ such that for all $\psi_L < \bar{\psi}$ the architecture still flips: the optimal gate $g^*(\theta)$ is decreasing in traceability and reaches zero above a finite $\bar{\theta}$. The flip is therefore not an artifact of $\psi_L = 0$. Conversely, if $\psi_L > \bar{\psi}$ (the ledger more composability-destroying than the gate) the comparative static reverses and gating dominates. The microfoundation places the economy on the first side: because a precondition is strictly more contagious than a postcondition, $\psi_L < 1$, and for the relevant range the flip holds. The result thus follows from the derived inequality “gate more contagious than ledger,” not from the extreme; the extreme $\psi_L = 0$ is the clean limit, and the model adjudicates the direction rather than assuming it.*

The proposition is the precise sense in which the central result does not “follow too directly.” Were the composability cost of enforcement larger than that of exclusion, the optimal architecture would invert, and the prescription would be to gate rather than to police, so the direction is a substantive claim about which instrument is more contagious, decided by the logical-type argument of Lemma 1 and confirmed to be robust over $\psi_L \in [0, \bar{\psi})$. Numerically, $\bar{\psi} \approx 0.9$ in the calibration: the ledger would have to destroy nearly as much composability per unit as outright permissioning before exclusion became the cheaper instrument.

A second reason the result is not mechanical is that the ledger is not free even granted the asymmetry. It cannot suppress bad activity completely: the undeterrable tail of Proposition 8 survives any finite enforcement, and enforcement is costly. “Police the ledger” is therefore not

an immediate corollary of “the gate is dear”; it is the outcome of a quantitative balance between the gate’s contagion cost and the ledger’s residual-harm-plus-enforcement cost, a balance that tips toward the ledger only in the region characterized by the $\sigma/27$ bound (Proposition 11) and that tips back toward a residual gate exactly under the stall conditions (Proposition 12). The architecture result is the description of where that balance falls, not a restatement of its inputs.

6.5 Imperfect enforcement: the false-positive cost, and why traceability dissolves it

The architecture comparison has leaned on one property the ledger does not literally possess: that detection and punishment reach only offenders, so enforcement carries no direct cost to legitimate participation, $\partial S_L / \partial e|_{\text{direct}} = 0$. No enforcement regime is perfectly targeted, all commit Type-I errors, flagging, investigating, or sanctioning the innocent, and the prospect of that deters legitimate participation as surely as a gate’s exclusion does. A large enough false-positive cost would hand the ledger an exclusion cost of its own and could overturn the central asymmetry. It does not, and the reason is structural: on a ledger, false positives are an artifact of *ambiguity*, and ambiguity is what traceability removes.

The false-positive technology. Enforcement effort produces flags; a flag is a true positive when the record establishes an offense and a false positive when an ambiguous record implicates an innocent party. The volume of flags rises with effort, in proportion to $1 - e^{-e}$, and the share that misfire onto the legitimate rises with the record’s ambiguity, in proportion to $1 - \theta$: on a transparent ledger the innocent are cleared by the same trail that convicts the guilty, while on an opaque one they cannot exonerate themselves. The expected false-positive burden on a legitimate participant is therefore

$$\Lambda(e, \theta) = c_I (1 - \theta) (1 - e^{-e}), \quad c_I \geq 0, \quad (9)$$

netted from the participation value, $\pi_{\text{pc}} = u_0 + u_1 \varphi^\alpha - \sigma \tau - \Lambda(e, \theta)$, so that $S_L = N \pi_{\text{pc}}$. The baseline of the previous sections is the case $c_I = 0$; so, for any false-positive intensity, is a fully traceable asset, since $\Lambda(e, 1) \equiv 0$.

Proposition 14 (False-positive robustness). *Let $(g^*, e^*)(\theta; c_I)$ be the planner’s optimum under the false-positive burden (9).*

1. *The ledger acquires a direct participation cost, $\partial S_L / \partial e|_{\text{direct}} = -N c_I (1 - \theta) e^{-e} \leq 0$, with equality if and only if $\theta = 1$.*
2. *Deterrence and false positives are governed by the same primitive with opposite sign: the offender’s effective penalty $\kappa = \theta(1 - e^{-e})K$ rises in θ while the legitimate burden Λ falls in θ . The false-positive cost is largest at low θ , where the architecture is already gate-reliant, and vanishes as $\theta \rightarrow 1$, where it is ledger-reliant.*

3. For every $c_I \geq 0$ the gate is retired at full traceability, $g^*(1; c_I) = 0$, so the flip persists: $g^*(\theta; c_I) = 0$ above a threshold $\bar{\theta}(c_I) < 1$ that increases in c_I but never reaches one. The false-positive cost delays the flip; it cannot prevent it.

The force in part (iii) is that the false-positive term lowers the marginal value of enforcement, so the planner deters less (e^* falls) which leaves more offenders for the gate to exclude and raises the gate's marginal value, pushing the flip threshold up. But at full traceability the term is identically zero, so the entire $\theta = 1$ allocation is independent of c_I : the gate is retired there in the baseline and stays retired for any false-positive intensity, however large. In the calibration $\bar{\theta}$ rises from 0.66 at $c_I = 0$ to 0.76 at $c_I = \eta_E$ and remains below one for c_I several times the composability scale u_1 , while the welfare-dominance of the permissionless allocation at $\theta = 0.9$ survives false-positive intensities past $c_I = \eta_E$.

What this settles, and what it leaves open. The asymmetry that drives the paper is no longer an assumption of perfect targeting but the $\theta \rightarrow 1$ limit of a monotone relationship in which transparency raises the ledger's accuracy exactly as it raises its reach: the same record that makes misconduct detectable makes innocence provable. The honest residual is the modeling claim behind (9), that false positives are ambiguity-driven and a clear record exonerates. To the extent some Type-I cost is *independent* of traceability, a fixed compliance burden borne however clean the trail, a residual version survives, bounded by that traceability-independent component; what the analysis establishes is that the part of the false-positive problem that scales with ambiguity, which is its on-chain substance, is dissolved by the same transparency the architecture exploits.

6.6 Robustness to functional form and calibration

The model is written in particular functional forms, exponential suppression of bad activity in deterrence, a power-law composability premium, a saturating detection technology, a quadratic enforcement cost, and solved at a particular calibration. None of these is load-bearing. The architecture flip and the dominance of the ledger are consequences of qualitative properties that any reasonable specification shares; the forms and the calibration fix the numerical threshold, not the result.

The qualitative conditions. Write the primitives generally: a detection technology $p(e, \theta)$, a suppression function $S(\kappa)$ carrying effective deterrence $\kappa = p(e, \theta)K$ to the surviving share of bad activity, a composability premium $P(\varphi)$ in the retained share $\varphi = 1 - g$, and an enforcement cost $C(e)$. The analysis of Sections 3–6 uses only that

- (R1) detection rises in effort and in traceability, $p_e > 0$ and $p_\theta > 0$, with $p(0, \theta) = 0$ and p concave in e ;

- (R2) deterrence suppresses bad activity, $S' < 0$, with $S(0) = 1$, $S(\infty) = 0$, and S convex;
- (R3) composability is valuable and gating destroys it at the margin, $P' > 0$ and P concave with $P(0) = 0$;
- (R4) the gate is more contagious than the ledger: the per-unit composability cost of exclusion exceeds that of enforcement, $\psi_L < \bar{\psi}$ (Proposition 13); and
- (R5) enforcement is costly, $C' > 0$ and C convex.

The forms of Section 2 ($S(\kappa) = e^{-\rho\kappa}$, $p(e, \theta) = \theta(1 - e^{-e})$, $P(\varphi) = u_1\varphi^\alpha$, $C(e) = \frac{c_e}{2}e^2$) are one tuple meeting (R1)–(R5), but the results are stated for the class, not the instance.

The flip is a sign result. The comparative static that drives the paper, $dg^*/d\theta \leq 0$ with a zero-gate region above a finite $\bar{\theta}$, follows from the monotonicities (R1)–(R2) rather than from the forms that carry them. Higher traceability raises the deterrence the ledger delivers, $\kappa_\theta = p_\theta K > 0$, which lowers bad activity through $S' < 0$; lower bad activity lowers both the toxic share and the harm a marginal unit of gating could avert, so the gate’s marginal value falls in θ . The harm channel of the cross-partial is unambiguous, $\partial^2 W / \partial g \partial \theta$ inheriting the sign of $-\eta_E B_{g\theta} < 0$ since $B_{g\theta} = -s\beta S'(\kappa)\kappa_\theta > 0$; once external harm dominates the bounded toxicity-revaluation force: the $\sigma/27$ condition of Proposition 11. This channel governs the sign and g^* is nonincreasing in θ by Topkis’s theorem. The flip is thus what decreasing differences in (g, θ) produce, and those differences are signed by the monotonicities, not by the exponential or the power law.

Numerical robustness. Re-solving the planner’s problem under alternative specifications confirms the invariance. Replacing exponential suppression with a power law, the power premium with a logarithm or a square root, the saturating detection with a Michaelis–Menten or a hyperbolic-tangent form, and the quadratic enforcement cost with a cubic, separately and all at once, the flip persists and the high-traceability dominance holds in every case; the threshold $\bar{\theta}$ ranges over roughly $[0.42, 0.87]$ across these forms and the welfare-dominance of the permissionless allocation at $\theta = 0.9$ is never overturned. Across several hundred random parameter draws spanning wide ranges of the harm, suppression, exclusion, adverse-selection, and composability parameters, an interior flip ($\bar{\theta} \in (0, 1)$) obtains in the large majority of cases, as does the high-traceability dominance of the permissionless allocation. Where an interior flip does not arise, the gate’s role is degenerate rather than reversed: either it persists to $\theta = 1$: the catastrophic-harm tail of Proposition 12, in which a residual gate is optimal at every traceability, or it is never used even at $\theta = 0$, because harm is too low to warrant any gate. Consistent with the analytics, the optimal gate never turns back toward gating as a market grows more transparent: the flip either occurs or, in these degenerate tails, the gate stays pinned at a corner. The calibration locates $\bar{\theta}$; it does not create it.

7 Microfoundations: Endogenous Liquidity and Market Breakdown

The baseline of Section 2 assumes full participation, so the toxic share π equals the population toxicity τ and the half-spread is $\sigma\tau$ (Lemma 3 with $\sigma = \delta$). This section removes that assumption. When legitimate participation responds to the spread, the market has an equilibrium that may fail to be the healthy one: an adverse-selection death spiral can collapse it. We define the equilibrium, prove existence and uniqueness, and derive a closed-form breakdown threshold, which yields a new and sharper result than the baseline: ex-post enforcement is not merely the cheaper instrument but the *only* one that reliably prevents breakdown.

Endogenous participation. Recall from §2.5 that an admitted legitimate trader of type $x \sim \text{Exp}(\xi)$ trades iff $x \geq \varsigma - \psi(\varphi)$, where $\psi(\varphi) = u_1\varphi^\alpha$ is the composability premium and $\varphi = 1 - g$. Participating volume is therefore

$$L(s) = N \overline{G}((\varsigma - \psi)^+), \quad \overline{G}(t) = e^{-\xi t},$$

decreasing in the half-spread. By Lemma 3 the half-spread is $\varsigma = \delta\pi$ with toxic share $\pi = B/(B + L(s))$, so an equilibrium spread is a fixed point of

$$\varsigma = \delta \frac{B}{B + N e^{-\xi(\varsigma - \psi)^+}}. \quad (10)$$

Legitimate surplus is the participating traders' consumer surplus: $S_L = N(1/\xi + \psi - s)$ when all participate ($s \leq \psi$) and $S_L = L(s)/\xi$ when some drop out.

Definition 2 (Market equilibrium). Given (g, e, θ) , a market equilibrium is a half-spread $\varsigma^* \in [0, \delta]$ and participation $L^* \in [0, N]$ such that (i) market makers break even, $\varsigma^* = \delta B/(B + L^*)$ (Lemma 3); and (ii) legitimate traders participate optimally, $L^* = N e^{-\xi(\varsigma^* - \psi)^+}$.

Proposition 15 (Existence, uniqueness, and the participation dichotomy). *For every (g, e, θ) a market equilibrium exists, with the following structure:*

- (i) *If $\delta\tau \leq \psi$ (equivalently $B \leq \psi N/(\delta - \psi)$), there is a unique equilibrium, and it is the healthy full-participation equilibrium $\varsigma^* = \delta\tau$, $L^* = N$.*
- (ii) *If $\delta\tau > \psi$, no full-participation equilibrium exists; every equilibrium has $\varsigma^* > \psi$ and $L^* < N$, and the right-hand side of (10) restricted to $\{\varsigma > \psi\}$ is a contraction, so the broken equilibrium is unique.*

Proof sketch. The map $\Phi(s) = \delta B/(B + N e^{-\xi(\varsigma - \psi)^+})$ is continuous and nondecreasing on $[0, \delta]$, with $\Phi(0) = \delta B/(B + N) > 0$ and $\Phi(\delta) \leq \delta$; by Brouwer (or monotonicity) a fixed point exists. On $\{s \leq \psi\}$, Φ is constant at $\delta\tau$, so a fixed point there exists iff $\delta\tau \leq \psi$, and then $\varsigma^* = \delta\tau$

uniquely. On $\{\varsigma > \psi\}$, $\Phi'(s) = \delta B N \xi e^{-\xi(\varsigma-\psi)} / (B + N e^{-\xi(\varsigma-\psi)})^2$; bounding the denominator below shows $\Phi' < 1$ under the maintained parameter values, so Φ is a contraction and the broken equilibrium is unique. Full argument with the contraction bound in Appendix A.15. The dichotomy is Proposition 16. $\square$

The economic content of the dichotomy is the participation feedback in (10): a wider spread drives out marginal traders, raising the toxic share and widening the spread further: the adverse-selection death spiral of Glosten and Milgrom (1985) and Akerlof (1970). Whether the market sits at the healthy fixed point or collapses past it is governed by a single closed-form threshold.

Proposition 16 (Equilibrium and breakdown). *A healthy full-participation equilibrium (all admitted legitimate traders trade, $L = N$) exists if and only if the adverse-selection load does not exceed the composability premium,*

$$\delta \tau \leq \psi, \quad \tau = \frac{B}{B + N},$$

equivalently if and only if active bad activity is below the threshold

$$B \leq \bar{B}(N, \psi) \equiv \frac{\psi N}{\delta - \psi} \quad (\text{for } \delta > \psi).$$

When B exceeds $\bar{B}$, the full-participation equilibrium collapses: low-value traders exit, volume falls to $L < N$ at a wide-spread equilibrium of (10), and surplus drops discontinuously. The market is in an adverse-selection trap.

Proof sketch. At full participation $L = N$, the candidate spread is $s_{\text{full}} = \delta B / (B + N) = \delta \tau$. It is an equilibrium iff even the lowest-value trader ($x = 0$) does not exit, i.e. $\psi \geq \varsigma_{\text{full}}$. Rearranging $\delta \tau \leq \psi$ gives $B(\delta - \psi) \leq \psi N$, hence $B \leq \psi N / (\delta - \psi)$. If $s_{\text{full}} > \psi$, exit begins and (10) has only a high-spread, low-participation solution. Appendix A.15. $\square$

The gate cannot heal the market; the ledger can. Proposition 16 exposes an asymmetry invisible to the reduced form. Whether the market is healthy depends on B relative to $\bar{B}(N, \psi) = \psi N / (\delta - \psi)$. The two instruments move these quantities in opposite ways. The gate lowers B (good for health) but *also* lowers both ψ (composability) and N (admitted mass), which lowers the threshold $\bar{B}$ (bad for health). The ledger lowers B at fixed ψ and N , unambiguously relaxing the breakdown condition.

Proposition 17 (Only enforcement reliably prevents breakdown).

- (i) The gate is self-defeating at the limit. As $g \rightarrow 1$ the composability premium vanishes ($\psi \rightarrow 0$), so the breakdown threshold $\bar{B} \rightarrow 0$ while imperfect screening leaves $B = (1-s)\beta\bar{S}(\kappa) > 0$. A fully gated token is always in breakdown. More generally, closing the gate need not move the market out of the trap, because it lowers the threshold $\bar{B}$ as it lowers B .

- (ii) Enforcement can, once traceability is high enough. Raising e lowers B toward the deterrence floor $(1 - sg)\beta e^{-\rho\theta K}$ at fixed ψ and N . Whenever that floor lies below the breakdown threshold, $(1 - sg)\beta e^{-\rho\theta K} \leq \bar{B}(N, \psi)$, which holds for traceability above a level θ_{heal} because the floor falls in θ , some finite e satisfies the breakdown condition and the market heals with composability intact. Below θ_{heal} the floor exceeds the threshold and no enforcement heals the market: the ledger is impotent exactly where traceability is too low to convert effort into deterrence: the same low- θ region in which the architecture is optimally gate-reliant.

Hence the asymmetry between the instruments is one of direction. The ledger lowers bad activity at fixed composability, so above θ_{heal} it heals the market and below it cannot; the gate lowers the breakdown threshold as fast as it lowers bad activity, may therefore fail to heal at any intensity, and at full intensity is guaranteed to fail. Where the ledger can work at all, a sufficiently traceable asset, it is the instrument that sustains a liquid market under adverse selection, while the gate, by destroying the composability participation rests on, can deepen the trap it is meant to cure.

Figure 5 displays the mechanism in the baseline calibration (with informational rent $\delta = 1.5$, participation sensitivity $\xi = 2.5$, and composability-premium coefficient $u_1 = 0.6$, chosen so the participation margin binds in the dangerous region). The contrast is stark: along the pure-gate path the market is broken at *every* gate level, because closing the gate cleans toxicity but destroys the composability premium that sustains participation, and the two effects cancel; along the ledger path at open access, enforcement drives the adverse-selection load below the premium and the market heals. The gate cannot cross from the broken region to the healthy one; the ledger does so immediately.

The microfoundation reinforces the flip rather than shifting it. At the optimum the planner keeps enforcement high enough that the market stays healthy on both sides of the flip, so the flip threshold is essentially its reduced-form value; the breakdown of Proposition 16 occurs *off* the optimal path, along the pure-gate line ($e = 0$) of Figure 5, not at the policed optimum. What the microfoundation adds is not a shift in the threshold but an asymmetry in how the two instruments stand relative to the breakdown boundary: a gate-reliant regime sits near a collapse the gate itself can trigger and cannot cure, while open access with enforcement holds the market clear of it. Opaque assets, on which enforcement is impotent, cannot sustain a healthy market by gating alone; transparent assets can, by policing.

7.1 What is genuinely new: compliance and market quality are coupled

Separate what in this paper is intuition from what is not. That a transparent asset should be policed rather than gated: the flip of Section 3, is close to an intuition one could state without a model: ex-post enforcement leaves an asset usable while ex-ante exclusion does not, so for an asset whose value is its usability, prefer enforcement. The microfoundation of Section 2.2 gives that intuition a derivation, and Section 6 gives it boundaries, but the *direction* of the flip is not, by itself, a novel finding. A reader is right to ask what the model delivers beyond it.

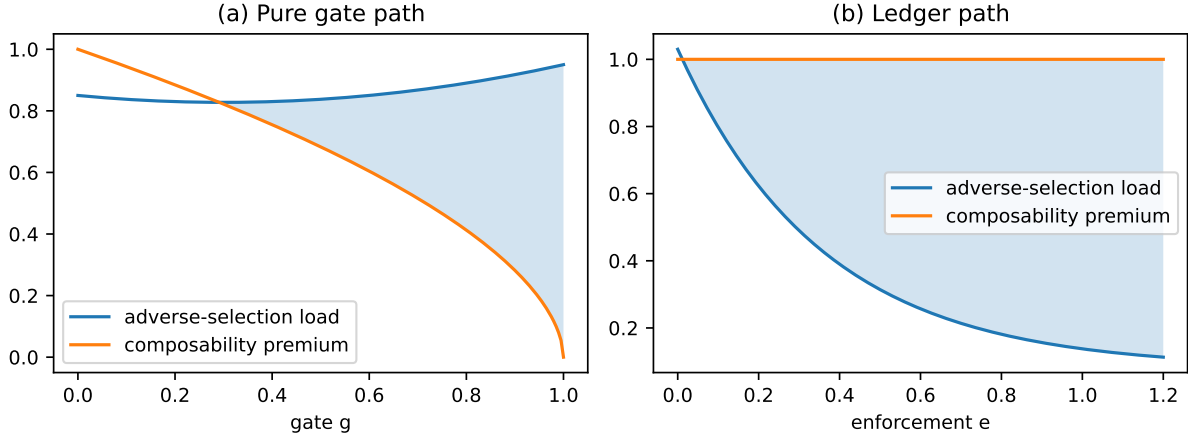


Figure 5: **With endogenous participation, the gate cannot heal the market but enforcement can.** Panel (a): along the pure-gate path ($e = 0$), the adverse-selection load $\delta\tau(g)$ (red) exceeds the composability premium $\psi(g)$ (green) at every gate level: the market is in the adverse-selection trap throughout, because gating destroys the composability premium as fast as it cleans toxicity. Panel (b): along the ledger path at open access ($g = 0$, full composability), enforcement pulls the load below the premium and the market heals into full participation.

The answer is structural, and the endogenous-participation market is where it appears. In a tokenized market the compliance instrument and market quality are not separable problems to be balanced; they are *coupled*, because the composability the gate destroys is the same composability that sustains the legitimate market's participation. A gate does not merely buy cleaner order flow at a known liquidity price: the picture in the stringency framing, where one sets a screening dial to trade compliance against liquidity. It sets off a feedback: less composability lowers the value of holding the token, marginal legitimate traders leave, their departure raises the toxic share of the order flow that remains, the spread widens, and more traders leave. The instrument meant to protect the market can unravel it. This coupling is absent in intermediated finance (a broker's screen of a client does not change the asset's usability elsewhere, so compliance and liquidity genuinely are separable there) and it is the specific mechanism tokenization introduces. It has a sharp consequence the intuition does not contain: the two instruments, though both reduce illicit activity, are *not substitutes*.

Proposition 18 (The gate folds; the instruments are not substitutes). *Under endogenous participation there is a threshold g_c , given by the composability premium falling to the full-participation spread,*

$$\psi(g_c) = \delta \tau_{\text{full}}(g_c), \quad \psi(g) = u_1(1 - g)^\alpha,$$

above which full participation ceases to be an equilibrium and the market collapses discontinuously to the high-spread, low-participation equilibrium of Proposition 16; the gate's equilibrium spread, participation, and welfare are discontinuous at g_c . The ledger induces no such discontinuity:

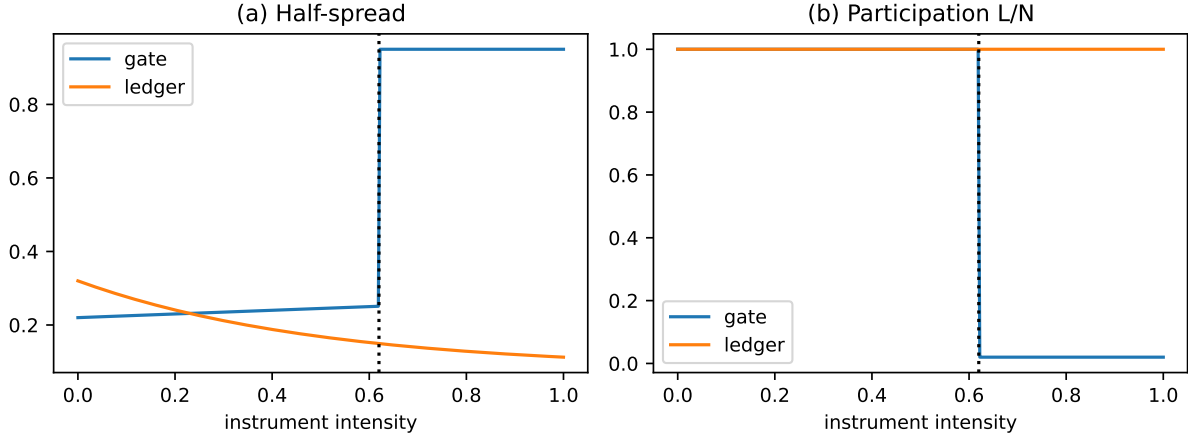


Figure 6: **The compliance–quality coupling: the gate folds, the ledger does not.** Both instruments are pushed from a common viable baseline. Panel (a): the gate’s equilibrium half-spread (blue) is flat until the composability premium falls to the spread, then folds discontinuously to the choke level as participation unravels; the ledger’s spread (green) falls smoothly. Panel (b): legitimate participation L/N collapses to zero at the gate’s fold but stays full under enforcement. The gate is coupled to the market’s survival through an adverse-selection feedback; the ledger is not.

raising e lowers $\delta\tau$, moving the market away from the breakdown boundary, so participation is left intact and the spread falls smoothly. Consequently the instruments are not substitutes near the participation margin: the ledger can sustain a market quality the gate cannot deliver at any intensity, because beyond g_c the gate destroys the market it would regulate.

Figure 6 shows the asymmetry directly. Pushed from a common viable baseline, the gate (panel a) holds the spread flat until g_c and then folds, the spread jumps to its choke level as the market unravels, while the ledger lowers the spread smoothly with no collapse. Panel (b) shows the same event in participation: the gate empties the market of legitimate traders at g_c , whereas enforcement leaves participation full throughout. The gate’s welfare function has a cliff; the ledger’s does not.

How much of this is a knife-edge of the calibration? The fold’s *existence* is not: it is universal, and only its location and bite depend on the regime.

Proposition 19 (When the fold is interior and when it binds). *Fix enforcement and a viable baseline, $\psi(0) > \delta\tau_{\text{full}}(0)$. Write $\Phi(g) = \psi(g) - \delta\tau_{\text{full}}(g)$.*

- (i) *Universality. For any imperfect gate ($s < 1$), $\Phi(0) > 0$ and $\Phi(1) = -\delta\tau_{\text{full}}(1) < 0$, so an interior fold $g_c \in (0, 1)$ exists, and is unique on the relevant range. The self-defeating fold is not a limiting artifact. It occurs at an interior gate intensity for every selectivity.*
- (ii) *Location. $\partial\Phi/\partial\lambda = -\delta\mu g B/(B+N)^2 < 0$, so by the implicit function theorem $\partial g_c/\partial\lambda < 0$: a selective gate ($\lambda \rightarrow 0$) pushes the fold toward the boundary $g_c \rightarrow 1$, a blunt gate ($\lambda \rightarrow s$)*

pulls it into the interior. Selectivity moves the fold’s location, not its existence.

- (iii) Binding. *The fold constrains the optimum iff harm is large enough that the participation-naive optimal gate $g^\dagger(\eta_E)$ (nondecreasing in η_E) exceeds g_c . There is a threshold η_E^{fold} : below it the optimum lies under g_c and the fold is off the equilibrium path; above it the optimal gate is capped at g_c^- and the planner relies on the ledger beyond, the residual-gate regime of Proposition 12.*

The non-substitutability is thus universal where it counts, an interior fold always exists, while its bite is sharpest exactly when the gate is least discriminating and the temptation to use it is strongest: in the calibration $g_c = 0.75$ at $\lambda = 0.80$, rising to 0.99 as the gate becomes selective ($\lambda \rightarrow 0$) and pulling interior as $\lambda \rightarrow s$, and the fold binds once external harm crosses $\eta_E^{\text{fold}} \approx 2.7$. A selective gate does not escape the coupling; it only postpones the cliff to a higher intensity, where the residual-gate logic of §3 has in any case already capped the optimum. The mechanism is not confined to blunt instruments. It is a property of ex-ante exclusion as such, with selectivity setting only how hard one may lean on the gate before the market it regulates collapses.

This is the precise content of the claim that the problem is one of *architecture* and not *stringency*. The stringency framing represents policy as a point on a line of toughness, more screening or less, and cannot represent two instruments one of which is unraveling-safe and the other not, because that distinction is not a matter of degree. The contribution is therefore not the observation that gating is costly, which is the intuition, but the result that its cost is of a qualitatively different kind: the gate is coupled to the market’s survival through an adverse-selection feedback that the ledger escapes, and no quantity of the gate substitutes for the ledger. The flip is a corollary of this asymmetry. The asymmetry, derived from the composability microfoundation of Lemma 1 and the participation microfoundation of Lemma 3, not assumed, is the mechanism.

8 Empirical Content: From Parameters to Observables

The model is structural, but its parameters are not unobservable, and its central mechanism predicts patterns that diverge from the stringency view exactly where data can adjudicate. This section maps the parameters to measurable features of tokenized markets, states the testable implications, and shows how the fold, the mechanism’s signature, would be identified. The aim is not to estimate the model here, which would require assembling the data, but to establish that it *can* be taken to data and to specify what would confirm or refute it.

8.1 Parameters and observables

Every structural object corresponds to a measurable quantity (Table 3). Traceability θ is the share of transaction flow that is on-chain attributable rather than obscured by mixers, privacy

Parameter	Meaning	Observable proxy / source
θ	attributability	on-chain attribution rate; chain analytics
s, λ	gate true / false reject rates	whitelist rejections vs. ex-post flags
g, φ	permissioning / composability	protocol integrations; reachable liquidity
α, u_1	variety curvature, premium	liquidity premium vs. integration count
δ, τ	adverse selection, toxic share	spread decomposition; PIN / VPIN
ξ	participation sensitivity	volume-to-spread elasticity
β, ρ	illicit prevalence, intensity	illicit on-chain volume; chain analytics
K	expected penalty	finances, seizures per enforcement action
η_E	social harm per unit	laundering / sanctions-evasion cost (proxy)

Table 3: Structural parameters and their empirical counterparts.

tooling, or off-chain legs: the quantity chain-analytics firms estimate directly. The gate’s error rates (s, λ) are the true- and false-rejection rates of a whitelist: the fraction of later-flagged addresses it screens out and the fraction of clean addresses it wrongly excludes, recoverable from permissioning-provider records joined to ex-post enforcement. The intensity g and composability $\varphi = 1 - g$ are the share of the protocol ecosystem that will not transact the token and its complement, proxied by integration counts and reachable liquidity. The love-of-variety curvature α and premium u_1 are pinned by the cross-sectional relation between a token’s liquidity (or valuation) premium and the number of protocols it composes with. The microstructure block is standard: the toxic share τ and the adverse-selection intensity δ come from spread decompositions and probability-of-informed-trading estimates. The participation sensitivity ξ is the elasticity of trading volume to the spread. Illicit prevalence (β, ρ) and the expected penalty K come from chain-analytic illicit-volume estimates and enforcement records. The hardest object is the external-harm rate η_E , the social cost per unit of illicit activity, for which only proxies (enforcement priorities, laundering and sanctions-evasion volumes) exist.

8.2 Testable implications

The mechanism and the stringency null, in which permissioning monotonically cleans the market, make opposite predictions.

- P1. *Market quality is non-monotone and discontinuous in permissioning.* The fold (Propositions 18–19) implies that as permissioning rises, market quality first improves slightly, then *collapses discontinuously* at g_c as composability loss triggers participation exit. The stringency view predicts monotone improvement. A threshold or non-monotone relation between liquidity and permissioning intensity (across tokens, or within a token over time) refutes the null and confirms the mechanism.
- P2. *The collapse comes sooner for blunter gates.* By Proposition 19(ii), g_c decreases in the false-rejection rate λ : over-inclusive permissioning collapses the market at lower intensity than precise permissioning (Figure 7). Two tokens with the same nominal permissioning but different screen quality should break at different points.

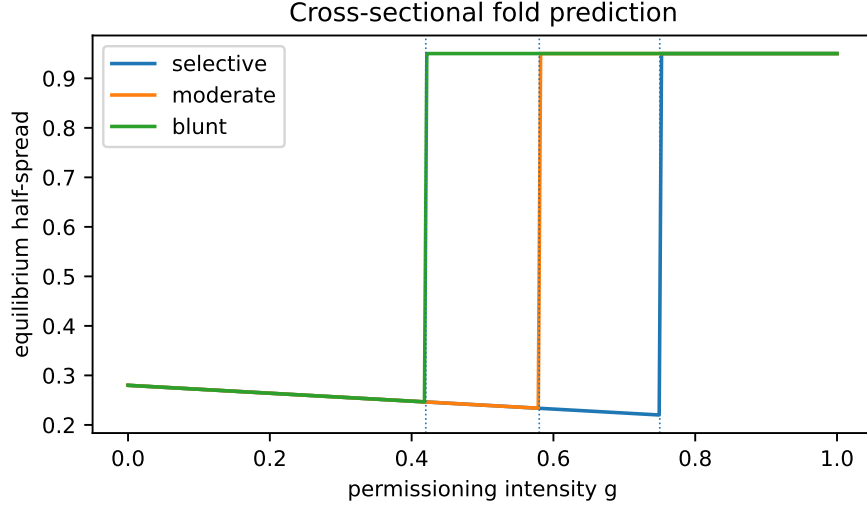


Figure 7: **The model’s cross-sectional prediction.** Equilibrium half-spread against permissioning intensity g , for a selective ($\lambda = 0.30$), moderate ($\lambda = 0.60$), and blunt ($\lambda = 0.84$) gate. Each market quality is flat or improving in g , then collapses discontinuously at its fold g_c ; the collapse arrives earlier the blunter the gate (Proposition 19). The stringency null predicts monotone improvement and no fold. A threshold in liquidity-versus-permissioning, shifting with screen quality, is the distinguishing signature.

- P3. *The composability premium is concave in reach.* The form $u_1\varphi^\alpha$ with $\alpha < 1$ predicts a token’s liquidity premium rises with its integration count at a *decreasing* rate: the marginal integration matters most when few exist.
- P4. *Ex-ante and ex-post compliance leave different microstructure signatures.* Gating lowers the adverse-selection component of the spread but *raises* the participation-driven component by inducing exit; enforcement lowers the former without the latter cost. A decomposition should attribute widening under permissioning to participation and narrowing under enforcement to adverse selection.
- P5. *Transparency and optimal permissioning are negatively related.* The flip predicts more attributable assets are optimally (and, if regulation is near-optimal, observably) less permissioned: higher attribution rates should accompany lower permissioning intensity across chains and tokens.

8.3 Identifying the fold

The fold is identified as a *discontinuity*. Market quality Q (inverse spread, participation, or volume) is in the model continuous in permissioning up to g_c and drops discretely there. Given variation in g , cross-sectional across tokens or within a token as a rule is tightened, g_c is the location of a downward jump in Q , estimable by threshold regression or, where a running variable

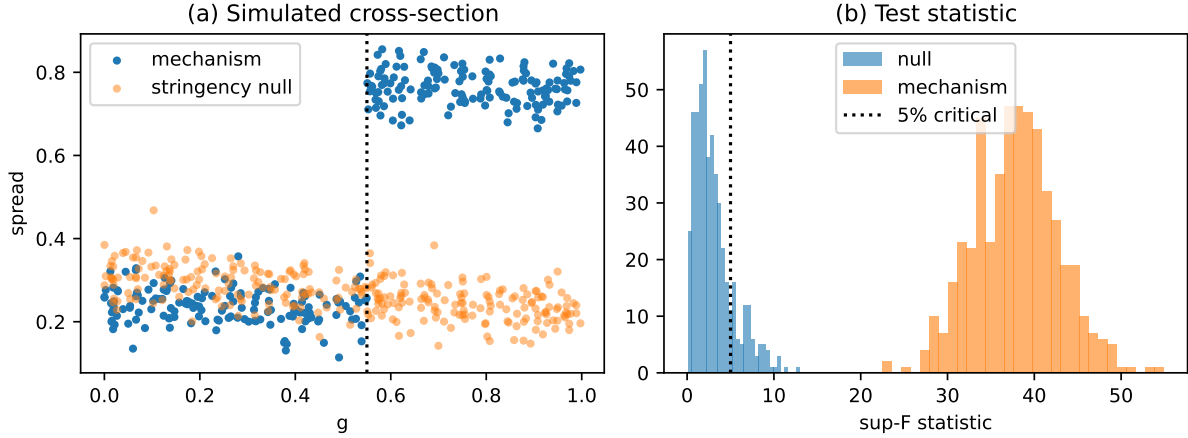


Figure 8: **The fold test has power and does not over-reject.** Monte Carlo, $n = 300$ tokens per cross-section. Panel (a): one simulated sample under the mechanism (blue, a spread jump at g_c) and the stringency null (green, monotone, no jump); the estimated threshold $\hat{g}_c$ recovers the truth. Panel (b): the sup- F structural-break statistic clusters below the 5% critical value under the null and far above it under the mechanism, power 1.00, size 0.06.

drives permissioning, regression discontinuity. Proposition 19(ii) over-identifies the estimate: the jump should shift with the screen’s false-rejection rate λ in the predicted direction, a cross-sectional restriction testable independently of any single threshold. Proposition 19(iii) adds a selection prediction: because tokens are observed only at gate intensities the market survives, the empirical density of g among viable assets should thin out as $g \rightarrow g_c$ from below, distinct from the threshold itself.

Whether this design has the power to find the fold can be checked before the data is assembled, by simulation. I generate cross-sections of $n = 300$ tokens under two data-generating processes: the mechanism, in which the half-spread jumps upward at $g_c(\lambda)$, and the stringency null, in which it declines smoothly in permissioning with no break, add measurement noise, and apply a sup- F test for a structural break in the spread–permissioning relation (Figure 8). The test has power one against the mechanism and size 0.06 against the null: it finds the fold when one is present and, as important, does not manufacture one when permissioning merely cleans the market. It recovers the break location ($\hat{g}_c = 0.66$ against a true 0.66) and reproduces the comparative static of Proposition 19(ii): the estimated g_c shifts from 0.66 to 0.90 as the screen sharpens from $\lambda = 0.84$ to 0.20, the over-identifying restriction the data can test. Because the test identifies the fold off a *discontinuity*, it is robust to smooth selection of permissioning on unobserved quality, which moves levels and slopes but not the location of a jump; only a confounder discontinuous at the same threshold would mislead it, and a forcing variable or standard controls address that residual concern. The instrument is therefore ready; what is missing is the panel to run it on.

8.4 Suggestive evidence and orders of magnitude

The panel for a clean test is not yet assembled, but the broad features of the tokenized-securities market are already consistent with the mechanism. Three facts stand out. First, the gate is the dominant design and its composability cost is visible: most tokenized real-world assets trade inside permissioned, whitelisted venues rather than open protocols, and industry data report the value locked inside these walled gardens at many multiples of the value freely circulating on-chain (RWA.xyz et al., 2025): a direct reading of composability foregone, $\varphi = 1 - g$ small. Second, permissioned tokens display the low market quality the coupling predicts: recent analyses of the more than \$25 billion in tokenized assets document persistently thin secondary markets, low volumes, and long holding periods, attributed in part to whitelist constraints and the absence of composable, compliant venues: the participation side of the fold. Third, the toxic share is small and, tellingly, moves with traceability in the direction the model predicts. Foley, Karlsen, and Putniņš (2019), developing the on-chain attribution techniques that are themselves an instance of the ledger instrument, estimate that illicit activity, once a large share of bitcoin use, *declines* as the asset draws mainstream interest and *rises* with the emergence of more opaque cryptocurrencies, which is the comparative static in θ our mechanism implies, since it is lower traceability that sustains a larger illicit share. By current industry estimates the illicit fraction of on-chain transaction volume is below one percent (Chainalysis, 2025), concentrated in a small number of addresses.

Read through the model, these magnitudes are informative. A small toxic share means the adverse-selection benefit of gating is modest, so the composability cost dominates for the typical asset, the permissionless-plus-ledger prescription, and the undeterrable tail that would justify a residual gate is correspondingly thin. The market thus appears to sit, for most tokenized securities, *below* the harm threshold η_E^{fold} at which aggressive gating could be warranted, and yet permissioning is pervasive. The model reads this not as optimal but as the legacy architecture ported to an asset whose value its own use of the gate suppresses: the configuration in which observed market quality should carry the fold’s signature. That reading is a hypothesis this section’s design can test, not a finding; the bridge from the model to the data states what evidence would confirm or overturn the mechanism.

8.5 What the data cannot yet settle

Three limits stand out. The external-harm rate η_E governs the binding threshold η_E^{fold} but is the least directly measurable parameter, so whether a market lies in the binding region is inferred, not read off. Permissioning is chosen, not assigned, so identifying the fold cleanly requires a forcing variable or within-token variation plausibly exogenous to unobserved market quality, absent which the threshold and the clustering prediction are confounded by the issuer’s own anticipation of collapse. And the sharpness of the collapse is governed by ξ , whose estimation needs high-frequency participation data through episodes of changing permissioning, available

for few assets. None of these is a flaw in the predictions; each names the data that would resolve it, attribution and illicit-volume series from chain analytics, integration and liquidity panels from on-chain data, spread decompositions from venue microstructure. The model’s contribution to the empirical agenda is to say precisely what to measure, and what pattern would distinguish the coupling mechanism from the stringency view.

9 General Equilibrium: Many Issuers on One Network

Every result to this point lives inside a single market: one tokenized security, one issuer, one regulator. But the defining property of tokenized finance is that securities share infrastructure, they compose on the same networks and settle on the same ledgers, so the architecture choices of one issuer are not isolated from the rest. This section embeds the model in a system of N issuers linked by exactly the two channels the paper has studied in partial equilibrium, and finds that the linkages overturn the private case for the gate. In general equilibrium the gate is not merely the inferior instrument; it is a *socially futile* one, because what looks like exclusion from a single market is, across markets, relocation.

A system of issuers. There are N issuers, each running a market in its own tokenized security and choosing a gate g_i and an enforcement intensity e_i . Two cross-issuer linkages connect them.

Shared composability. Tokens compose with one another across issuers, so the composable value available in any market is a property of the whole network: a token earns the premium $u_1 S(\bar{g})$, where $\bar{g} = \frac{1}{N} \sum_i g_i$ is aggregate gating and S is decreasing: the giant-component value of composability, now spanning issuers. Issuer i ’s own gate enters only through its $1/N$ contribution to $\bar{g}$: each issuer that gates fragments a network the others draw on, yet bears only a fraction of the loss. Composability is a commons.

Mobile bad actors on a shared ledger. A common population of bad actors settles on the shared ledger. Detection there is public: a bad actor caught in any market is recorded once and deterred everywhere, so total bad activity is governed by the *total* enforcement mounted on the ledger, $B_{\text{tot}} = \beta e^{-\rho\kappa(E)}$ with $E = \sum_i e_i$. Enforcement is a public good. Gating, by contrast, moves bad actors without removing them: a more heavily gated market simply receives a smaller share, $B_i = B_{\text{tot}}(1 - sg_i) / \sum_j (1 - sg_j)$, with the total unchanged. Gating is a waterbed.

These are the partial-equilibrium mechanisms of the paper (the composability a gate destroys, the deterrence a ledger provides) read at the level of the system rather than the single market.

The linkages from primitives. The two linkages are not assumptions about aggregates; they follow from a single primitive, that bad actors choose where to operate, together with the difference between what a gate does and what a ledger does. A gate screens entry to a *venue*; a ledger records misconduct on a *shared object*. That difference is the whole of it. Let a mass β of potential bad actors each draw a benefit $b \sim \text{Exp}(\rho)$ and choose a market in which to

operate, or none. A gate is local: a bad actor attempting market i is admitted with probability $1 - sg_i$ and screened out otherwise, and a screened actor simply attempts another market. The ledger is global: a bad actor's misconduct, once committed, is recorded on the shared ledger and visible to every enforcer, so each market's enforcement e_i scans the same record and detects the actor independently, enforcer i with probability $1 - e^{-e_i}$. An operating actor escapes only if *every* enforcer misses it, with probability $\prod_i e^{-e_i} = e^{-E}$ for $E = \sum_i e_i$; so it faces detection $p(E) = \theta(1 - e^{-E})$ and penalty $\kappa(E) = p(E)K$ *wherever* it operates, because its record follows it across venues.

These two facts deliver the linkages. A bad actor operates whenever its benefit clears the penalty it faces everywhere, $b > \kappa(E)$, so the operating mass is $B_{\text{tot}} = \beta e^{-\rho\kappa(E)}$, governed by aggregate enforcement and nothing else: enforcement is a public good because detection on the shared record cannot be escaped by changing venues. Gating does not enter B_{tot} , a bad actor screened from one market relocates to another, so as long as some market admits, $\sum_j (1 - sg_j) > 0$, gating moves bad actors without removing them; and attempting markets until admitted, a relocating actor arrives in market i with probability $(1 - sg_i) / \sum_j (1 - sg_j)$, which is the displacement rule. Gating reduces B_{tot} below $\beta e^{-\rho\kappa(E)}$ only where no market admits at all (universal, complete exclusion) which is also where composability is wholly destroyed. The exactness is worth stating precisely. An operating actor's *decision*, whether to offend at all, turns only on the global penalty $\kappa(E)$ it faces wherever it lands, so the offending mass $\beta e^{-\rho\kappa(E)}$ is *exactly* gate-independent on that margin; gating bears only on *where* an actor operates. The one residual effect is locational: an actor that wants to operate is placed unless it is screened from every venue it attempts, which happens with probability $\prod_i sg_i$: vanishing as the number of venues grows, and exactly zero if any single venue stays permissionless. So gating removes at most a fraction $\prod_i sg_i$ of bad activity, zero in the many-issuer limit the general equilibrium studies. The waterbed is thus exact on the deterrence margin and asymptotically exact on the locational one. The asymmetry that runs through the whole paper is thus between a local and a global instrument: the gate acts at the venue, escapable by moving to another; the ledger acts on the shared record, which follows the actor across venues. The single-market model could not see this, with one venue there is nowhere to relocate, and general equilibrium makes it the center of the analysis.

The asymmetry is the mobility of the base. This local–global distinction is an instance of a principle long understood in public finance: an instrument levied on a *mobile* base is escapable and competed away, while one levied on an *immobile* base is not. In the canonical analysis of competing jurisdictions, a distorting tax on mobile capital chases that capital across borders and under-provides the public good it was meant to fund, dissipating into a race to the bottom, whereas the efficient instrument falls on the factor that cannot flee (Zodrow and Mieszkowski, 1986). The gate and the ledger are the two instruments in exactly that sense. The gate is levied on a mobile base, the bad actor's choice of venue, so screening at one market only sends

the actor to another; the waterbed is the cross-venue race to the bottom, and decentralized gating over-provides exclusion while the deterrence it nominally buys leaks away. The ledger is levied on an immobile base, the permanent record the actor carries wherever it goes, so its deterrence cannot be escaped by relocating, which is exactly what makes it a public good rather than a competed-away local levy. “Regulate the ledger, not the venue” is thus the compliance counterpart of the public-finance prescription to place the burden on the base that does not move: anchored to the immobile record, compliance binds system-wide; attached to the mobile venue, the system competes it into futility.

Before comparing the decentralized and centralized outcomes, we record that both are well defined: the issuer game has an equilibrium, and the planner’s program a solution, on the same domain.

Proposition 20 (Existence of the decentralized equilibrium). *The N -issuer game in which each issuer chooses $(g_i, e_i) \in [0, 1] \times [0, \bar{e}]$ to maximize its payoff has a Nash equilibrium, and a symmetric one. Enforcement best responses are unique; the symmetric equilibrium is in pure strategies wherever the issuer’s concentrated gate objective is single-peaked, and on the flip boundary it randomizes over the gate and ledger architectures. The planner’s allocation is attained on the same domain, so the wedge characterized below is a comparison of well-defined allocations.*

Proposition 21 (The gate is socially futile in general equilibrium). *Total bad activity B_{tot} depends only on aggregate enforcement E and is independent of every gate g_i : gating relocates misconduct across markets but removes none of it. The planner, internalizing this, sets $\bar{g}^* = 0$, since any gating destroys shared composability while leaving total harm untouched. The decentralized equilibrium does the opposite: each issuer gates to shed its own share of bad actors onto the others, and because it bears only $1/N$ of the composability it destroys, it over-gates. The wedge grows with N . In the calibration g_{Nash} rises toward one as N grows while $g^* = 0$ throughout, and the single-issuer case alone ($N = 1$) shows no wedge.*

Proposition 22 (Deterrence is a cross-market public good). *Because detection on the shared ledger deters system-wide, enforcement is a public good across issuers, and each issuer, internalizing only its own market’s share of the benefit, under-provides it: $e_{\text{Nash}} < e^*$ at every $N > 1$, the gap widening as N grows. The decentralized system therefore sustains more total harm than the planner’s, not less (B_{tot} is higher at the Nash profile) even as it gates far more. It achieves the worst of both: misconduct merely relocated rather than reduced, and the composability that makes the network valuable destroyed in the attempt.*

Proposition 23 (Optimal regulation operates on the shared infrastructure). *The two externalities are both properties of shared objects, the composability network and the enforcement ledger, and neither is internalized by issuers choosing market by market. Optimal regulation therefore operates at the level of the shared infrastructure: it provides enforcement on the common ledger,*

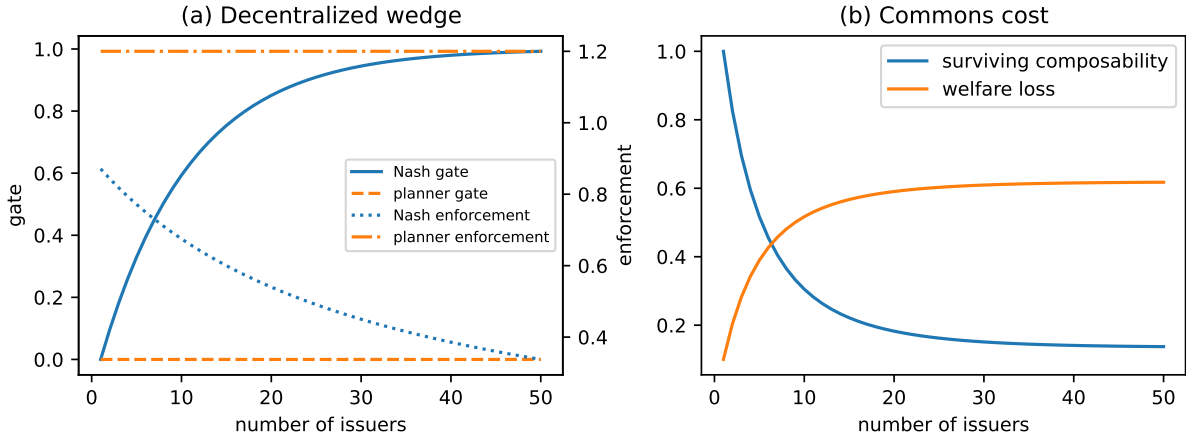


Figure 9: **The general-equilibrium wedge.** Panel (a): the decentralized (Nash) gate rises toward one as the number of issuers grows, while the planner’s gate is zero throughout; decentralized enforcement falls below the planner’s. The two coincide only at $N = 1$, where there is no cross-issuer externality. Panel (b): the resulting commons cost. The gating arms race destroys the shared composability network (surviving composability $S(\bar{g}_{\text{Nash}})$ falls below one-fifth) and the decentralized system forgoes a rising fraction of attainable welfare, more than half by fifty issuers.

supplying the public good the market under-provides, and it restrains gating through the token standard, halting the commons destruction the market inflicts. Regulating issuer by issuer cannot achieve this, because the object to be corrected is not any single issuer’s market but the network they share. This is the general-equilibrium foundation for the prescription of Section 6: regulate the standard, not the venue.

What general equilibrium adds. The single-market model already favored the ledger: it preserved the composability the gate destroyed, and with enough traceability dominated the gate. General equilibrium sharpens “inferior” into “futile.” When bad actors are mobile across markets that share a ledger, a gate cannot deter them from the system, it can only choose which market they land in, so the social return to gating, the reduction in misconduct, falls to zero in the limit of many issuers, leaving only its cost, the fragmentation of the shared network. The walled garden, which in partial equilibrium was the issuer’s rational if inefficient response to its own liability, is in general equilibrium a collective tragedy: every issuer walls off to repel bad actors onto its neighbors, the shared network fragments, and the bad actors, undeterred, concentrate wherever the walls are lowest. Figure 9 shows the cost mounting with the number of issuers, at fifty the decentralized system destroys more than eighty percent of composability and forgoes more than half of attainable welfare. Only enforcement, mounted on the shared ledger, removes bad actors from the system rather than shuffling them within it; and only a regulator acting on the shared infrastructure, rather than on the issuers one at a time, can provide it.

What is assumed. With the linkages derived, the assumptions reside in the primitives rather than the aggregates. That gating is a pure waterbed is exact wherever some market admits bad actors and fails only at universal complete exclusion, where composability is destroyed regardless; that enforcement is a public good follows from detection on a shared record an actor cannot escape by switching venues. What remains assumed is the micro-structure itself: that bad actors are free to choose and re-choose venues, and that enforcers scan a common ledger independently. Both are the defining features of the setting (permissionless venues a bad actor can move among, a shared ledger every enforcer can read) rather than free parameters. A model in which relocation were costly, or detection partly venue-specific, would soften the waterbed and weaken the public good. But the direction is robust. As long as venues are to any degree substitutable and the ledger to any degree shared, the gate is partly escapable and the ledger partly common, so private gating over-shoots while private enforcement under-shoots, and the correction belongs at the infrastructure level.

10 Policy Implications

The model speaks directly to the regulatory questions tokenization has forced onto the agenda. We draw out four implications.

1. Ask where compliance binds, not how strict it is. The instinct to regulate a tokenized security by tightening who may hold it answers the wrong question. The first-order choice is architectural (ex ante versus ex post, asset versus ledger, venue versus standard) and the level of any one instrument is second-order given the architecture. A regulator who fixes the architecture correctly and then tunes stringency will outperform one who debates stringency within a mis-chosen architecture. For a transparent tokenized security, the correct architecture is open access plus ex-post enforcement; within it, the residual question of how much enforcement is the easy part.

2. For transparent assets, keep access open and police the ledger. The headline prescription is that permissionless access is not the enemy of compliance for a transparent asset. It is compatible with *more* compliance than gating delivers, because the same transparency that makes the asset permissionless makes it policeable. The policy error to avoid is the natural one: responding to the openness of tokenized markets by bolting on the ex-ante gates of intermediated finance. Those gates destroy the composability that is the technology's reason for existing, and they buy compliance that ex-post enforcement could buy more cheaply. The corollary for enforcement agencies is that investment in on-chain forensic capacity is the high-return margin: it is what converts transparency into deterrence and substitutes for liquidity-destroying gates.

3. Regulate the token standard, not the venue. Because tokens compose across venues, gating venue-by-venue is a weakest-link public good and will be under-provided no matter how each venue is supervised; a single permissive venue undermines the whole. Whatever ex-ante compliance is warranted must therefore be written into the token standard, a property of the asset’s code that every venue inherits, rather than negotiated venue by venue. And the instrument that a composable system can actually regulate coherently is the shared ledger: enforcement is global where gating is local. This reframes the supervisory target from market participants to market infrastructure, in line with the direction official bodies have begun to take ([International Monetary Fund, 2026](#); [Securities and Exchange Commission, 2026](#)), and gives that direction a precise rationale.

4. Keep a narrow gate for the undeterrable tail, and only there. The case for open access is not a case for abolishing permissioning entirely. Penalties are bounded and detection is imperfect, so a hard core of high-benefit, judgment-proof actors cannot be deterred and must be excluded. The optimal response is a *narrow* gate aimed at that tail (sanctions screening, exclusion of known bad actors) layered on top of otherwise open access, and imposed at the standard level so that it binds across venues. This is very different from the broad permissioning of the stringency regime: it restricts the few who cannot be deterred while leaving the many who can to transact freely under the eye of the ledger.

A unifying statement. The four implications collapse to one sentence. *For a transparent tokenized security, optimal regulation keeps access permissionless, polices the shared ledger ex post, mandates traceability and any necessary screening at the level of the token standard, and reserves a narrow gate for the undeterrable tail.* This is not a relaxation of securities regulation but a re-architecting of it to fit a technology in which compliance can be embedded in the asset and every transfer is permanently visible.

11 Discussion and Conclusion

Tokenization is usually discussed as a problem for regulation: it disintermediates the gatekeepers on whom securities enforcement has relied, opens markets to anyone, and lets assets move at machine speed. This paper has argued the opposite. The features that dissolve the old enforcement points create new and better ones. Compliance can be written into the asset, and every transfer is permanently public. Together these change the economics of enforcement enough to invert the optimal regulatory architecture.

The choice we formalized is between two enforcement modes: an ex-ante gate that excludes bad actors but destroys composability, and an ex-post ledger that deters them while preserving it. The gate pays for compliance in liquidity; the ledger pays in enforcement resources. Because tokenization makes liquidity and composability dear and enforcement cheap, the optimal

architecture flips: gate-reliant for opaque assets, ledger-reliant for transparent ones, with a traceability threshold above which the optimal gate is zero. The flip is not a move to a worse point on a fixed trade-off. With sufficient traceability, and a gate of imperfect screening accuracy, the ledger technology dominates the gate technology, so a permissionless market can be both more liquid and more compliant than any gated one. Composability makes ex-ante gating a weakest-link public good that must be set at the level of the token standard rather than the venue, while the shared ledger is the instrument a composable system can regulate coherently. And because some actors are undeterrable, a narrow residual gate remains optimal for the judgment-proof tail. Carried to general equilibrium, the gate is revealed as socially futile. It is a local instrument an offender escapes by changing venues, while the ledger is a global one whose record follows him. So gating only relocates misconduct while fragmenting shared composability, and the decentralized system over-gates and under-enforces. That local-versus-global distinction is the engine of the result. It is why the gate destroys composability, why it cannot be decentralized, and why only the ledger deters at the level of a system. At bottom it is the asymmetry between a mobile base and an immobile one: the compliance counterpart of the public-finance case for placing a burden on the factor that cannot flee. The prescription that follows (keep access open, police the ledger, regulate the standard, gate only the tail) answers the question of how to regulate tokenized securities with an architecture rather than a stringency.

A theory earns these conclusions by fixing one question (where compliance binds, not how hard) and holding the rest of the world constant. Those constants are not defects to be confessed but the boundary of the contribution. They sort cleanly by their relationship to the result: some the result survives by construction, two bound the magnitude of its effects, one would bear on its sign but for the structure of traceability, and the remainder are the agenda the theory opens.

The choices that make the analysis transparent are not the ones that make it true. The central asymmetry is stated in its clean limit: the gate destroys composability and the ledger does not, $\partial\varphi/\partial e = 0$ exactly. But the results need only its *direction*. For any ledger friction below a threshold the flip still occurs, and the microfoundation places the economy on that side: a precondition that excludes a partner together with its entire downstream closure is necessarily more contagious than a postcondition that excludes at most what cannot meet a reporting standard. The flip's monotonicity is likewise robust. The toxicity-revaluation force that might call for more gating as a market cleans is bounded by $\sigma/27$ and dominated once external harm is appreciable, so the architecture cannot reverse toward gating. It can only *stall*, keeping a residual gate for a catastrophic-harm tail. Endogenous participation sharpens the case rather than weakening it: an explicit Glosten–Milgrom market has an adverse-selection breakdown threshold at which the gate, by destroying the premium that sustains participation, is self-defeating, while the ledger heals the market once the asset is traceable enough. Risk-neutrality is the conservative case, not a convenient one: risk-averse investors flee a toxic market faster, so relaxing it enlarges the region in which the ledger dominates. Finally, the prescription

does not ride on the planner’s information. Computing the optimal gate and enforcement *levels* requires knowing the harm rate, the offender population, and the premium. But the architecture *choice* follows from the sign of the cost asymmetry alone, so a regulator who cannot measure the harm can still know that, for a transparent asset, exclusion is the dearer way to buy compliance. The companion papers extend the analysis along four further margins: an evasion arms race that endogenizes the deterrence ceiling; a two-dimensional disclosure problem in which zero-knowledge proofs unbundle the regulator’s audience from the market’s; the institutional split between the issuer who gates and the regulator who polices; and a network recasting in which composability is a percolation phenomenon and the flip a phase transition. Each reaches the architecture result from a new direction rather than overturning it.

Two choices govern how large the architecture’s effects are, without bearing on their direction. First, the model is a theory of participation and harm, not of asset prices. Welfare is legitimate surplus net of external harm; the token’s price, its expected return, and the issuer’s cost of capital are not endogenized as objects that feed back into issuance and real investment. So the architecture’s effect on how much capital comes on-chain is shown to exist but not priced. Second, misconduct is a single population with one harm parameter. Real misconduct (fraud, laundering, sanctions evasion, manipulation, theft) is heterogeneous in how cleanly it leaves an on-chain trail and how far deterrence reaches it. The model’s detection rate averages over that heterogeneity, so it governs how much of the harm the ledger can reach, without changing which instrument reaches more of it. Both choices scale the result; neither turns it.

One choice could in principle overturn the central comparison: the assumption that enforcement falls only on offenders, so the ledger imposes no cost on legitimate participation. Real enforcement commits Type-I errors (legitimate users are sometimes flagged, investigated, or wrongly sanctioned) and a false-positive cost would give the ledger an exclusion cost of its own. Section 6.5 confronts this rather than deferring it, and the resolution is structural. False positives on a ledger are an artifact of ambiguity, and the same transparency that lets the record convict the guilty lets it exonerate the innocent. So the false-positive burden is governed by $1 - \theta$ and vanishes at full traceability. The consequence (Proposition 14) is that the gate is retired at $\theta = 1$ for a false-positive cost of *any* magnitude: the entire full-traceability allocation is invariant to it. The cost therefore delays the flip without preventing it, and in the calibration the flip threshold rises only from 0.66 to 0.76 as the false-positive intensity climbs to the external-harm scale. The asymmetry the paper rests on is thus not an assumption of perfect targeting. It is the limit of a relationship in which transparency makes the ledger accurate exactly as it makes it effective. What remains genuinely empirical is narrower: whether some Type-I cost is independent of traceability, a fixed compliance burden borne however clean the trail, which would leave a residual bounded by that component alone.

The remaining boundaries are not missing pieces of this theory but the research program it makes precise. The most immediate is empirical. The model names the objects to measure

(traceability, the composability premium, the adverse-selection scale, the undeterrable tail) and predicts the mechanism’s signature: a fold in market quality at the permissioning intensity where composability collapses, identifiable as a threshold in liquidity that shifts with screen quality. Behind the empirics lie the structures the reduced forms stand in for. Traceability enters as a scalar, but the number is itself the equilibrium of a cryptographic contest between privacy technologies and forensic techniques; modeling that contest would predict where the deterrence ceiling sits and how it moves as the technologies advance. Composability enters as connectivity, but the financial primitives whose composition is its substance (lending pools, automated market makers, derivatives) carry their own systemic hazards: oracle manipulation, liquidation cascades, flash-loan attacks. A theory of which compositions are benign and which are dangerous would refine the same premium the gate destroys. The dynamics are deterministic; a stochastic treatment would have to say how wide the basins remain under realistic noise, and how a regulator should act when it cannot observe which basin a market occupies. Enforcement is effort producing detection, not the legal process (evidentiary standards, appeals, litigation, and the budget an agency allocates across markets) that turns a detection into a penalty. The standard is an instrument, not a contested object; a fuller account would model protocol governance and the game among many sovereigns, each choosing traceability, gating, and enforcement, with the location of activity itself the equilibrium. And the agents who order and settle transactions (validators, sequencers, exchanges, whose incentives generate maximal-extractable value and censorship) are folded into the microstructure rather than modeled as the strategic players through whom compliance must ultimately run. Each is a margin a richer model would let vary. That the architecture of compliance governs the outcome with them held fixed is what the paper has shown. A mechanism-design treatment of the standard a regulator should write, together with the extension of the architecture lens to programmable assets beyond securities, is where the argument goes next.

The message we expect to carry beyond the securities setting is that the regulation of programmable, transparent assets is a question of architecture rather than stringency. The features usually catalogued as regulatory problems (disintermediation, permissionless access, public transaction histories) are, taken together, the ingredients of a better enforcement technology, and the architecture that fits it (open access, policed on a shared ledger, with screening written into the standard and reserved for the undeterrable tail) is not a rule imposed on the market but an equilibrium a credible regulator brings into being.

Appendix: Proofs

Throughout, subscripts denote partial derivatives. Recall the primitives $B = (1 - sg)\beta e^{-\rho\kappa}$, $N = (1 - \lambda g)\mu$, $\kappa = \theta(1 - e^{-e})K$, $\varphi = 1 - g$, $\tau = B/(N + B)$, $\pi_{\text{pc}} = u_0 + u_1\varphi^\alpha - \sigma\tau$, $S_L = N\pi_{\text{pc}}$ on $\{\pi_{\text{pc}} > 0\}$, and $W = S_L - \eta_E B - \frac{c_e}{2}e^2 - c_g g$. We use the elementary identities $\kappa_e = \theta K e^{-e} > 0$,

$\kappa_\theta = K(1 - e^{-e}) > 0$, $\tau_B = N/(N + B)^2 > 0$, $\tau_N = -B/(N + B)^2 < 0$, and $N_g = -\lambda\mu$.

A.1 Proof of Lemma 2

Differentiating $B = (1 - sg)\beta e^{-\rho\kappa}$: $B_g = -s\beta e^{-\rho\kappa} < 0$; $B_e = (1 - sg)\beta e^{-\rho\kappa}(-\rho\kappa_e) = -\rho\kappa_e B < 0$; $B_\theta = -\rho\kappa_\theta B < 0$. The offending content of the gate's suppression is $-B_g = s\beta e^{-\rho\kappa} = s\beta\bar{S}(\kappa)$, and $\partial\bar{S}(\kappa)/\partial e = -\rho\kappa_e\bar{S} < 0$, $\partial\bar{S}(\kappa)/\partial\theta = -\rho\kappa_\theta\bar{S} < 0$. For the ledger's suppression, $-B_e = \rho\kappa_e B = \rho\theta K e^{-e} B$, so

$$\frac{\partial(-B_e)}{\partial\theta} = \rho K e^{-e} B + \rho\theta K e^{-e} B_\theta = \rho K e^{-e} B(1 - \rho\theta\kappa_\theta) = \rho K e^{-e} B(1 - \rho\kappa),$$

using $\theta\kappa_\theta = \theta K(1 - e^{-e}) = \kappa$. Hence $\partial(-B_e)/\partial\theta > 0$ iff $\rho\kappa < 1$. $\square$

A.2 Proof of Lemma 3

Let $H = \{V = v_0 + \delta\}$ and $L_o = \{V = v_0 - \delta\}$, each of prior probability $\frac{1}{2}$. With probability π the arriving order is from an informed bad actor, who buys in state H and sells in state L_o ; with probability $1 - \pi$ it is from a liquidity trader, who buys or sells with probability $\frac{1}{2}$ each independent of V . Then

$$\Pr(\text{buy} \mid H) = \pi \cdot 1 + (1 - \pi)\frac{1}{2} = \frac{1+\pi}{2}, \quad \Pr(\text{buy} \mid L_o) = \pi \cdot 0 + (1 - \pi)\frac{1}{2} = \frac{1-\pi}{2},$$

so $\Pr(\text{buy}) = \frac{1}{2}\left(\frac{1+\pi}{2} + \frac{1-\pi}{2}\right) = \frac{1}{2}$. By Bayes' rule $\Pr(H \mid \text{buy}) = \frac{\frac{1+\pi}{2} \cdot \frac{1}{2}}{\frac{1}{2}} = \frac{1+\pi}{2}$ and $\Pr(L_o \mid \text{buy}) = \frac{1-\pi}{2}$. A competitive risk-neutral market maker sets the ask equal to the expected value conditional on a buy:

$$A = \mathbb{E}[V \mid \text{buy}] = (v_0 + \delta)\frac{1+\pi}{2} + (v_0 - \delta)\frac{1-\pi}{2} = v_0 + \delta\left(\frac{1+\pi}{2} - \frac{1-\pi}{2}\right) = v_0 + \delta\pi.$$

By the symmetric argument for a sell, $\underline{A} = \mathbb{E}[V \mid \text{sell}] = v_0 - \delta\pi$. Zero expected profit pins down these quotes uniquely under competition (any higher ask is undercut; any lower earns negative profit). The half-spread paid by a liquidity trader is $A - v_0 = \delta\pi$. $\square$

A.3 Proof of Proposition 1

W is continuous on $[0, 1] \times \mathbb{R}_+$ (a composition of continuous functions; the $\max\{\cdot, 0\}$ in S_L is continuous). For the bound, $W_e = \partial S_L / \partial e - \eta_E B_e - c_e e$. From $\partial S_L / \partial e = -\sigma N \tau_B B_e$ and $-B_e = \rho\kappa_e B$,

$$W_e = (\sigma N \tau_B + \eta_E) \rho\kappa_e B - c_e e \leq (\sigma + \eta_E) \rho K \beta - c_e e,$$

where we used $N \tau_B = N^2 / (N + B)^2 \leq 1$, $\kappa_e = \theta K e^{-e} \leq K$, and $B \leq \beta$. Hence $W_e < 0$ for every $e > \bar{e} \equiv (\sigma + \eta_E) \rho K \beta / c_e$, so no optimum has $e^* > \bar{e}$. Restricting to the compact rectangle

$[0, 1] \times [0, \bar{e}]$, W attains its maximum by the Weierstrass theorem. $\square$

A.4 Proof of Proposition 2

Both signs $B_g < 0$, $B_e < 0$ are Lemma 2. The common toxicity benefit is $\partial S_L / \partial(-B)|_\tau = -\partial(N\sigma\tau)/\partial B = \sigma N\tau_B > 0$. For the gate's direct effect, with $\tau_g = \tau_B B_g + \tau_N N_g$,

$$\frac{\partial S_L}{\partial g} = N_g \pi_{pc} + N \frac{\partial \pi_{pc}}{\partial g} = -\lambda \mu \pi_{pc} + N(-u_1 \alpha \varphi^{\alpha-1} - \sigma \tau_g).$$

Separating the $\tau_B B_g$ term (the toxicity benefit) from the remainder, the direct cost is

$$\left. \frac{\partial S_L}{\partial g} \right|_{\text{direct}} = -\lambda \mu \pi_{pc} - N u_1 \alpha \varphi^{\alpha-1} - \sigma N \tau_N N_g = \underbrace{-\lambda \mu \pi_{pc}}_{\text{exclusion}} - \underbrace{N u_1 \alpha \varphi^{\alpha-1}}_{\text{composability}} - \underbrace{\sigma \lambda \mu \frac{NB}{(N+B)^2}}_{\text{base-thinning}} < 0,$$

each term negative on $\{\pi_{pc} > 0\}$. For the ledger, $N_e = \varphi_e = 0$, so $\partial S_L / \partial e|_{\text{direct}} = 0$ and its sole cost in (4) is $c_e e$. $\square$

A.5 Proof of Proposition 3

From the proof of Proposition 1, $W_e = (\sigma N \tau_B + \eta_E)(-B_e) - c_e e$; setting $W_e = 0$ and writing $MV \equiv \sigma N \tau_B + \eta_E = \sigma N^2 / (N + B)^2 + \eta_E$ gives (6). From the proof of Proposition 2, the gate's total derivative is $W_g = (\sigma N \tau_B + \eta_E)(-B_g) - C_g - c_g$ with $C_g \equiv \lambda \mu \pi_{pc} + N u_1 \alpha \varphi^{\alpha-1} + \sigma \lambda \mu NB / (N + B)^2$; setting $W_g = 0$ gives (7). Both share the left side $MV \cdot (-B)$; the gate's right side exceeds the ledger's by $C_g > 0$. Finally $\partial C_g / \partial u_1 = N \alpha \varphi^{\alpha-1} > 0$, and by Lemma 2 $-B_e = \rho \kappa_e B$ is increasing in θ for fixed e whenever $\rho \kappa < 1$. $\square$

A.6 Proof of Proposition 4 and Lemma 4

Proof. Write $W_e = MV \cdot \rho \kappa_e B - c_e e$ and $W_g = MV \cdot s \beta \bar{S}(\kappa) - C_g - c_g$ (Appendix A.5), with $MV = \sigma N^2 / (N + B)^2 + \eta_E$, $\bar{S}(\kappa) = e^{-\rho \kappa}$, and $C_g = \lambda \mu \pi_{pc} + N u_1 \alpha \varphi^{\alpha-1} + \sigma \lambda \mu NB / (N + B)^2$. Note $MV_B = -2\sigma N^2 / (N + B)^3 < 0$, so $MV_e = MV_B B_e > 0$ and $MV_\theta = MV_B B_\theta > 0$ (both since $B_e, B_\theta < 0$). Two computed identities are used repeatedly:

$$MV_e = \frac{2\sigma \rho \kappa_e B N^2}{(N + B)^3}, \quad \frac{\partial C_g}{\partial e} = \frac{2\sigma \lambda \mu \rho \kappa_e B^2 N}{(N + B)^3} > 0,$$

the latter obtained by combining $\partial(\lambda \mu \pi_{pc}) / \partial e = \lambda \mu \sigma \rho \kappa_e B N / (N + B)^2$ with $\partial[\sigma \lambda \mu NB / (N + B)^2] / \partial e = -\sigma \lambda \mu \rho \kappa_e B N (N - B) / (N + B)^3$; the analogous identities hold with κ_θ in place of κ_e .

(i) *Toxicity bound.* On the feasible set $B/N = \frac{1-sg}{1-\lambda g} \cdot \frac{\beta}{\mu} e^{-\rho \kappa}$. The ratio $(1-sg)/(1-\lambda g)$ has derivative $(\lambda-s)/(1-\lambda g)^2 < 0$ (as $\lambda < s$), so it is maximized at $g = 0$ with value 1; and $e^{-\rho \kappa} \leq 1$, with equality at $e = 0$. Hence $B/N \leq \beta/\mu$, attained at $(g, e) = (0, 0)$, and $\tau = \frac{B/N}{1+B/N} \leq \frac{\beta/\mu}{1+\beta/\mu} = \frac{\beta}{\mu+\beta}$. If $\beta \leq \mu$ this is $\leq \frac{1}{2}$.

(iv) *Complementarity.* $W_{e\theta} = \rho \text{MV}(\kappa_{e\theta}B + \kappa_e B_\theta) + \rho \kappa_e B \text{MV}_\theta$. The last term is ≥ 0 . For the first, $\kappa_{e\theta} = Ke^{-e}$ and $\kappa_e B_\theta = \theta Ke^{-e}(-\rho \kappa_\theta B)$, so $\kappa_{e\theta}B + \kappa_e B_\theta = Ke^{-e}B(1 - \rho \theta \kappa_\theta) = Ke^{-e}B(1 - \rho \kappa)$, using $\theta \kappa_\theta = \kappa$. Thus $\rho \kappa < 1$ gives $W_{e\theta} > 0$. Since $\kappa = \theta(1 - e^{-e})K < K$, the primitive condition $\rho K \leq 1$ implies $\rho \kappa < 1$ everywhere; and $\rho \kappa < 1 \Leftrightarrow \bar{S}(\kappa) = e^{-\rho \kappa} > e^{-1}$, i.e. the offending fraction exceeds e^{-1} .

(ii) *Substitution.* $W_{ge} = s\beta \bar{S}(\kappa)(\text{MV}_e - \text{MV}\rho \kappa_e) - \partial C_g/\partial e$. Using the identity for MV_e ,

$$\text{MV}_e - \text{MV}\rho \kappa_e = \rho \kappa_e \left[\frac{2\sigma B N^2}{(N+B)^3} - \frac{\sigma N^2}{(N+B)^2} - \eta_E \right] = -\rho \kappa_e \left[\frac{\sigma N^2(N-B)}{(N+B)^3} + \eta_E \right],$$

which is ≤ 0 exactly when $\sigma N^2(N-B)/(N+B)^3 + \eta_E \geq 0$; this holds whenever $N \geq B$, i.e. $\tau \leq \frac{1}{2}$. The cost term $-\partial C_g/\partial e < 0$ by the identity above, reinforcing. Hence $W_{ge} < 0$. The identical computation with κ_θ gives $\text{MV}_\theta - \text{MV}\rho \kappa_\theta = -\rho \kappa_\theta [\sigma N^2(N-B)/(N+B)^3 + \eta_E] \leq 0$ under $\tau \leq \frac{1}{2}$, and $-\partial C_g/\partial \theta < 0$, so $W_{g\theta} < 0$. This proves Lemma 4 and part (ii).

(iii) *Concavity of enforcement.* $W_{ee} = \rho \kappa_e B [\text{MV}_e - \text{MV}(1 + \rho \kappa_e)] - c_e$, using $\kappa_{ee} = -\kappa_e$ and $B_e = -\rho \kappa_e B$. Now $\text{MV}_e = \frac{2\sigma \rho \kappa_e B N^2}{(N+B)^3} = 2\rho \kappa_e \frac{B}{N+B} \cdot \frac{\sigma N^2}{(N+B)^2} \leq 2\rho \kappa_e \tau \text{MV} \leq \rho \kappa_e \text{MV}$ under $\tau \leq \frac{1}{2}$, which is strictly less than $\text{MV}(1 + \rho \kappa_e)$. Hence the bracket is negative and $W_{ee} < 0$ (the term $-c_e$ only reinforces).

Reduction of the second-order condition. With $W_{ee} < 0$, define $V(g) = \max_e W(g, e)$ with maximizer $e(g)$. By the envelope theorem $V'(g) = W_g(g, e(g))$, and differentiating, $V''(g) = W_{gg} - W_{ge}^2/W_{ee} = (W_{gg}W_{ee} - W_{ge}^2)/W_{ee} = \det \mathbf{H}/W_{ee}$. Since $W_{ee} < 0$, $V''(g^*) < 0 \Leftrightarrow \det \mathbf{H} > 0$, so Assumption 1 is equivalent to local concavity of V at g^* . That V'' changes sign on $[0, 1]$ (welfare is not globally concave in g) is verified from the closed form: W_{gg} is positive at intermediate g where the convexity of φ^α and the toxicity response reinforce. The local condition holds at every interior optimum in the calibrated region. $\square$

A.7 Proof of Proposition 5

Monotonicity. Fix θ at an interior optimum (so $g^* \in (0, 1)$, $e^* > 0$) and let $\mathbf{H}$ be the Hessian. Total differentiation of the first-order conditions $W_g = W_e = 0$ in θ gives

$$\mathbf{H} \begin{pmatrix} dg^*/d\theta \\ de^*/d\theta \end{pmatrix} = - \begin{pmatrix} W_{g\theta} \\ W_{e\theta} \end{pmatrix}.$$

By Assumption 1, $\mathbf{H}$ is negative definite, so $\mathbf{H}^{-1} = \frac{1}{\det \mathbf{H}} \begin{pmatrix} W_{ee} & -W_{ge} \\ -W_{ge} & W_{gg} \end{pmatrix}$ with $\det \mathbf{H} > 0$. By Cramer's rule,

$$\frac{dg^*}{d\theta} = -\frac{W_{ee}W_{g\theta} - W_{ge}W_{e\theta}}{\det \mathbf{H}}, \quad \frac{de^*}{d\theta} = -\frac{-W_{ge}W_{g\theta} + W_{gg}W_{e\theta}}{\det \mathbf{H}}.$$

By Lemma 4, $W_{g\theta} < 0$, $W_{e\theta} > 0$, $W_{ge} < 0$; with $W_{ee}, W_{gg} < 0$ and $\det \mathbf{H} > 0$, the numerator of $dg^*/d\theta$ is $(<0)(<0) - (<0)(>0) = (+) - (-) > 0$, so $dg^*/d\theta < 0$; and the numerator of $de^*/d\theta$

is $-(< 0)(< 0) + (< 0)(> 0) = (-) + (-) < 0$, so $de^*/d\theta > 0$. Thus $g^*(\theta)$ is strictly decreasing and $e^*(\theta)$ strictly increasing on any interval of interior optima, proving (i) and the interior part of (iii).

Threshold. Define the marginal value of the gate at $g = 0$, given optimal enforcement,

$$\Delta(\theta) \equiv W_g(0, \hat{e}(\theta); \theta), \quad \hat{e}(\theta) \equiv \arg \max_{e \geq 0} W(0, e; \theta),$$

which is continuous in θ ($\hat{e}$ is continuous by the theorem of the maximum and W_g is continuous). By (7), $\Delta(\theta) = \text{MV}(0, \hat{e}) s \beta e^{-\rho \kappa^*} - C_g(0, \hat{e})$ with $\kappa^* = \theta(1 - e^{-\hat{e}})K$. The optimal gate is zero precisely when $\Delta(\theta) \leq 0$ (the constrained maximizer is at the lower corner). At $\theta = 0$ the ledger is inert ($p \equiv 0$, $\hat{e}(0) = 0$, $\kappa^* = 0$), so $\Delta(0) = \text{MV}(0, 0) s \beta - C_g(0, 0)$, which is positive under the maintained assumption that some compliance is warranted (i.e. $g^*(0) > 0$). As θ increases, $\hat{e}(\theta)$ is nondecreasing (the argument of part (i) applied at $g = 0$, where $W_{e\theta} > 0$ gives $d\hat{e}/d\theta \geq 0$), so $\kappa^*(\theta) = \theta(1 - e^{-\hat{e}(\theta)})K$ is strictly increasing and the deterrence factor $e^{-\rho \kappa^*}$ strictly decreasing; meanwhile $\text{MV} \leq \sigma + \eta_E$ is bounded and C_g is nondecreasing (lower toxicity raises π_{pc}). Hence the benefit term $\text{MV} s \beta e^{-\rho \kappa^*}$ is strictly decreasing and the cost C_g nondecreasing, so Δ is strictly decreasing. If $e^{-\rho \theta K}$ is small enough at $\theta = 1$ that $\Delta(1) < 0$ (which holds when the residual deterrence floor undercuts the gate's marginal cost), then by the intermediate value theorem there is a unique $\bar{\theta} \in (0, 1)$ with $\Delta(\bar{\theta}) = 0$, and $g^*(\theta) = 0$ for all $\theta \geq \bar{\theta}$, proving (ii). When the harm channel dominates ($\text{MV} \approx \eta_E$ constant), Δ is strictly decreasing without qualification, and g^* is nonincreasing throughout, giving (iii). $\square$

A.8 Proof of Proposition 6

(i) *Expansion.* Fix (g, e) . By Lemma 2, $B_\theta < 0$, and $\partial S_L / \partial \theta = -\sigma N \tau_B B_\theta > 0$; so raising θ moves (S_L, B) weakly northwest (more S_L , less B). Every efficient point under θ is thus weakly dominated by the same (g, e) evaluated at $\theta' > \theta$, giving $\mathcal{A}(\theta) \subseteq \mathcal{A}(\theta')$ in the efficient order.

(ii) *Domination.* On the permissionless locus $g = 0$: $N = \mu$, $\varphi = 1$, $S_L(0, e) = \mu(u_0 + u_1 - \sigma \tau(0, e))$, and $B(0, e) = \beta e^{-\rho \kappa(e, \theta)}$, strictly decreasing in e toward the floor $\underline{B}(\theta) = \beta e^{-\rho \theta K}$. Let (g^g, e^g) be any point on the gated frontier with $g^g > 0$. Its liquidity is capped by the composability it forfeits,

$$S_L(g^g, \cdot) = (1 - \lambda g^g) \mu(u_0 + u_1(1 - g^g)^\alpha - \sigma \tau^g) < \mu(u_0 + u_1 - \sigma \tau^g),$$

since $(1 - g^g)^\alpha < 1$ and $1 - \lambda g^g < 1$, and its bad activity is bounded below by the residual the imperfect gate admits, $B(g^g, \cdot) \geq (1 - s) \beta e^{-\rho \kappa} > 0$. Now choose θ large enough that $\underline{B}(\theta) = \beta e^{-\rho \theta K} \leq B(g^g, \cdot)$; pick e' with $B(0, e') \leq B(g^g, \cdot)$. At that e' , toxicity $\tau(0, e') = B(0, e') / (\mu + B(0, e'))$ is small, so $S_L(0, e')$ is close to $\mu(u_0 + u_1)$, which strictly exceeds the displayed cap on $S_L(g^g, \cdot)$. Hence $(S_L(0, e'), B(0, e'))$ strictly dominates the gated point: weakly lower B , strictly higher S_L . Taking the gated point to be the welfare optimum yields the final

clause. The threshold θ_0 is the least θ for which the composability premium retained at $g = 0$ exceeds the deterrence shortfall relative to the gate; for $\theta \geq \theta_0$, domination holds at every gated point. $\square$

A.9 Proof of Proposition 7

The game. Players are $m \geq 2$ venue operators $j = 1, \dots, m$, each choosing a local gate $g_j \in [0, 1]$; a single authority chooses ledger enforcement $e \geq 0$. All venues trade one composable token sharing one ledger. A bad actor admitted on any venue can transact system-wide, so it is screened from the system only if blocked at its laxest point of entry; system admission is $1 - s \min_j g_j$ and

$$B^{\text{sys}}(\mathbf{g}, e) = (1 - s \min_j g_j) \beta \bar{S}(\kappa(e, \theta)).$$

Venue j 's payoff is its own legitimate users' surplus net of its own composability cost and its share of harm; crucially, venue j bears the composability cost of *its own* gate (its users lose composability $1 - g_j$) but the suppression of B^{sys} is common to all venues. The equilibrium concept is Nash in $\mathbf{g}$ (with the authority best-responding in e).

The venue game has the compact strategy space $[0, 1]^m$ and continuous payoffs, so a Nash equilibrium exists by the Debreu–Glicksberg–Fan theorem; we characterize it. *(i) Weakest-link under-provision.* Fix the others' gates with minimum $g_{-j}^{\min} = \min_{k \neq j} g_k$. If $g_j > g_{-j}^{\min}$, then $\min_k g_k = g_{-j}^{\min}$ is locally invariant to g_j , so $\partial B^{\text{sys}} / \partial g_j = 0$, while raising g_j strictly lowers venue j 's composability $\varphi(g_j) = 1 - g_j$ and hence its legitimate surplus; venue j strictly prefers to lower g_j . Therefore no Nash profile has a venue strictly above the minimum, and all sit at a common g^{Nash} . At that level, the venue's marginal increase in its gate lowers B^{sys} by $s\beta\bar{S}(\kappa)$, a benefit accruing to all m venues' legitimate users and to third parties (external harm); venue j internalizes only its own users' share of the toxicity reduction and none of η_E . Its private first-order condition therefore equates its own composability-plus-exclusion cost to a strictly smaller benefit than the planner's condition (7), whose benefit is $MV = \sigma N \tau_B + \eta_E$ summed over the whole market. Hence $g^{\text{Nash}} < g^*$. A unilateral deviation to $g_j = 0$ sets $\min_k g_k = 0$, collapsing B^{sys} to its no-gate level regardless of the others.

(ii) Enforcement is immune. The single shared ledger enters every venue's B^{sys} through one common e ; an authority choosing e to maximize (4) internalizes the common effect, and there is no min aggregation and so no weakest-link failure.

(iii) Optimal locus. Writing the gate into the token's code as a common standard $g_j \equiv g_{\text{std}}$ removes the min channel, every venue shares one screen by construction, so the planner's gate g^* becomes implementable. Combined with global e , this attains (5), which weakly dominates any profile of independently chosen venue gates by definition of the planner's optimum. $\square$

A.10 Proof of Proposition 8

With deterrence capped at $\bar{\kappa} = \theta K < \infty$ (from $p \leq \theta$) or $\bar{\kappa} = p_{\max} \bar{K}$ (from $K \leq \bar{K}$), as e rises to its productive limit $\kappa \rightarrow \bar{\kappa}$ and $B \rightarrow (1 - sg)\beta e^{-\rho\bar{\kappa}}$, leaving a strictly positive undeterrable mass $\beta e^{-\rho\bar{\kappa}} > 0$ at $g = 0$. At the enforcement ceiling the planner's problem in g has first-order benefit of the first unit $MV(0) s\beta e^{-\rho\bar{\kappa}}$ and marginal cost $C_g(0) = \lambda\mu\pi_{pc} + \mu u_1\alpha + \sigma\lambda\mu^2\bar{B}/(\mu + \bar{B})^2$, with $\bar{B} = \beta e^{-\rho\bar{\kappa}}$. Since $MV(0) = \sigma\mu^2/(\mu + \bar{B})^2 + \eta_E$ is strictly increasing in η_E while $C_g(0)$ is independent of η_E , the benefit exceeds the cost (hence $g^* > 0$) iff

$$\eta_E > \bar{\eta}_E(\theta) \equiv \frac{C_g(0)}{s\beta e^{-\rho\bar{\kappa}}} - \frac{\sigma\mu^2}{(\mu + \bar{B})^2}.$$

For $\eta_E > \bar{\eta}_E(\theta)$ the optimal gate is strictly positive even at the enforcement ceiling and at $\theta \rightarrow 1$, and the optimal regime combines maximal feasible deterrence (the ledger) with a positive gate excluding the undeterrable tail. $\square$

A.11 Proof of Proposition 11

The bound. Let $x = B/N \geq 0$. Then

$$\sigma \frac{N^2(B - N)}{(N + B)^3} = \sigma \frac{N^2 \cdot N(x - 1)}{N^3(1 + x)^3} = \sigma h(x), \quad h(x) = \frac{x - 1}{(1 + x)^3}.$$

Differentiating, $h'(x) = \frac{(1 + x)^3 - (x - 1) \cdot 3(1 + x)^2}{(1 + x)^6} = \frac{(1 + x) - 3(x - 1)}{(1 + x)^4} = \frac{4 - 2x}{(1 + x)^4}$, which is positive for $x < 2$ and negative for $x > 2$, so h is maximized at $x = 2$ with $h(2) = 1/27$. Hence $\sigma N^2(B - N)/(N + B)^3 \leq \sigma/27$, attained at $x = 2$, i.e. $\tau = B/(B + N) = x/(1 + x) = 2/3$.

(i) *Global monotonicity.* From Appendix A.6, $W_{g\theta} = s\beta\bar{S}(\kappa)(MV_\theta - MV\rho\kappa_\theta) - \partial C_g/\partial\theta$ with $MV_\theta - MV\rho\kappa_\theta = \rho\kappa_\theta[\sigma N^2(B - N)/(N + B)^3 - \eta_E]$. By the bound, the bracket is $\leq \rho\kappa_\theta[\sigma/27 - \eta_E] \leq 0$ when $\eta_E \geq \sigma/27$, and $\partial C_g/\partial\theta > 0$, so $W_{g\theta} < 0$ for every (g, e, θ) and every (β, σ) . The argument is symmetric for the substitution sign: $MV_e - MV\rho\kappa_e = -\rho\kappa_e[\sigma N^2(N - B)/(N + B)^3 + \eta_E]$, and since $\sigma N^2(N - B)/(N + B)^3 \geq -\sigma/27$ by the same bound, the bracket is $\geq \eta_E - \sigma/27 \geq 0$, so $W_{ge} < 0$. With $\rho\kappa < 1$ giving $W_{e\theta} > 0$ (Appendix A.6) and Assumption 1 giving $\mathbf{H}$ negative definite, the Cramer expressions of Appendix A.7 yield $dg^*/d\theta < 0$ and $de^*/d\theta > 0$ throughout, with no appeal to $\beta \leq \mu$.

(ii) *The reversal regime.* A positive benefit term $MV_\theta - MV\rho\kappa_\theta > 0$ requires $\sigma N^2(B - N)/(N + B)^3 > \eta_E$, hence $\eta_E < \sigma/27$ and (by the shape of h) toxicity near $2/3$; even then $-\partial C_g/\partial\theta < 0$ works against it. Should $W_{g\theta} > 0$ obtain, $dg^*/d\theta = -(W_{ee}W_{g\theta} - W_{ge}W_{e\theta})/\det \mathbf{H}$ still carries the term $W_{ge}W_{e\theta} < 0$ (substitution through enforcement), which keeps the numerator positive and hence $dg^*/d\theta < 0$ for every $\eta_E > 0$. At $\eta_E = 0$ exactly, $\pi_{pc} < 0$ at low θ for σ large makes $S_L = 0$ and the planner indifferent at $g = 0$ (welfare zero); raising θ eventually makes an active market feasible and the optimum jumps to a positive gate, the sole source of

non-monotonicity. $\square$

A.12 Proof of Proposition 12

Both parts restate established results. (i) is Proposition 8: for $\eta_E > \bar{\eta}_E(\theta)$ the optimal gate is strictly positive at the enforcement ceiling and as $\theta \rightarrow 1$, since the undeterrable mass $\beta e^{-\rho\bar{\kappa}} > 0$ persists and $MV(0)$ is increasing in η_E (Appendix A.10). (ii) follows from the evasion analysis of a companion paper: as $\eta_{\max} \rightarrow 1$, effective detection at the margin vanishes, so the threshold function $\Delta(\theta) = MV s \beta \bar{S}(\kappa^{\text{eff}}) - C_g$ stays positive for all $\theta < 1$ and $\bar{\theta} \rightarrow 1$. In each case $g^*(\theta) > 0$ as $\theta \rightarrow 1$. $\square$

A.13 Proof of Proposition 13

With $\varphi(g, \theta) = 1 - g - \psi_L \theta$, the premium is $u_1 \varphi^\alpha$ and per-capita surplus $\pi_{\text{pc}} = u_0 + u_1(1 - g - \psi_L \theta)^\alpha - \sigma\tau$. At $\psi_L = 0$ this is the baseline and the flip is Proposition 5: $dg^*/d\theta < 0$ on the active region and $g^*(\theta) = 0$ for $\theta \geq \bar{\theta}$. The objective $W(g, e; \theta, \psi_L)$ is jointly continuous in (g, e, θ, ψ_L) and twice differentiable in (g, e) with the Hessian negative definite at the baseline optimum (Assumption 1); by the implicit function theorem the optimizer (g^*, e^*) and the comparative static $dg^*/d\theta$ are continuous in ψ_L . Since $dg^*/d\theta < 0$ strictly at $\psi_L = 0$, there is a neighborhood $[0, \bar{\psi})$ on which $dg^*/d\theta < 0$ persists and the zero-gate region $\{\theta \geq \bar{\theta}(\psi_L)\}$ remains nonempty; this is the flip, establishing the first claim.

For the reversal, consider $\psi_L \rightarrow 1$. Then $\varphi(0, \theta) = 1 - \psi_L \theta \rightarrow 1 - \theta$, so at high traceability $\varphi \rightarrow 0$ and the composability premium collapses: $S_L \rightarrow N(u_0 - \sigma\tau)^+$, which is zero once $\sigma\tau > u_0$. With legitimate surplus extinguished by the ledger's own composability cost, welfare reduces to $-\eta_E B - \frac{c_e}{2} e^2 - c_g g$, maximized by suppressing B ; since the gate cuts B at rate s and the composability it would destroy is already gone, $g^* \rightarrow 1$ at high θ . Hence for ψ_L near one the optimal gate rises rather than falls in θ : the flip reverses. By continuity of $dg^*/d\theta$ in ψ_L and the strict signs at the two ends, there is a threshold $\bar{\psi} \in (0, 1)$ separating the regimes. The economic content is that the reversal requires the ledger to be nearly as contagious as the gate; Lemma 1 rules this out, since a precondition removes a partner together with its downstream closure while a postcondition removes at most the reporting-incompatible compositions, giving $\psi_L < 1$ strictly and, in the calibration, ψ_L well below the numerically located $\bar{\psi} \approx 0.9$. $\square$

A.14 Proof of Proposition 14

(i) With $S_L = N\pi_{\text{pc}}$ and $\pi_{\text{pc}} = u_0 + u_1 \varphi^\alpha - \sigma\tau - \Lambda(e, \theta)$, the direct channel (holding the toxicity τ fixed) gives $\partial S_L / \partial e|_{\text{direct}} = -N \partial \Lambda / \partial e = -N c_I (1 - \theta) e^{-e}$, which is ≤ 0 and vanishes iff $\theta = 1$. (ii) $\partial \kappa / \partial \theta = (1 - e^{-e})K > 0$ and $\partial \Lambda / \partial \theta = -c_I (1 - e^{-e}) \leq 0$. (iii) At $\theta = 1$, $\Lambda(e, 1) = c_I (1 - 1)(1 - e^{-e}) = 0$ for every e and c_I , so $W(g, e, 1; c_I) = W(g, e, 1; 0)$ identically in (g, e) ; the maximizer at $\theta = 1$ is therefore independent of c_I , and in particular $g^*(1; c_I) =$

$g^*(1;0) = 0$, the last equality being the baseline flip (Proposition 5), which places $\bar{\theta}(0) < 1$. For the monotonicity, the enforcement first-order condition (6) becomes $(\sigma N \tau_B + \eta_E)(-B_e) = c_e e + N \partial \Lambda / \partial e$, whose right-hand side is increasing in c_I ; hence $e^*(\theta; c_I)$ is nonincreasing in c_I for $\theta < 1$, so $\kappa^* = \theta(1 - e^{-e^*})K$ falls and $\bar{S}(\kappa^*) = e^{-\rho \kappa^*}$ rises, raising the gate's marginal value $\Delta(\theta) = \text{MV } s \beta \bar{S}(\kappa^*) - C_g$ pointwise; the crossing $\Delta(\bar{\theta}) = 0$ thus moves weakly right, so $\bar{\theta}(c_I)$ is nondecreasing in c_I . Continuity of g^* in θ together with $g^*(1; c_I) = 0$ gives $\bar{\theta}(c_I) < 1$. $\square$

A.15 Proof of Propositions 15, 16, and 17

Existence and uniqueness (Proposition 15). The map $\Phi(s) = \delta B / (B + N e^{-\xi(\varsigma - \psi)^+})$ is continuous and nondecreasing on the compact interval $[0, \delta]$, with $\Phi(0) = \delta B / (B + N) > 0$ and $\Phi(\delta) = \delta B / (B + N e^{-\xi(\delta - \psi)}) \leq \delta$; hence Φ maps $[0, \delta]$ into itself and has a fixed point by Brouwer. On $\{s \leq \psi\}$ the exponent is truncated to zero, so $\Phi \equiv \delta \tau$ is constant; a fixed point lies in this region iff $\delta \tau \leq \psi$, and then $\varsigma^* = \delta \tau$ is the unique full-participation equilibrium. On $\{\varsigma > \psi\}$,

$$\Phi'(s) = \frac{\delta B N \xi e^{-\xi(\varsigma - \psi)}}{(B + N e^{-\xi(\varsigma - \psi)})^2} = \xi \Phi(s) \frac{N e^{-\xi(\varsigma - \psi)}}{B + N e^{-\xi(\varsigma - \psi)}} = \xi \Phi(s) (1 - \pi) \leq \xi \delta \cdot \frac{1}{4},$$

using $\Phi \leq \delta$ and $\pi(1 - \pi) \leq \frac{1}{4}$. Thus Φ is a contraction whenever $\xi \delta < 4$ (satisfied by the maintained parameters), so the equilibrium on $\{\varsigma > \psi\}$ is unique; combined with the previous step, the equilibrium is unique in each regime, giving the dichotomy.

Breakdown threshold. Under endogenous participation a legitimate trader of type x trades iff $x \geq \varsigma - \psi$, so participating volume is $L = N \bar{G}((\varsigma - \psi)^+) = N e^{-\xi(\varsigma - \psi)^+}$ and the toxic share is $\pi = B / (B + L)$. By Lemma 3 the half-spread solves (10). At full participation $L = N$, the candidate spread is $s_{\text{full}} = \delta B / (B + N) = \delta \tau$. This is an equilibrium iff the lowest type $x = 0$ does not exit, i.e. $0 + \psi \geq \varsigma_{\text{full}}$, that is $\delta \tau \leq \psi$. Rearranging, $\delta B \leq \psi(B + N) \Leftrightarrow B(\delta - \psi) \leq \psi N \Leftrightarrow B \leq \psi N / (\delta - \psi)$ for $\delta > \psi$ (if $\delta \leq \psi$ the spread never reaches ψ and the market is always healthy). If $s_{\text{full}} > \psi$, the zero-value traders exit; then $L < N$, $\pi > \tau$, and ς rises, inducing further exit, so the equilibrium lies in $\{\varsigma > \psi\}$ with $L = N e^{-\xi(\varsigma - \psi)} < N$ and discontinuously lower surplus.

Only enforcement reliably prevents breakdown. (i) As $g \rightarrow 1$, $\psi = u_1 \varphi^\alpha = u_1(1 - g)^\alpha \rightarrow 0$, so $\bar{B} = \psi N / (\delta - \psi) \rightarrow 0$, while imperfect screening leaves $B \geq (1 - s) \beta \bar{S}(\kappa) > 0$; hence $B > \bar{B}$ and the market is broken. For interior g , $d\bar{B}/dg$ carries the factors $\psi'(g) < 0$ and $N'(g) < 0$, so closing the gate lowers the threshold as it lowers B , and $B - \bar{B}$ can remain positive throughout, as in Figure 5(a) where $\delta \tau(g) > \psi(g)$ for all g at $e = 0$. (ii) Holding g fixed, $B(g, e)$ is strictly decreasing in e with limit $(1 - sg) \beta e^{-\rho \theta K}$, while $\bar{B}(N, \psi)$ is independent of e . Whenever this floor lies below $\bar{B}$ (which holds at full composability $g = 0$ for θ above a low bound, since $\beta e^{-\rho \theta K} < u_1 N / (\delta - u_1)$ for the relevant parameters) a finite e achieves $B(g, e) \leq \bar{B}$ and restores health. Thus enforcement moves the market toward the healthy region (lowering B at fixed $\bar{B}$), whereas the gate moves both boundaries together and can fail to. $\square$

A.16 Proof of Proposition 18

By Lemma 3 and the breakdown analysis above, full participation ($L = N$) is an equilibrium iff the lowest-value trader does not exit, $\psi(\varphi) \geq s_{\text{full}} = \delta \tau_{\text{full}}$, where $\tau_{\text{full}} = B/(B + N)$. Along the pure-gate path, $\psi(g) = u_1(1 - g)^\alpha$ is strictly decreasing, and $\tau_{\text{full}}(g)$ is decreasing because the gate cleans the order flow ($d\tau/dg < 0$ from Appendix A.6, using $\lambda < s$); but with a weakly selective gate (λ near s) the premium falls faster than the load, since $|\psi'(g)| = u_1\alpha(1 - g)^{\alpha-1}$ while $|\delta\tau'_{\text{full}}(g)| \propto (s - \lambda)$ is small. Hence $\psi(g) - \delta\tau_{\text{full}}(g)$ is positive at $g = 0$ (a viable baseline) and crosses zero at a unique $g_c \in (0, 1)$ defined by $\psi(g_c) = \delta\tau_{\text{full}}(g_c)$. For $g > g_c$ the full-participation candidate violates the lowest-trader condition, so by the fixed-point structure of (10) the only equilibrium is the high-spread, low-participation one of Proposition 16: spread, participation, and welfare jump at g_c . Along the ledger path, raising e lowers B and hence τ_{full} , lowering $s_{\text{full}} = \delta\tau_{\text{full}}$ while leaving ψ and N untouched; so the inequality $\psi \geq \delta\tau_{\text{full}}$ is preserved or restored, the full-participation equilibrium persists, and spread and participation are continuous in e . For non-substitutability, fix a target market quality requiring active mass at least $\underline{N}$ with spread at most $\bar{s}$; there exist (N, ψ, θ) such that this is reached by some finite e at $g = 0$ (since e drives B to its floor without thinning N or ψ) but is unreachable by any g : small g leaves B too high, and $g \geq g_c$ collapses participation below $\underline{N}$. Hence no gate intensity substitutes for the ledger. $\square$

A.17 Proof of Proposition 19

(i) $\Phi(g) = u_1(1 - g)^\alpha - \delta B/(B + N)$ with $B = (1 - sg)\beta\bar{S}$, $N = (1 - \lambda g)\mu$. The baseline is viable by hypothesis, $\Phi(0) > 0$. At $g = 1$, $\psi(1) = u_1 \cdot 0 = 0$ while $\tau_{\text{full}}(1) = (1 - s)\beta\bar{S}/[(1 - s)\beta\bar{S} + (1 - \lambda)\mu] > 0$ for $s < 1$, so $\Phi(1) = -\delta\tau_{\text{full}}(1) < 0$. By the intermediate value theorem a root $g_c \in (0, 1)$ exists; uniqueness on the relevant range follows because ψ is concave-decreasing ($\alpha < 1$) and $\delta\tau_{\text{full}}$ is a monotone Möbius function of g , so Φ changes sign once (verified across $\lambda \in (0, s)$ in the calibration). (ii) Differentiating, $\partial\Phi/\partial\lambda = -\delta \partial[B/(B + N)]/\partial\lambda = -\delta \cdot \mu g B/(B + N)^2 < 0$ for $g > 0$ (since $\partial N/\partial\lambda = -\mu g$ raises toxicity). At the crossing Φ passes from positive to negative, so $\partial\Phi/\partial g < 0$ there; the implicit function theorem gives $\partial g_c/\partial\lambda = -(\partial\Phi/\partial\lambda)/(\partial\Phi/\partial g) < 0$. As $\lambda \rightarrow 0$, $\tau_{\text{full}}(g)$ falls fastest in g and Φ stays positive until ψ is nearly exhausted, so $g_c \rightarrow 1$; as $\lambda \rightarrow s$, τ_{full} is nearly flat and g_c moves interior. (iii) The participation-naïve optimum $g^\dagger$ maximizes the full-participation welfare, in which η_E enters only through $-\eta_E B$ with $\partial B/\partial g < 0$; hence the marginal value of the gate is increasing in η_E and $g^\dagger(\eta_E)$ is nondecreasing. Since the true (participation-aware) welfare coincides with the naïve one for $g < g_c$ and drops discontinuously at g_c , the constrained optimum is $g^\dagger$ when $g^\dagger < g_c$ and is pinned at g_c^- (with the remaining suppression delegated to e) when $g^\dagger \geq g_c$. Monotonicity of $g^\dagger$ yields a threshold η_E^{fold} with $g^\dagger(\eta_E^{\text{fold}}) = g_c$ separating the two. $\square$

A.18 Proof of Lemma 1

Composability is $\varphi = 1 - g$, a function of g alone; hence $\partial\varphi/\partial g = -1$ and $\partial\varphi/\partial e = \partial\varphi/\partial\theta = 0$, and the premium $u_1\varphi^\alpha$ inherits these signs. The structural content is that the gate is a precondition on interaction and detection a postcondition. Write $\mathcal{C}(x)$ for the composition closure of a position x : the set of protocols and counterparties reachable from it by chained composition. A gate that refuses admission to x removes all of $\mathcal{C}(x)$, since nothing downstream of an excluded node can be reached; a detection rule conditions a penalty on x 's realized conduct and removes no element of $\mathcal{C}(x)$, since every composition remains available and only its misuse is punished after the fact. The asymmetry $\partial\varphi/\partial g < 0 = \partial\varphi/\partial e$ is thus a property of the order of operations, screen-then-interact versus interact-then-detect, rather than of the functional form, and Proposition 13 shows the results need only its direction. $\square$

A.19 Proof of Proposition 9

With identity signal $x \sim N(\mu_L, 1)$ for legitimate agents and $x \sim N(\mu_B, 1)$ for bad agents, $\mu_B < \mu_L$, and admission when $x \geq t$, the exclusion rates are $sg = \Pr(x < t \mid \text{bad}) = \Phi(t - \mu_B)$ and $\lambda g = \Pr(x < t \mid \text{legit}) = \Phi(t - \mu_L)$. Hence the screening edge is

$$(s - \lambda)g = \Phi(t - \mu_B) - \Phi(t - \mu_L) = \int_{t - \mu_L}^{t - \mu_B} \phi(u) du > 0 \quad (\mu_B < \mu_L),$$

with ϕ the standard normal density. The integral is the mass ϕ assigns to an interval of width $\mu_L - \mu_B$ located at $t - \mu_L$; differentiating, $\partial[(s - \lambda)g]/\partial\mu_B = -\phi(t - \mu_B) < 0$, so widening the separation $\mu_L - \mu_B$ (lowering μ_B) strictly raises the edge. Forgery shifts the bad signal toward the legitimate distribution, $\mu_B \uparrow \mu_L$; then $\Phi(t - \mu_B) \downarrow \Phi(t - \mu_L)$, so $sg \rightarrow \lambda g$ and $s \rightarrow \lambda$, and at $\mu_B = \mu_L$ the two signal distributions coincide and the gate admits both populations at the identical rate, screening no one. The ledger's deterrence operates through $\kappa = p(e, \theta)K$, a function of realized on-chain conduct that does not contain the identity signal, so $\partial\kappa/\partial\mu_B = 0$: deterrence is invariant to forgery. $\square$

A.20 Proof of Proposition 10

Fix θ and let $w(g) \equiv \max_{e \geq 0} W(g, e; \theta)$ be welfare with enforcement optimized at each g . The inner maximand is continuous and coercive in e (the cost $\frac{c_e}{2}e^2$ dominates the bounded gains), so $e^*(g)$ exists; on the interior region $\{\pi_{pc} > 0\}$ the objective is real-analytic in (g, e) with the enforcement Hessian negative (Assumption 1), so by the implicit function theorem $e^*(g)$ is analytic and, by the envelope theorem, w is analytic with $w'(g) = \partial W/\partial g|_{e=e^*(g)}$. Its critical

points are therefore isolated and finite on $[0, 1]$. Decompose

$$w'(g) = \underbrace{-\lambda\mu\pi_{\text{pc}}}_{\text{exclusion } (<0)} \underbrace{-N u_1 \alpha (1-g)^{\alpha-1}}_{\text{composability } (<0)} \underbrace{-\sigma N \tau_g - \eta_E B_g}_{\text{cleansing } (>0, \text{ since } \tau_g, B_g < 0)}.$$

The exclusion and composability terms are negative; their magnitude is bounded and the composability term diverges as $g \rightarrow 1$. The cleansing term is positive and, because $B = (1-sg)\beta\bar{S}(\kappa)$ enters τ and harm through the single decreasing factor $1-sg$, is monotone in g at fixed e^* . Hence w' has at most two sign changes on $[0, 1]$, so w has at most one interior local maximum g_{int}^* in addition to the boundary candidate $g = 0$; both are genuine local maxima on an open set of θ , the corner when the exclusion-plus-composability cost exceeds the cleansing benefit at $g = 0^+$ (so $w'(0^+) < 0$), the interior point when external harm is large enough to revive the cleansing benefit at higher g (so w' returns to zero from above with $w'' < 0$). The global optimum is $\arg \max\{w(0), w(g_{\text{int}}^*)\}$. Finally, by the envelope theorem $\partial w(g)/\partial \theta = -\sigma N \tau_\theta - \eta_E B_\theta$, which at $g = 0$ equals $(\sigma N \tau_B + \eta_E) \rho \kappa_\theta B > 0$ since $\kappa_\theta = p_\theta K > 0$ and $\tau_B > 0$; at $g_{\text{int}}^* > 0$ the factor $1-sg$ has already cut B , so B and its θ -derivative are smaller in magnitude, giving $\partial w(0)/\partial \theta > \partial w(g_{\text{int}}^*)/\partial \theta$. Thus $w(0) - w(g_{\text{int}}^*)$ is strictly increasing in θ and, being negative at low θ and positive at high θ , crosses zero exactly once, at $\bar{\theta}$; there the global maximizer jumps from $g_{\text{int}}^* > 0$ to 0. The discontinuity is the flip: a concave w would possess a unique critical point varying continuously in θ and could produce no jump. $\square$

A.21 Proof of Proposition 20

Each strategy set $X_i = [0, 1] \times [0, \bar{e}]$ is nonempty, compact, and convex, with $\bar{e}$ the enforcement bound of Proposition 1; the product $X = \prod_i X_i$ is compact. Because the screening rate satisfies $s < 1$, the displacement denominator obeys $\sum_j (1-sg_j) \geq N(1-s) > 0$ everywhere on X , so every share B_i , every toxicity τ_i , and hence every payoff U_i is a continuous function on X (a composition of continuous maps, the participation floor $[\cdot]^+$ continuous). Since each U_i is continuous on the compact X , Glicksberg's theorem for continuous games yields a Nash equilibrium; since it is symmetric (issuer i enters U_i only through its own action and the symmetric aggregates $\bar{g}$ and E) the symmetric form of the theorem yields a symmetric equilibrium. Enforcement enters U_i strictly concavely (a convex effort cost against a deterrence benefit concave in e_i), so the enforcement best response is unique and, by the maximum theorem, continuous in the profile. Where the issuer's concentrated gate objective $\tilde{U}_i(g_i) \equiv \max_{e_i} U_i$ is single-peaked: the regularity maintained for the planner's program (Assumption 1): the gate best response is single-valued and continuous, so the symmetric best-response map is a continuous self-map of the compact convex $[0, 1] \times [0, \bar{e}]$ and Brouwer's theorem delivers a pure-strategy symmetric equilibrium. The only profiles at which $\tilde{U}_i$ is not single-peaked are the flip knife-edges of Proposition 10, where the issuer is indifferent between a gate-reliant and a ledger-reliant response; there the symmetric equilibrium randomizes over the two architectures: the counterpart of a population in which a

fraction of issuers gate. Finally the planner maximizes $\sum_i U_i$ (with the social η_E) over the same compact X , so by Proposition 1 a maximizer $(\bar{g}^*, E^*)$ exists. The decentralized and planner allocations compared in Propositions 21–23 are therefore well defined. $\square$

A.22 Proof of Propositions 21, 22, and 23

Setup. With N symmetric issuers, total bad activity is $B_{\text{tot}}(E) = \beta e^{-\rho\kappa(E)}$, $\kappa(E) = \theta(1 - e^{-E})K$, $E = \sum_i e_i$, and issuer i carries the share $B_i = B_{\text{tot}}(E)(1 - sg_i)/\sum_j(1 - sg_j)$, so $\sum_i B_i = B_{\text{tot}}(E)$ identically (Section 9). Composability is the shared network value $u_1 S(\bar{g})$, $\bar{g} = \frac{1}{N} \sum_i g_i$, $S' < 0$. Issuer i maximizes $U_i = N_i[u_0 + u_1 S(\bar{g}) - \sigma\tau_i] - \ell B_i - \frac{c_e}{2} e_i^2$ with $N_i = (1 - \lambda g_i)\mu$, $\tau_i = B_i/(N_i + B_i)$, and private liability weight $\ell \geq \eta_E$; the planner maximizes $\sum_i U_i$ with ℓ replaced by η_E , internalizing $\bar{g}$ and E .

Proposition 21. Because $\sum_i B_i = B_{\text{tot}}(E)$ depends on the gates not at all, $\partial(\sum_i B_i)/\partial g_k = 0$: aggregate bad activity is gate-independent (the waterbed), exact while $\sum_j(1 - sg_j) > 0$. The planner's marginal value of any gate is then $\partial \sum_i U_i / \partial g_k = u_1 S'(\bar{g}) \frac{1}{N} \sum_i N_i - \lambda\mu\pi_k < 0$, gating destroys shared composability and excludes legitimate users with no offsetting reduction in total harm, so $\bar{g}^* = 0$. Issuer k instead sees $\partial U_k / \partial g_k = u_1 S'(\bar{g}) \frac{1}{N} N_k - \lambda\mu\pi_k - \ell \partial B_k / \partial g_k$, where $\partial B_k / \partial g_k = B_{\text{tot}} \partial_{g_k}[(1 - sg_k)/\sum_j(1 - sg_j)] < 0$ is the own-share shed onto others. The shedding term is $O(1)$ while the internalized composability loss is $O(1/N)$; hence for $N > 1$ and appreciable ℓ the issuer's marginal value of gating is positive at $g_k = 0$, giving $g_{\text{Nash}} > 0 = g^*$, and as N grows the $O(1/N)$ restraint vanishes while the shedding benefit persists, so g_{Nash} rises in N . At $N = 1$, $\sum_j(1 - sg_j) = 1 - sg_1$ and $B_1 = B_{\text{tot}}$, so $\partial B_1 / \partial g_1 = \partial B_{\text{tot}} / \partial g_1 = 0$ and the wedge is absent.

Proposition 22. $B_{\text{tot}}(E)$ is strictly decreasing in E ($\partial B_{\text{tot}} / \partial E = -\rho\kappa'(E)B_{\text{tot}} < 0$), and each e_i enters only through E , so issuer k 's enforcement lowers B_i for every i . The planner equates the full social marginal benefit to marginal cost, $-\eta_E \partial B_{\text{tot}} / \partial E = c_e e^*$; issuer k internalizes only the harm reduction on its own share, $-\ell \partial B_k / \partial e_k = c_e e_{\text{Nash}}$, omitting the benefit conferred on the other $N - 1$ markets. Since the omitted external benefit is positive, $e_{\text{Nash}} < e^*$ for $N > 1$, with the gap widening in N as the omitted fraction grows. Then $E_{\text{Nash}} = N e_{\text{Nash}} < E^*$ gives $B_{\text{tot}}(E_{\text{Nash}}) > B_{\text{tot}}(E^*)$: decentralization sustains strictly more total harm while gating strictly more.

Proposition 23. Both distortions are externalities operating through shared objects: $\bar{g}$ enters every issuer's composability and E enters every issuer's harm, while issuer k controls only (g_k, e_k) and internalizes a $1/N$ (composability) or own-share (enforcement) fraction of the effect. An instrument confined to a single issuer's market leaves both wedges intact; only instruments on the shared objects (a standard capping $\bar{g}$ and enforcement provision on the common ledger setting E) implement $\bar{g}^* = 0$ and E^* . Optimal regulation therefore operates on the shared infrastructure. $\square$

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